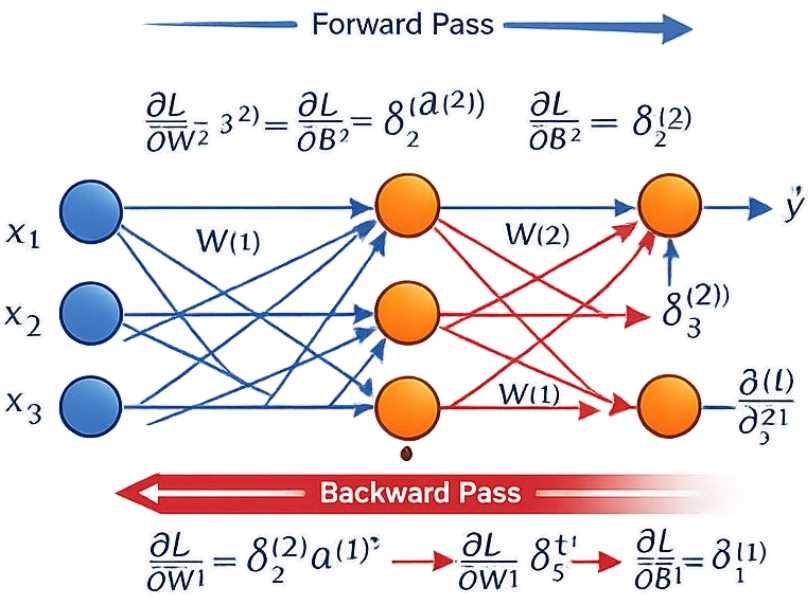
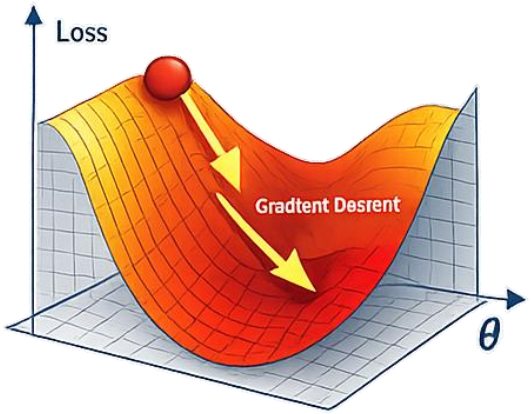
# Lecture

**March 3rd, 2026**

# Backpropagation in Neural Networks



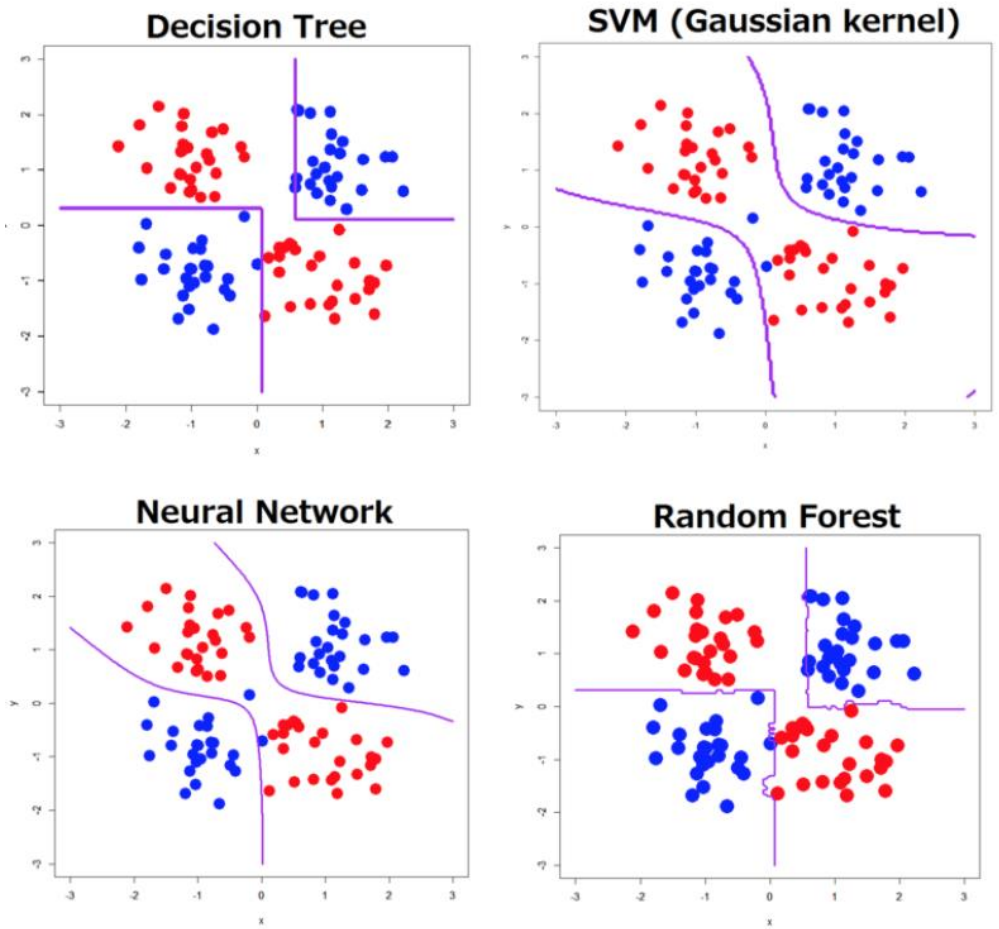


XU Exponential University of Applied Sciences | Data Science & AI

**ORHAN KONAK**

# Why Neural Networks?

- Many ML tasks require modeling **nonlinear relationships**
- Linear models often insufficient
- Neural networks = universal function approximators
- Used in vision, NLP, medicine, forecasting, etc.




Question: *How does it learn?*

# Agenda

- Neural Networks
- Gradient Descent
- Backpropagation

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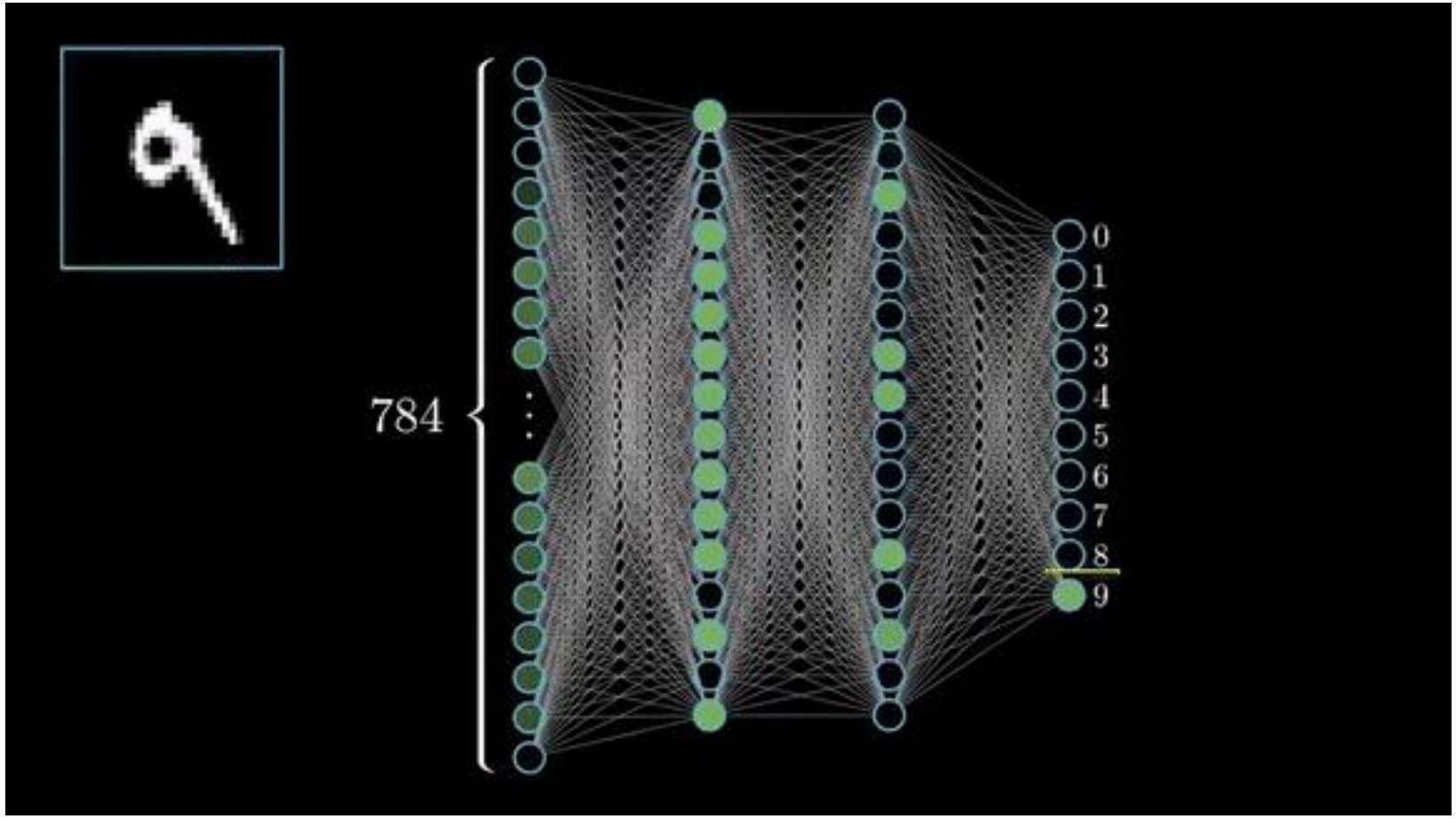
- **Neural Networks**
- Gradient Descent
- Backpropagation

# Neural Network Example

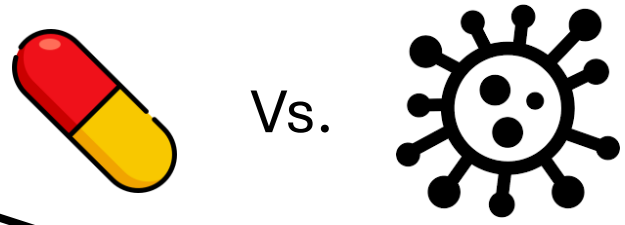



https://medium.com/analytics-vidhya/applying-ann-digit-and-fashion-mnist-13accfc44660

# Neural Network Example

- E.g. Drug Dosage

We gave drug to 3 different groups of people with 3 different **Dosages**



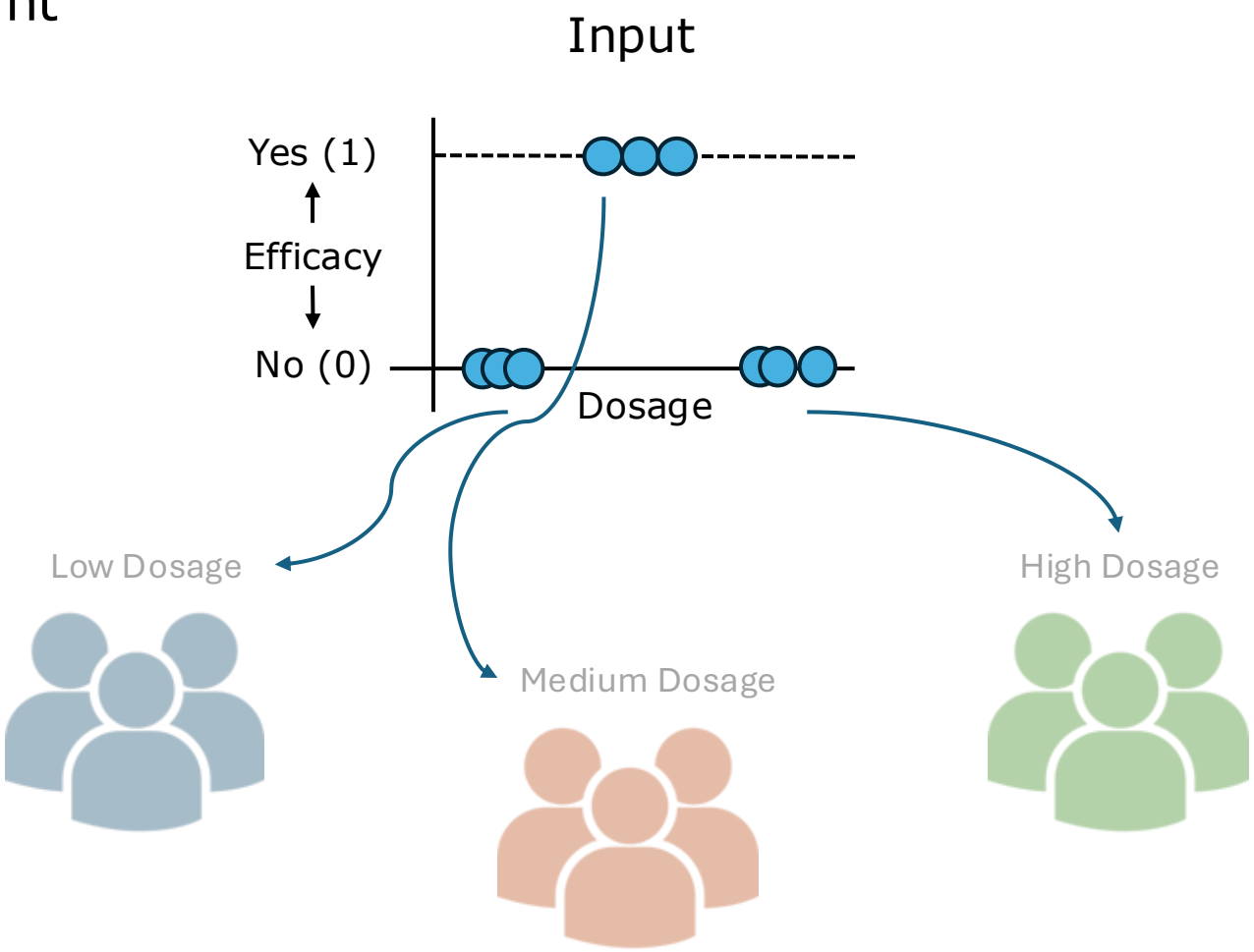

Low Dosage

Medium Dosage

High Dosage

# Neural Network Example

We can plot any point



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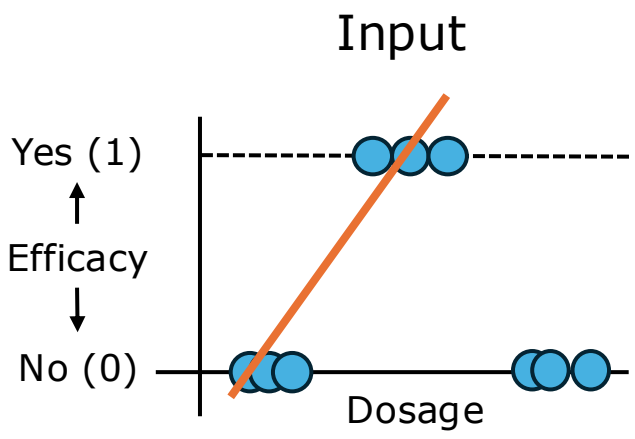
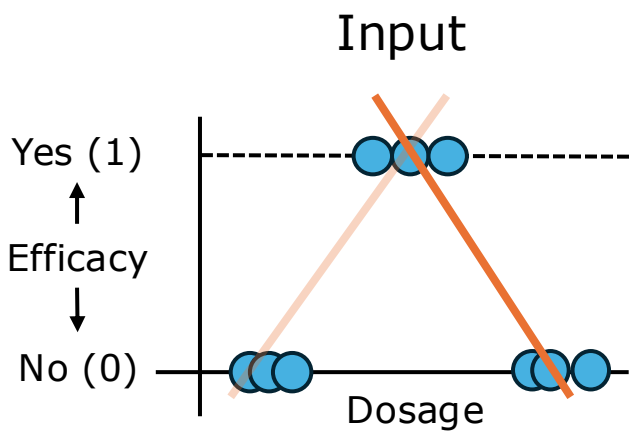



However, we can't just fit a **straight line** to the data to make predictions …

# Neural Network Example

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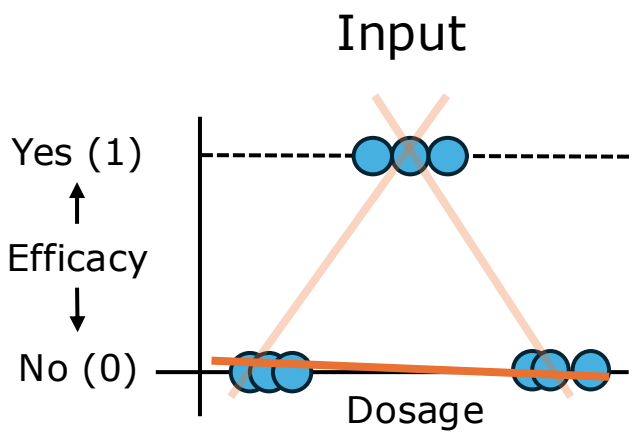


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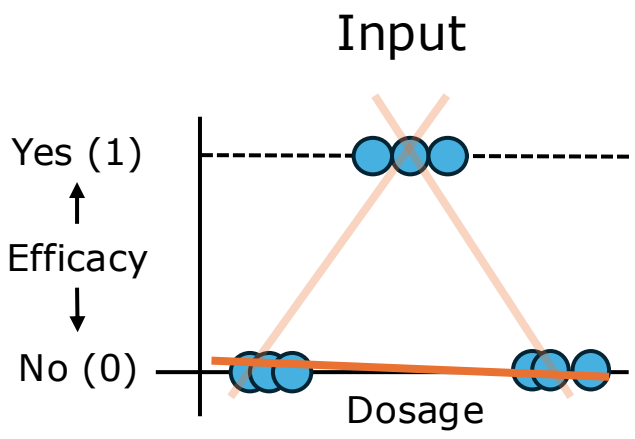


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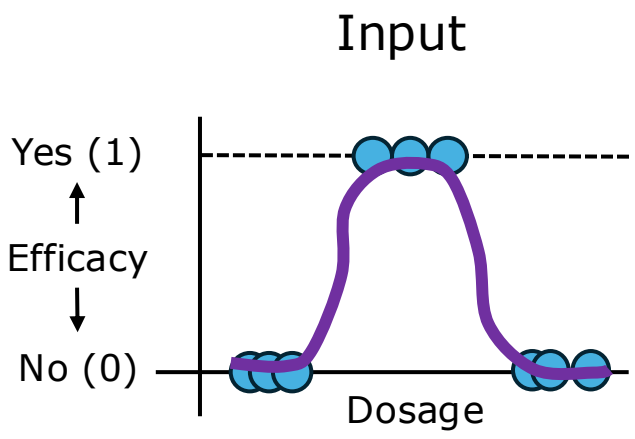
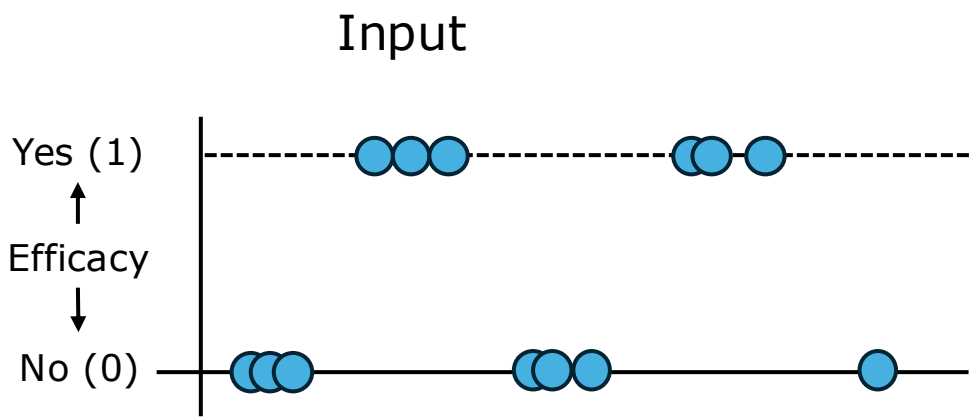



The good news is that a **Neural Network** can fit a **squiggle** to the data

# Neural Network Example




… even if we have a really complicated dataset like this

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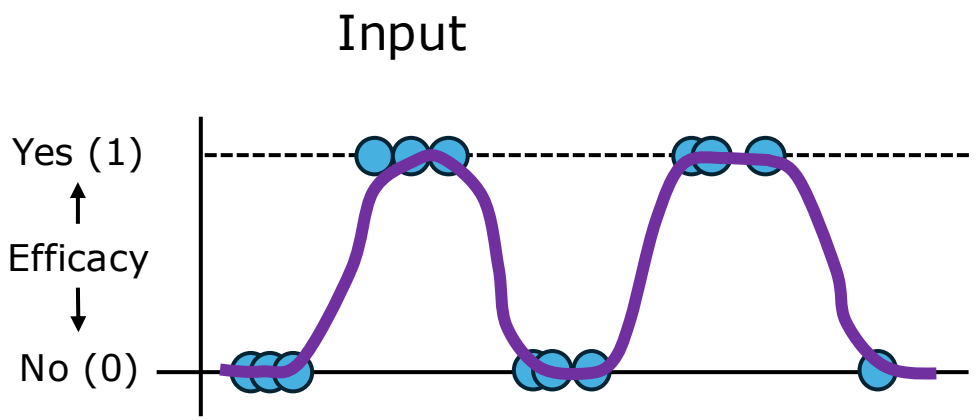


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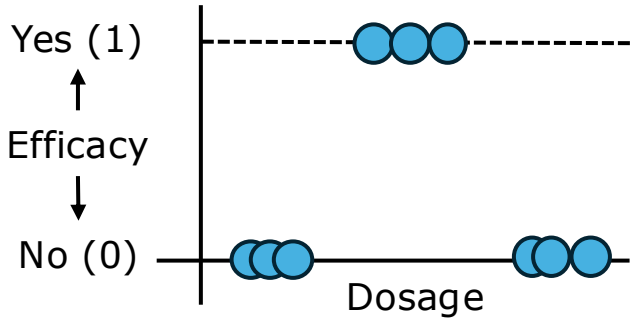
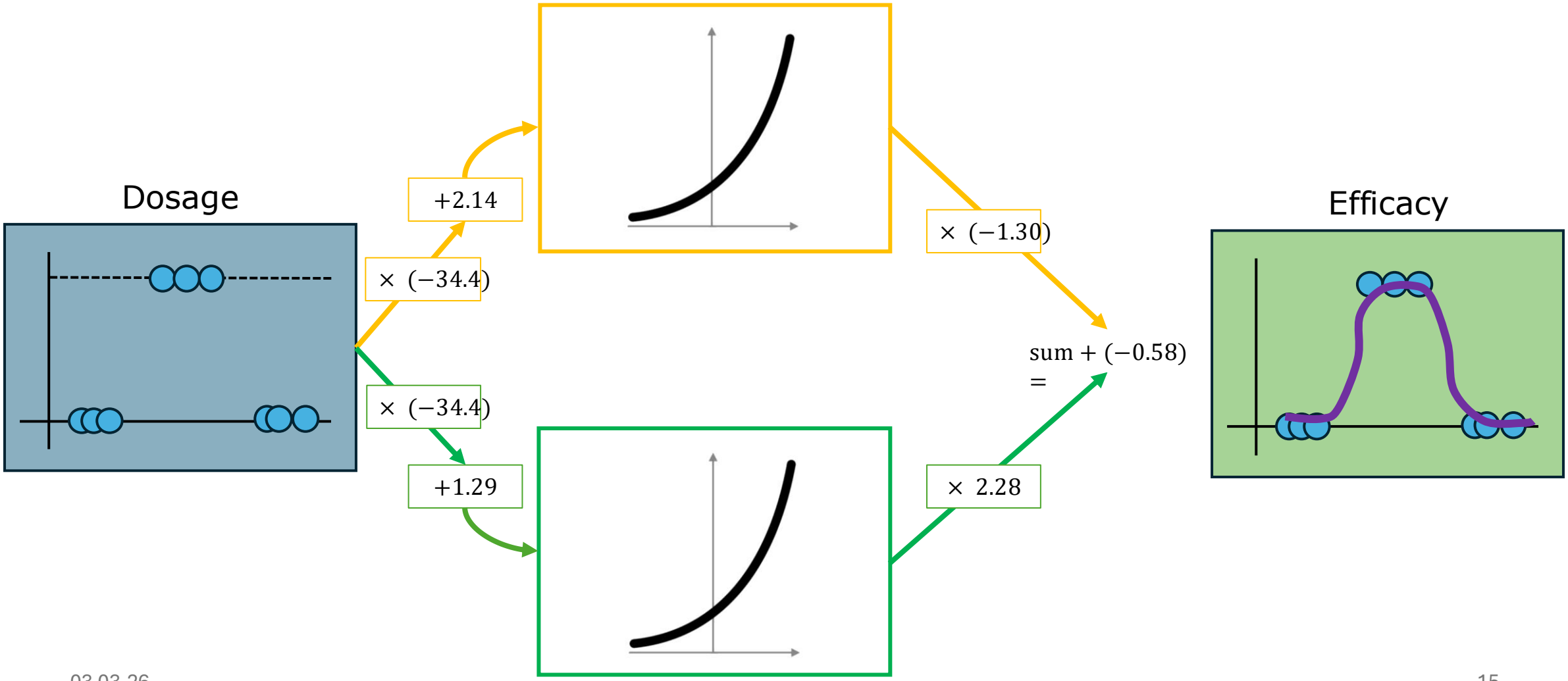



We can use this simple example dataset and turn it into a **Neural Network**

# Neural Network Example



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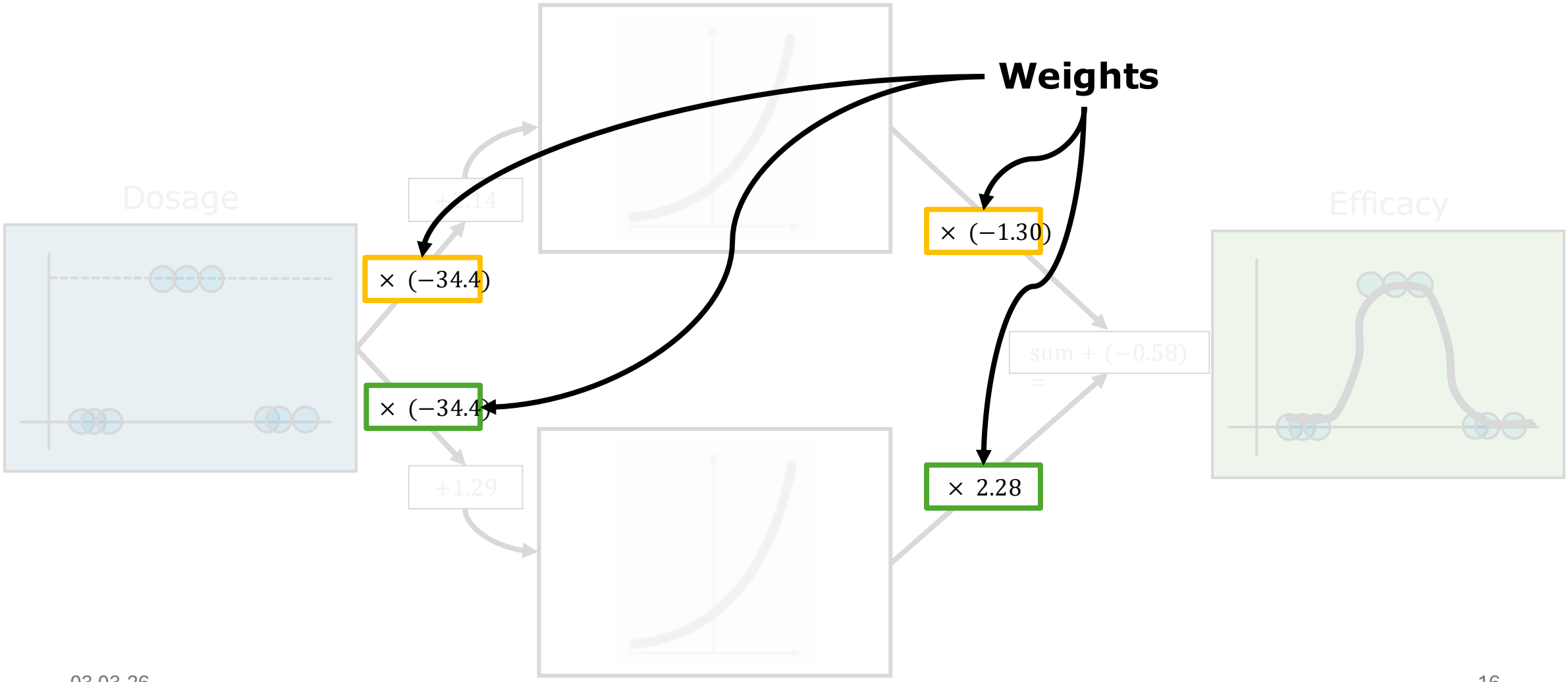

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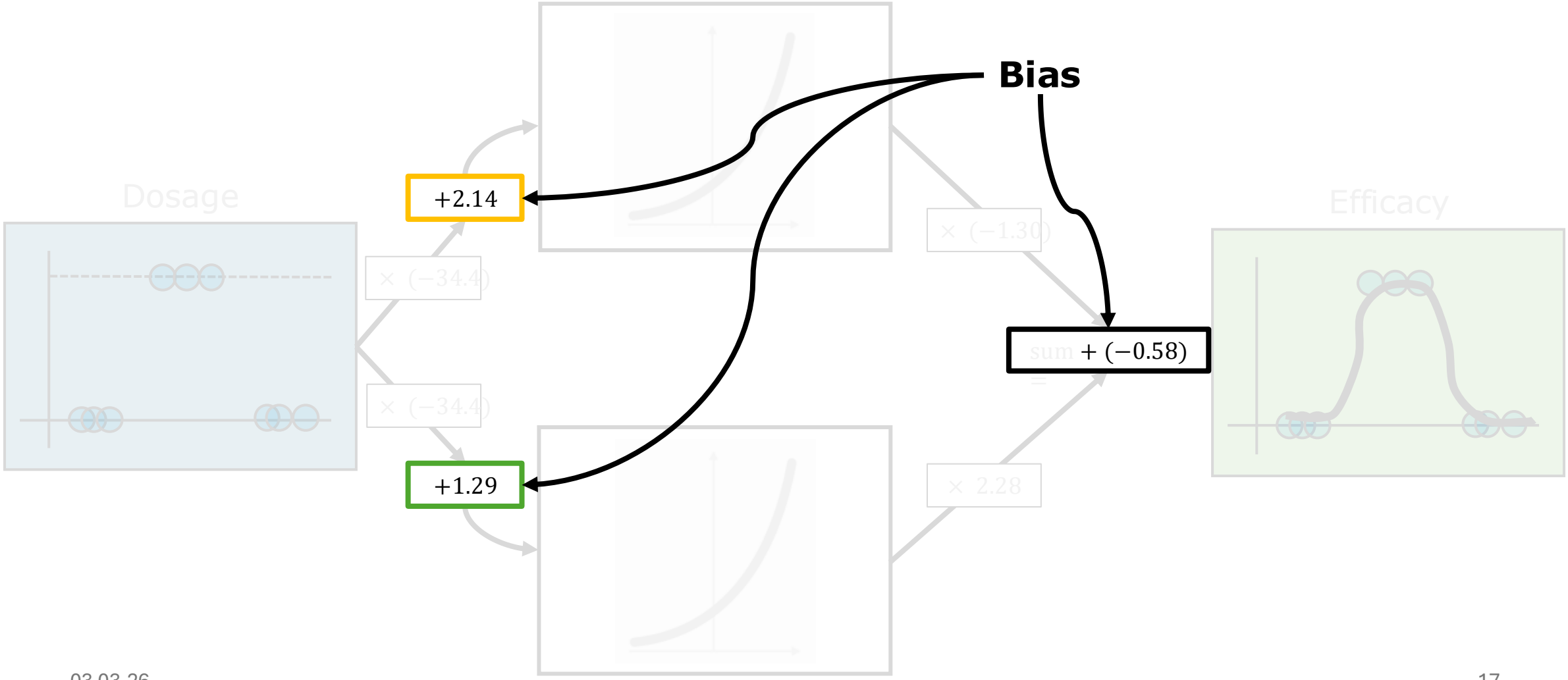

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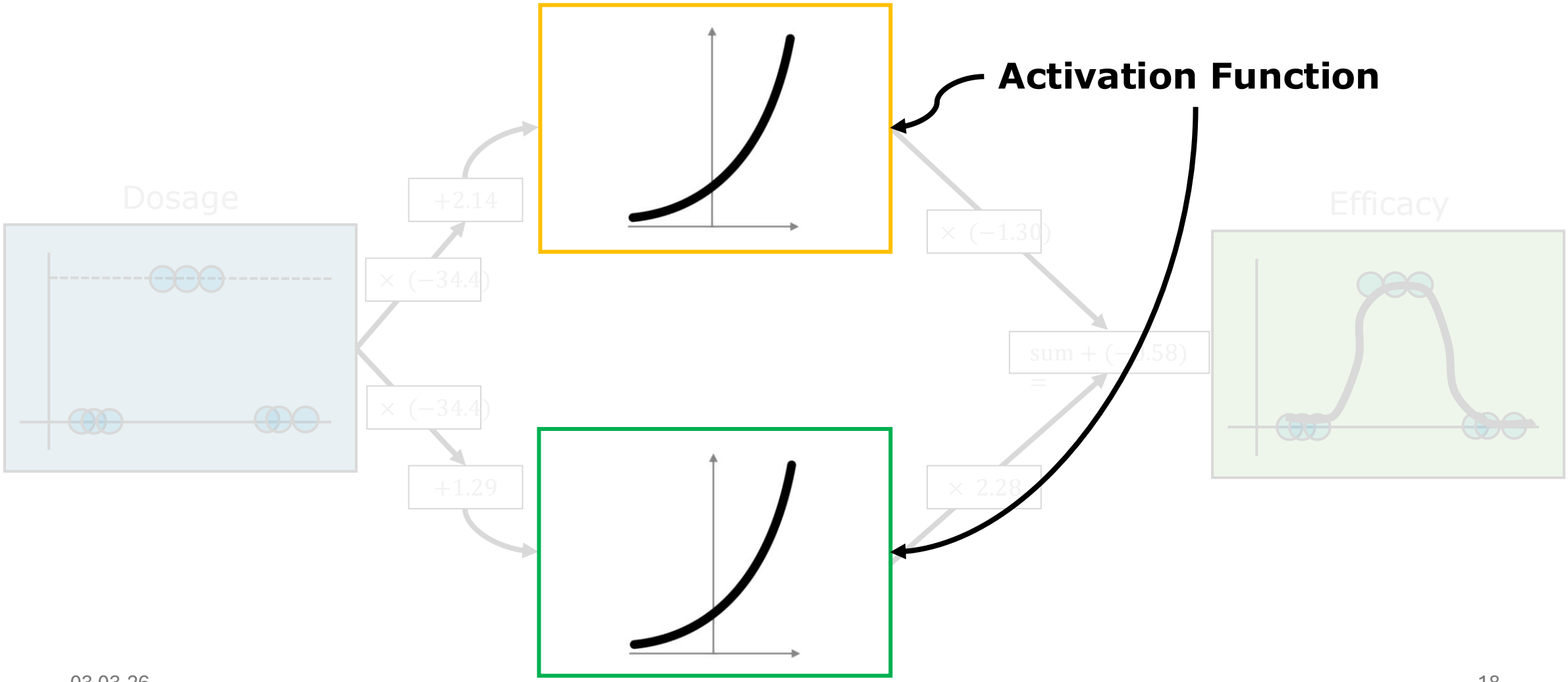
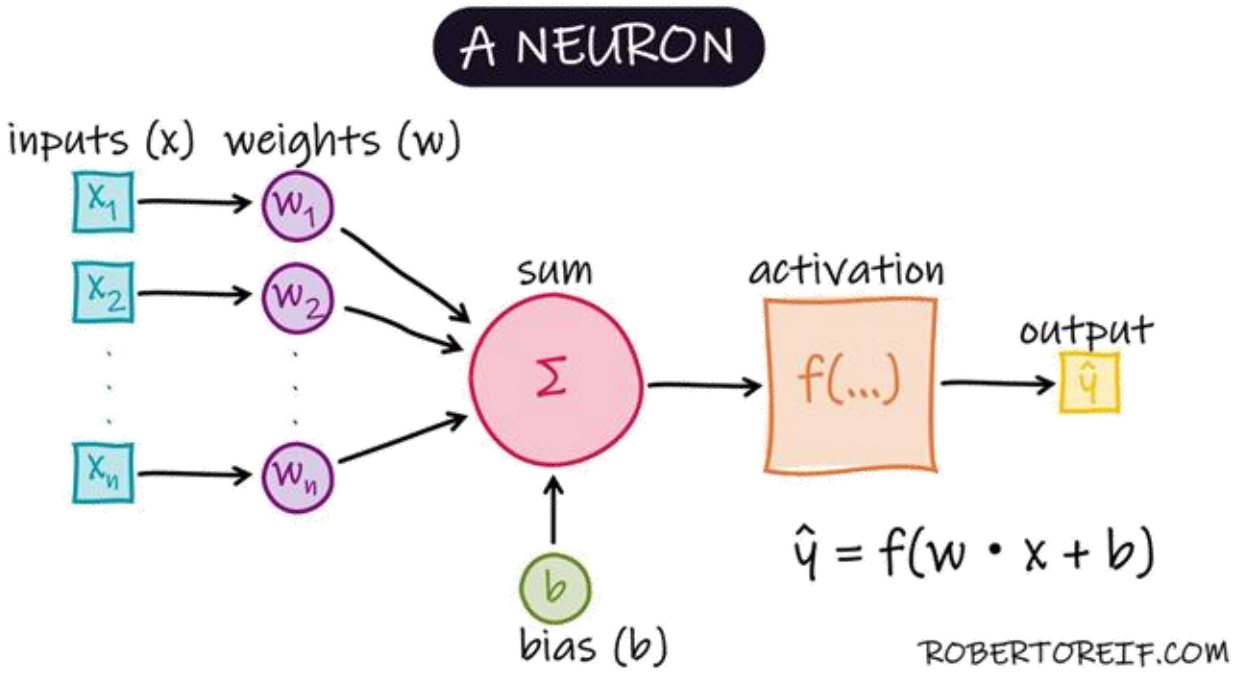


# What Is a Neural Network?

In mathematical notation:

- For one neuron:

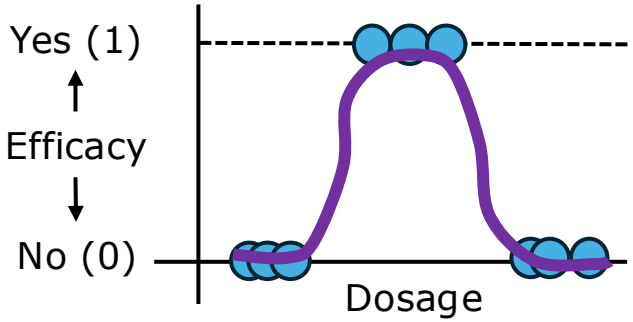
$$z = w^T x + b$$

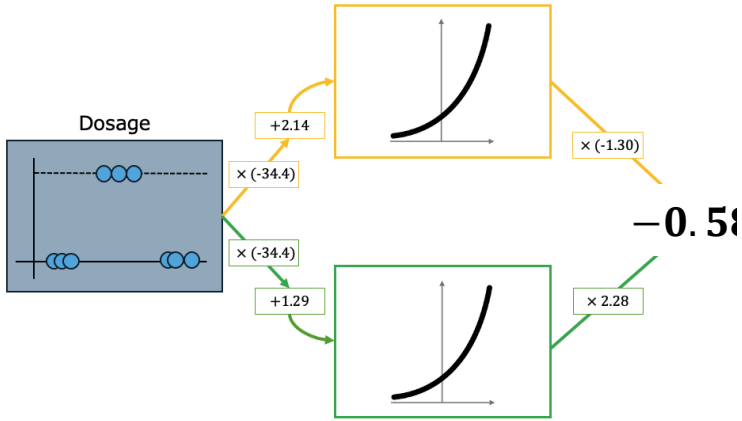



- Stack layers → deep network

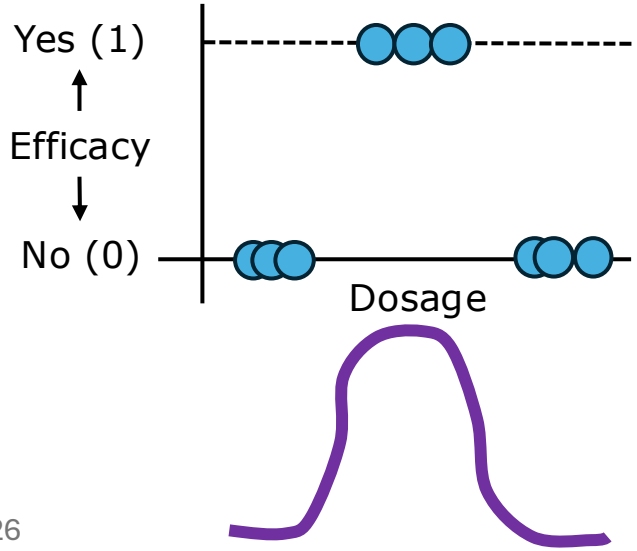
Question: *How do we calculate (optimize) the weights and biases?*

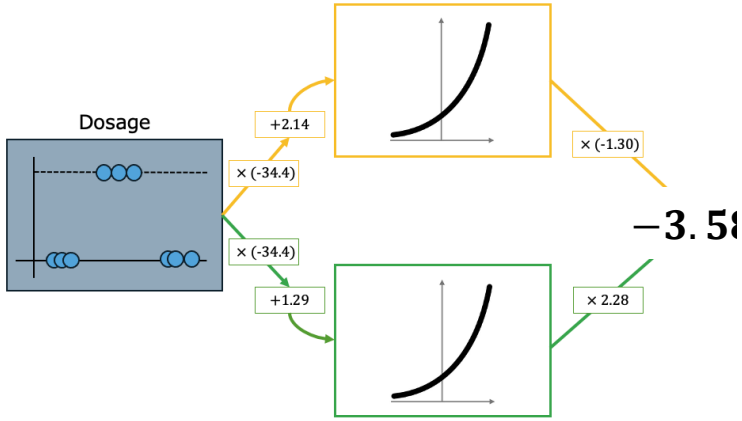
# The Learning Goal



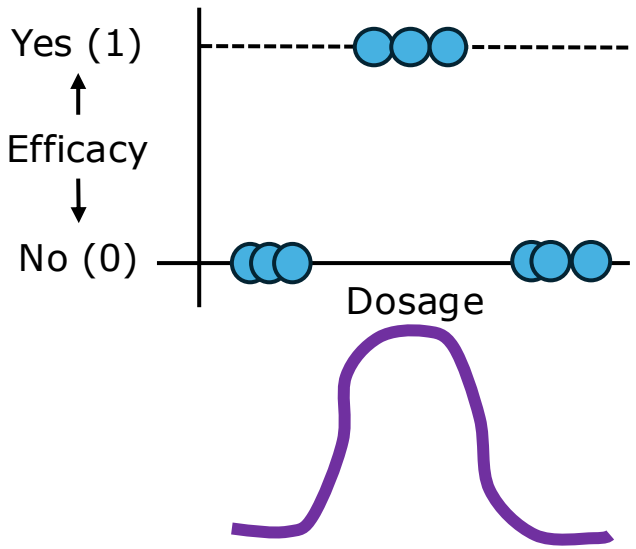

Let's imagine what happens to our outcome, when changing the last bias term **from this** …

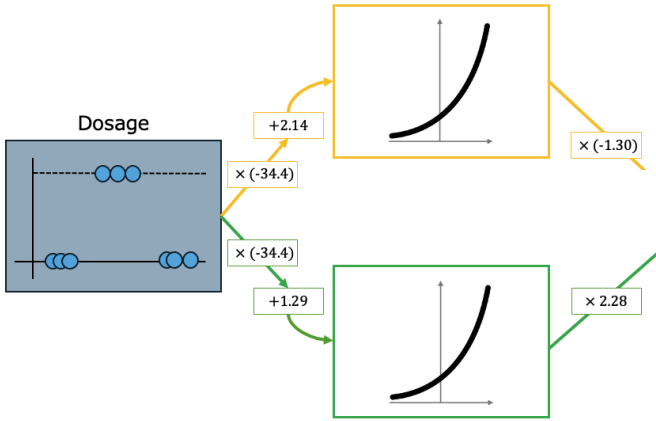



… **to this**

# The Learning Goal




$$b_3 = -3.58$$

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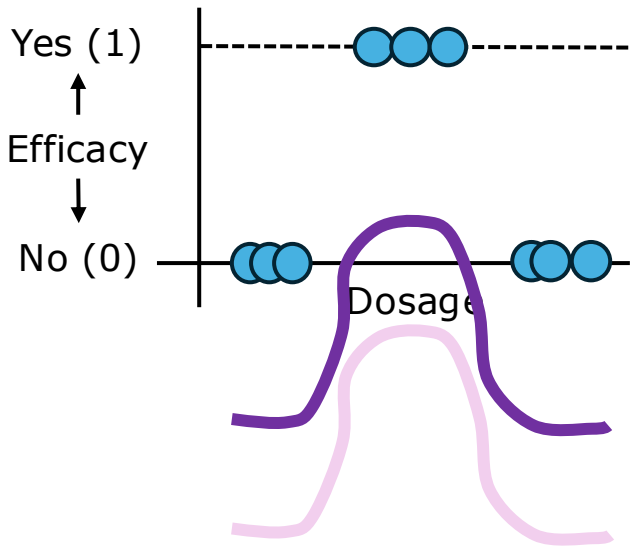


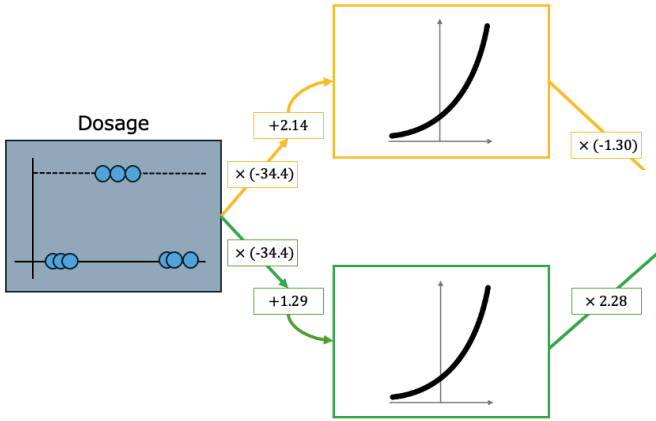


$$b_3 = -2.58$$

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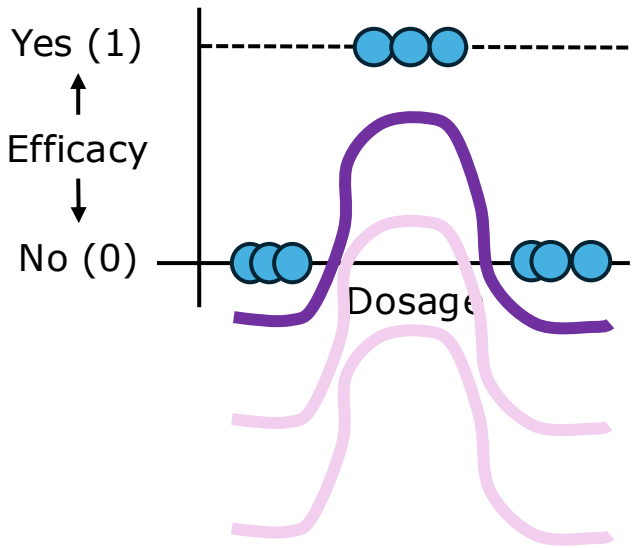


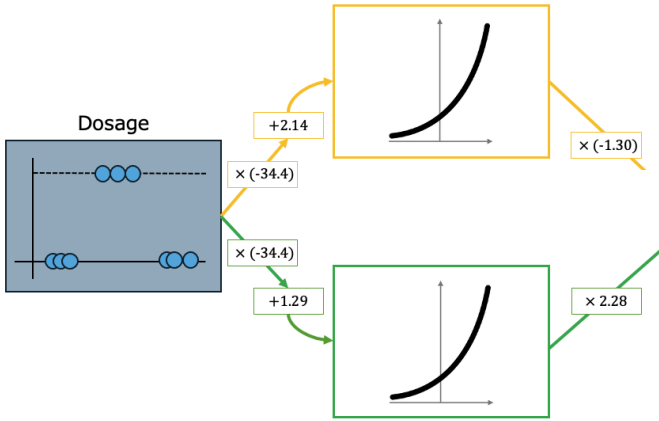


$$b_3 = -1.58$$

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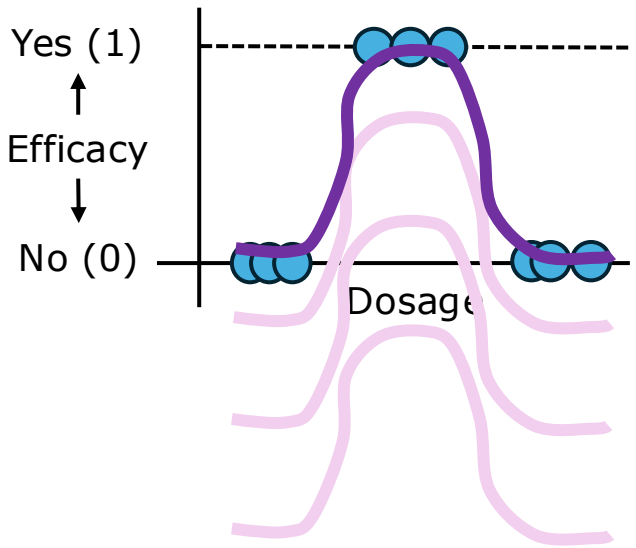


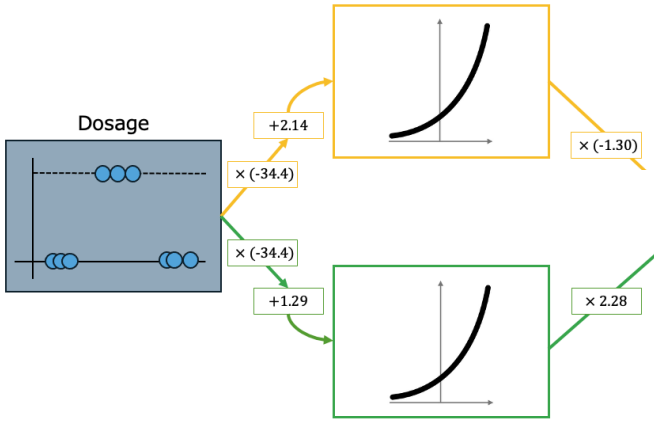


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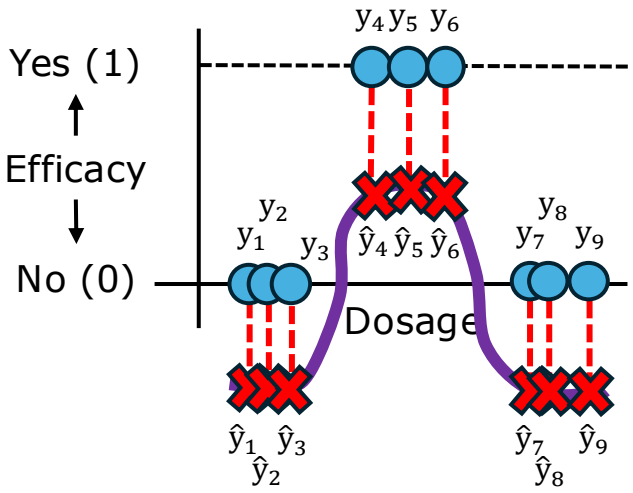



**Goal**: Find the best curve that represents our data

We can compute the error between our prediction $\hat{y}$ and the target $y$ (Ground Truth):

$$L = \frac{1}{2}\sum_{i=1}^{9}(y_i - \hat{y}_i)^2$$

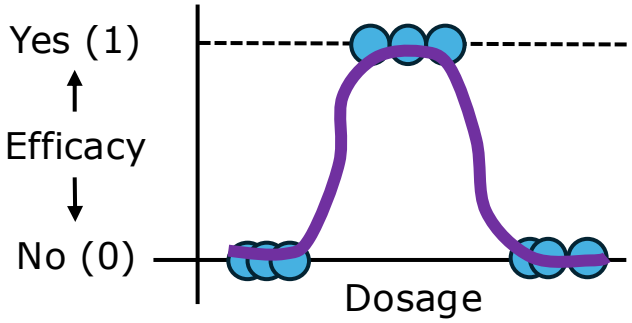
# The Learning Goal

In almost every ML task:
**Minimize Loss $\boldsymbol{L}$**

Example:

$$L = \frac{1}{2}(\boldsymbol{y} - \hat{\boldsymbol{y}})^2$$

- $y$ = Ground Truth
- $\hat{y}$ = Prediction
- Goal: adjust $w$, $b$ to minimize $L$

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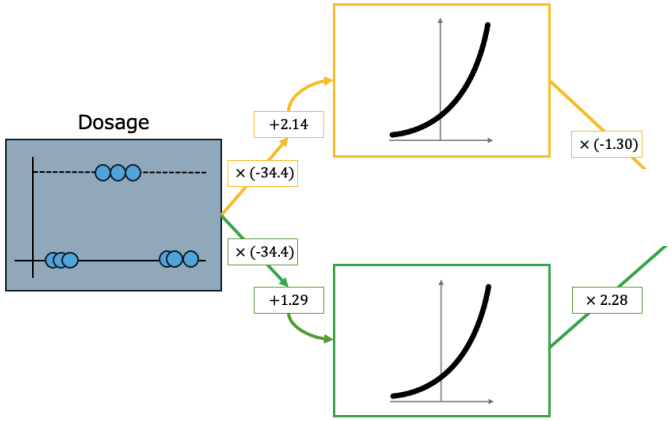


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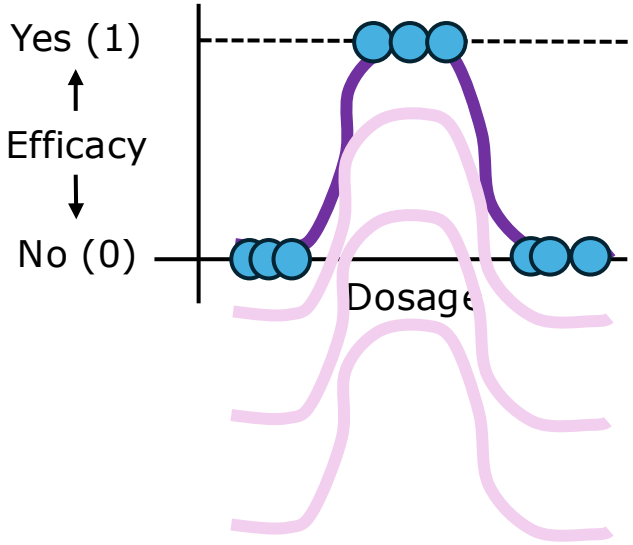


Plot

$$L(b_3) = \frac{1}{2}\sum_{i=1}^{9}(y_i - \hat{y}_i)^2$$

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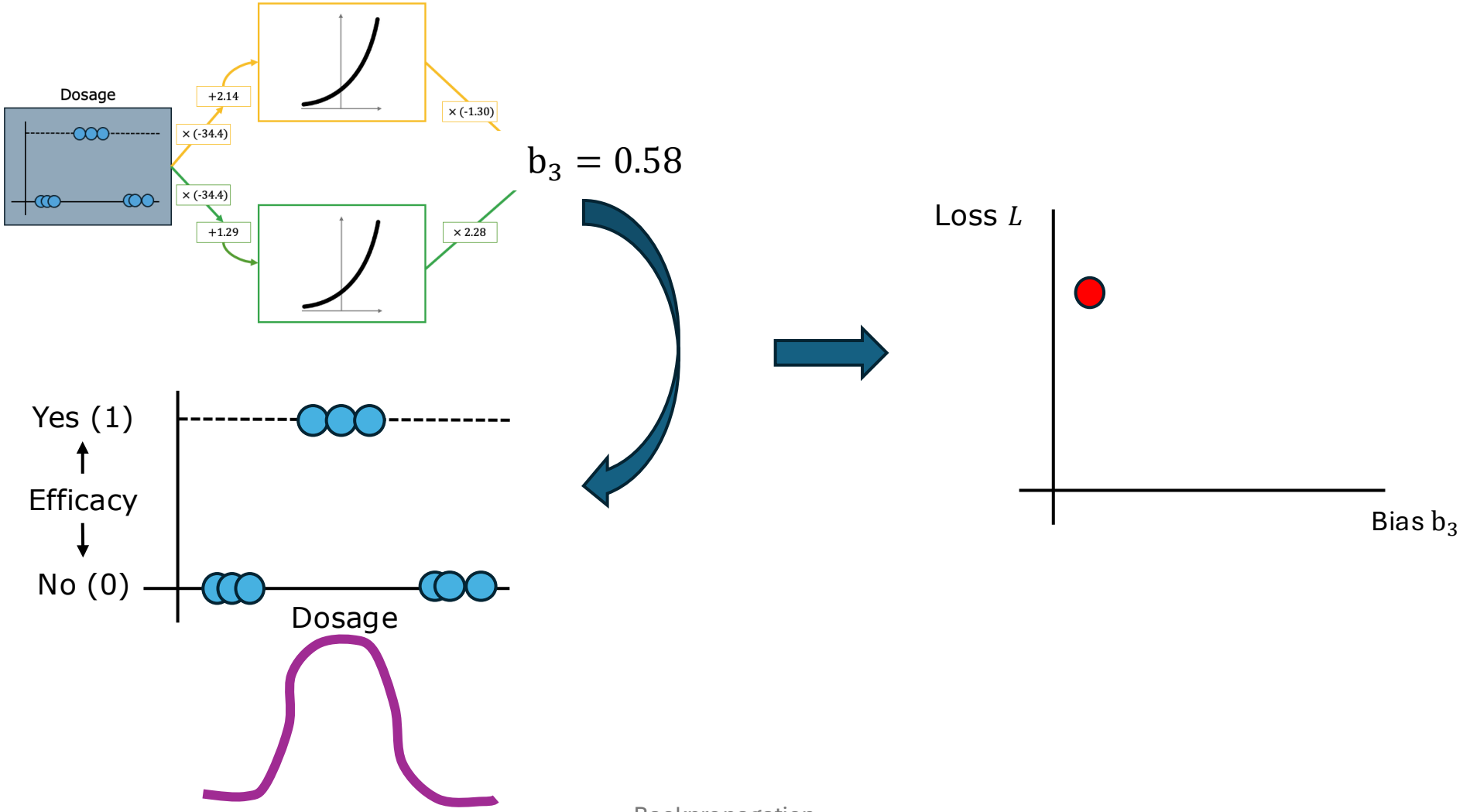

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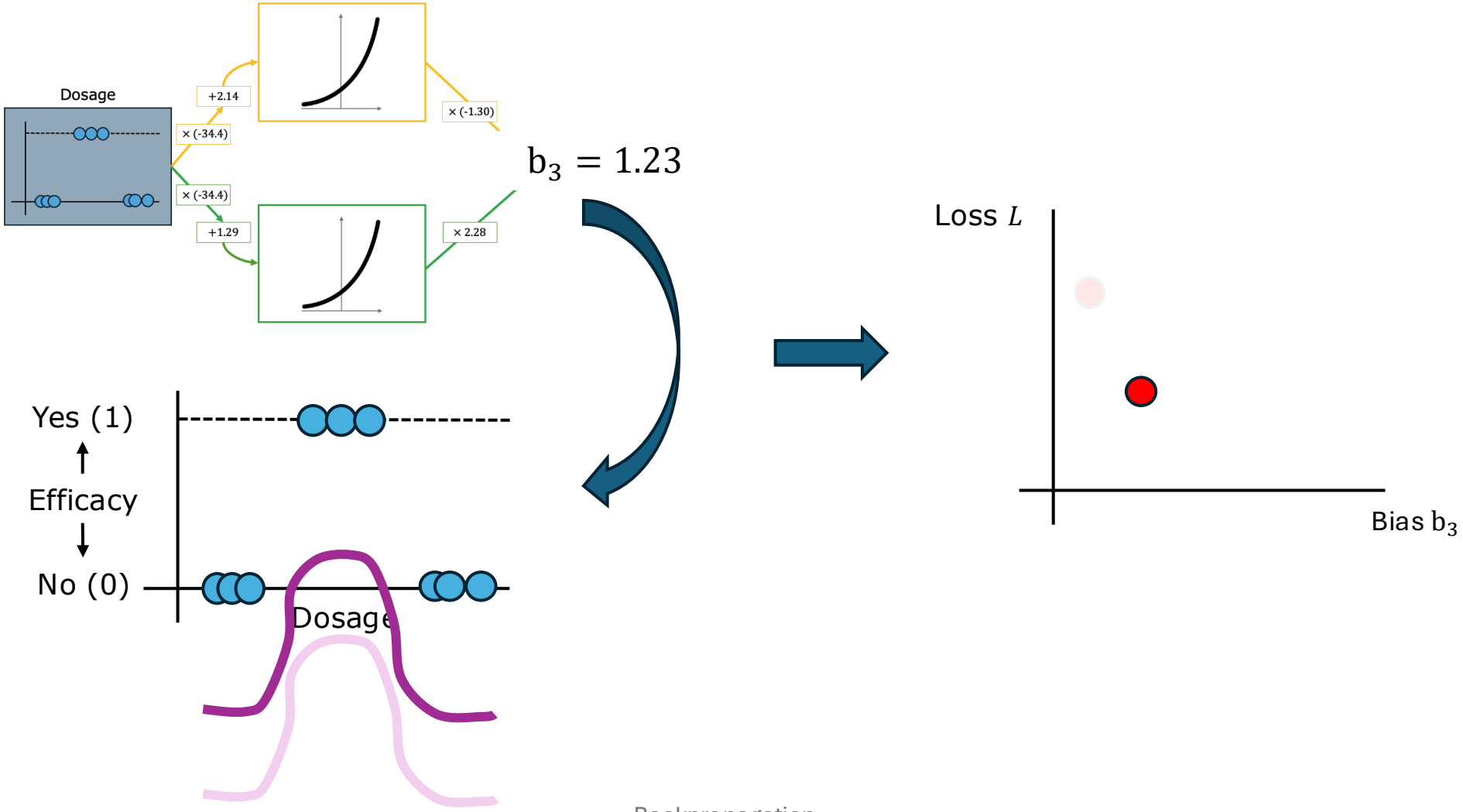

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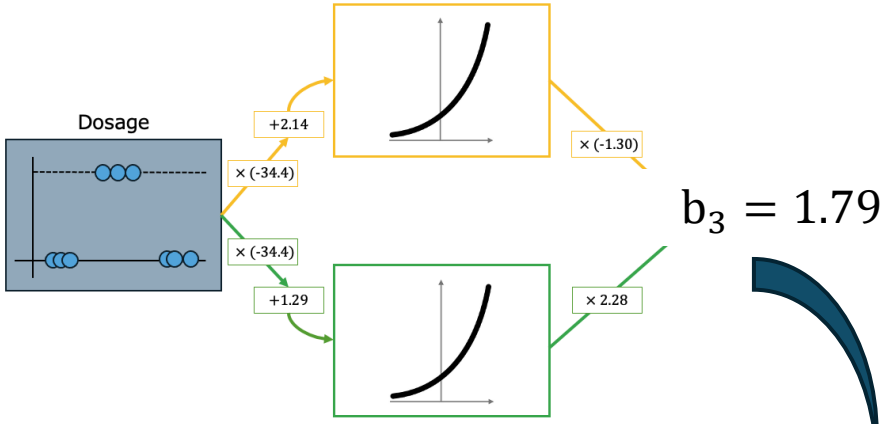
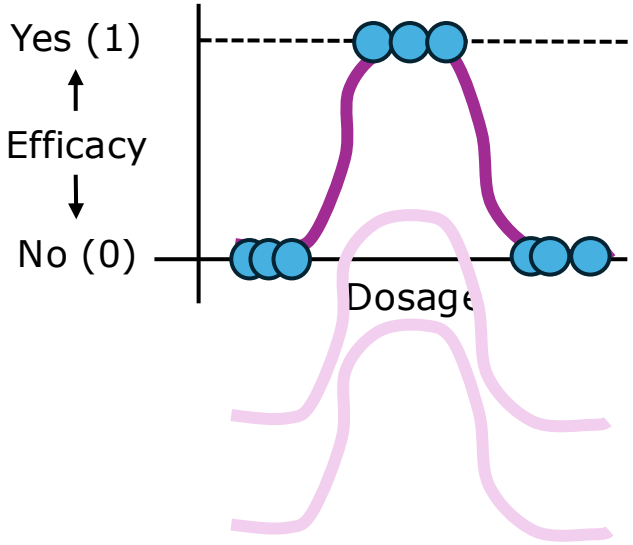
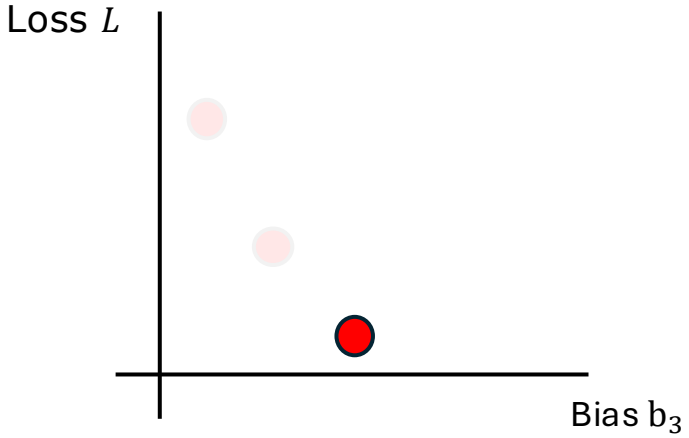



$b_3 = 1.79$



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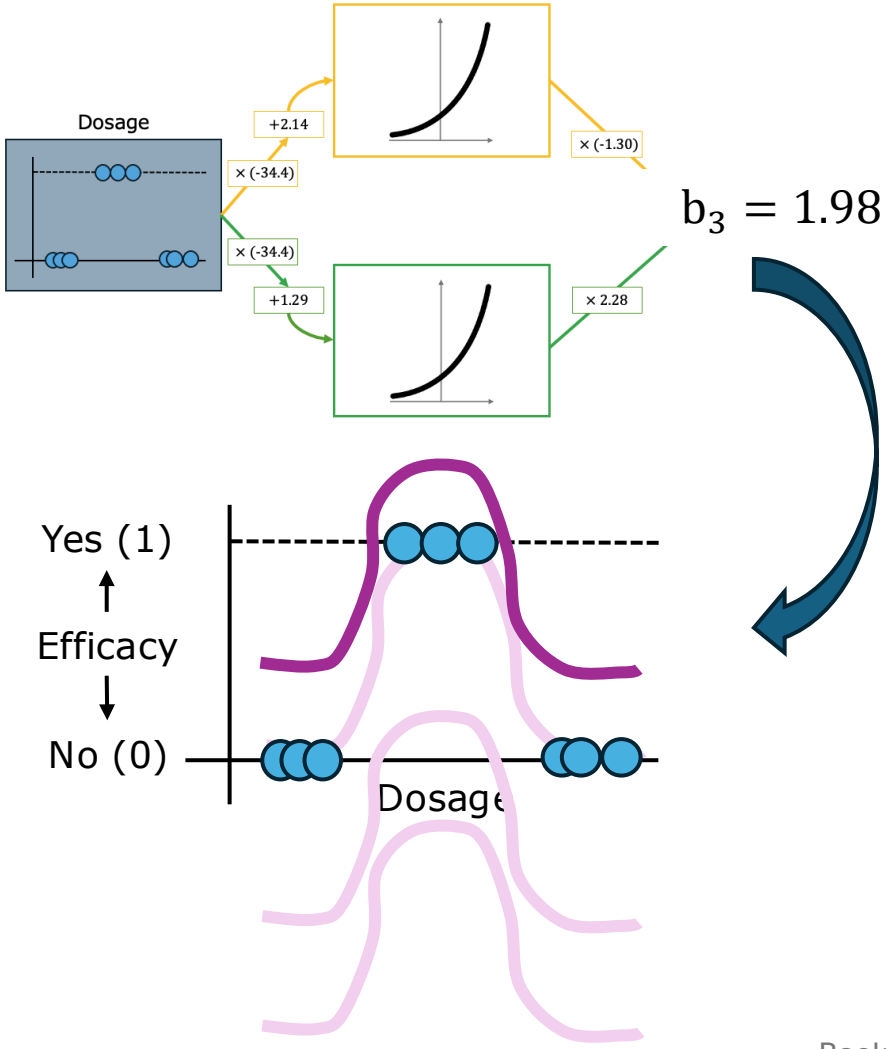
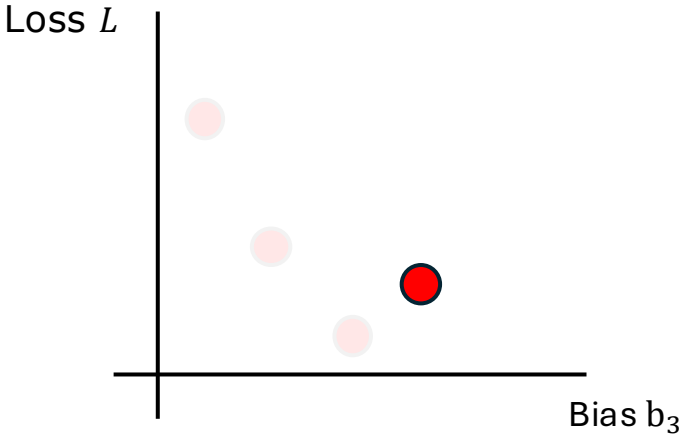



$b_3 = 1.98$



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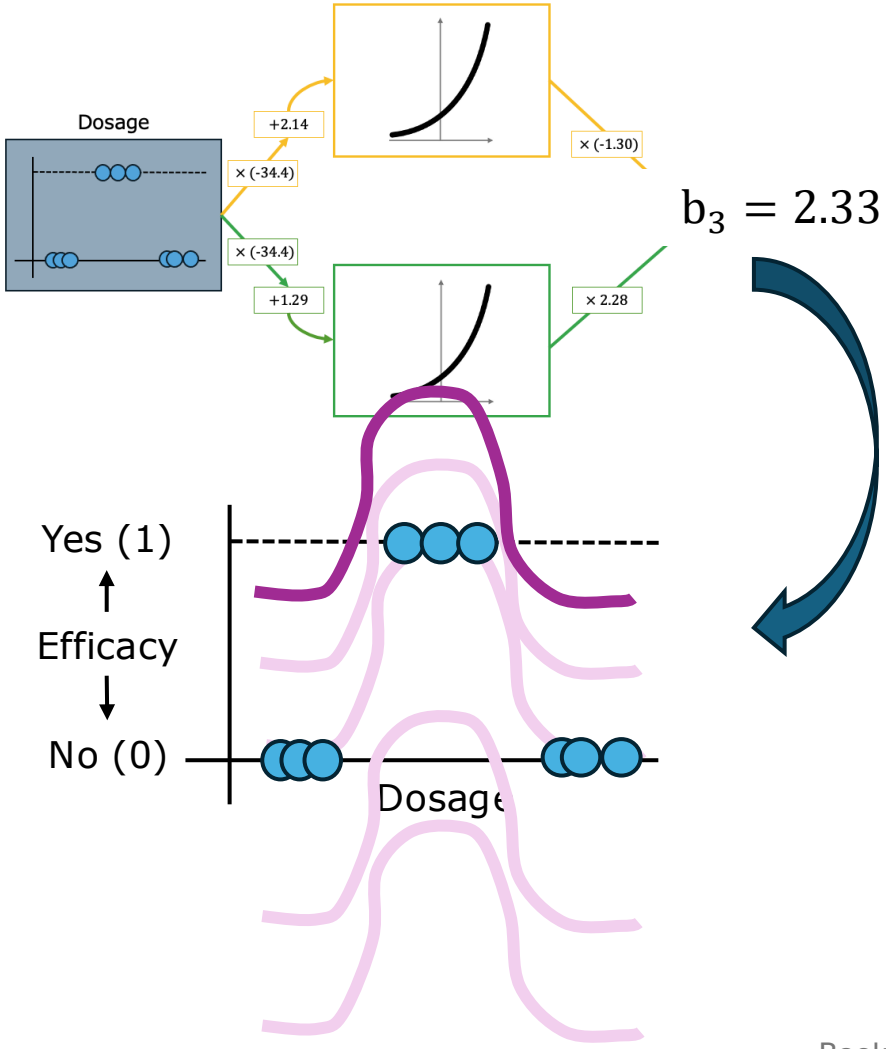
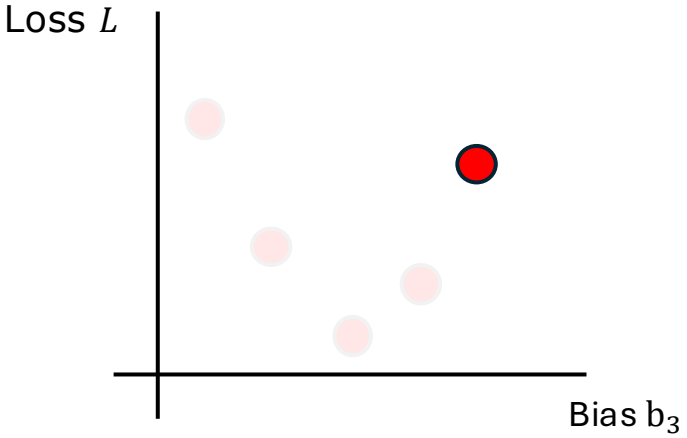



$b_3 = 2.33$



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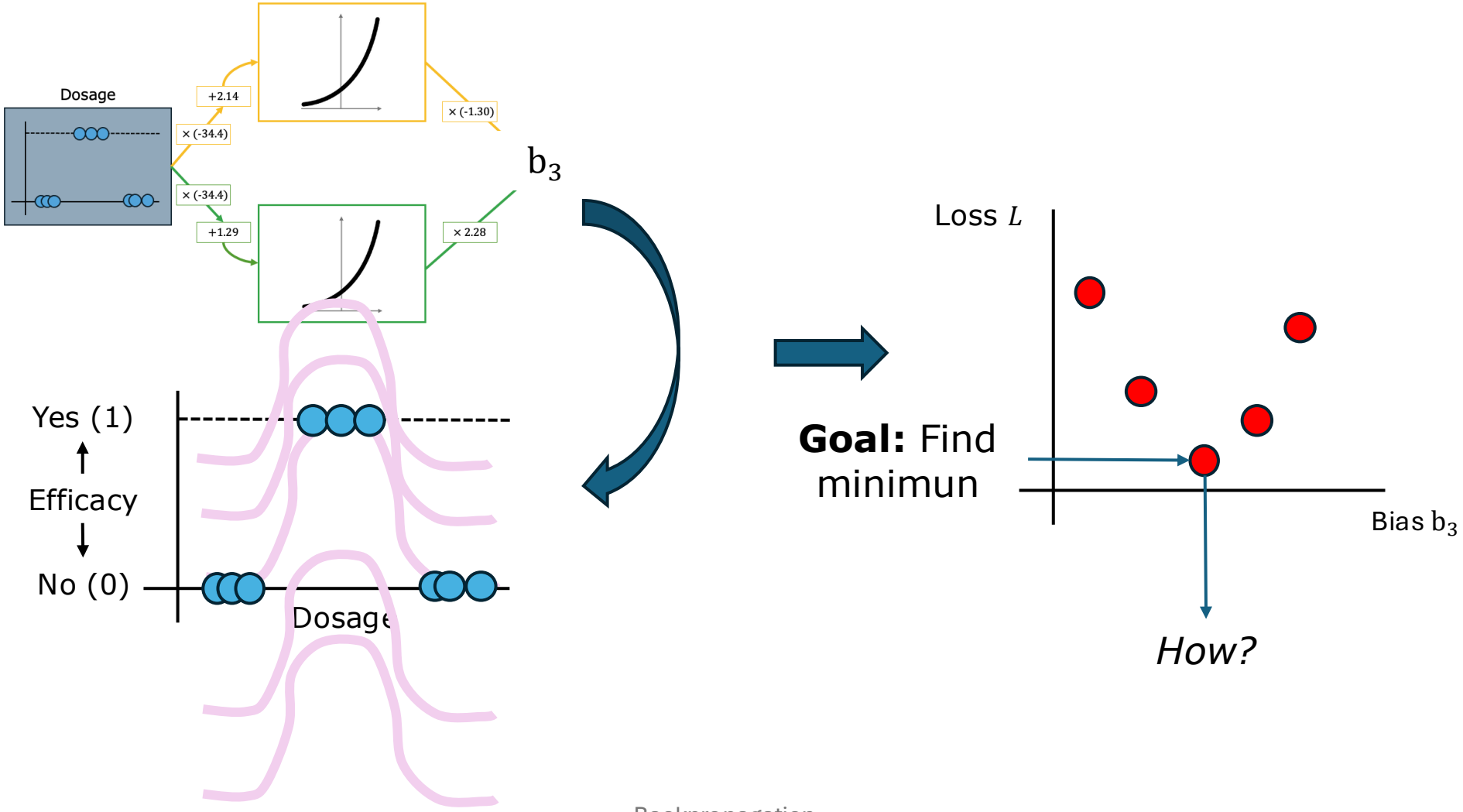


# Agenda

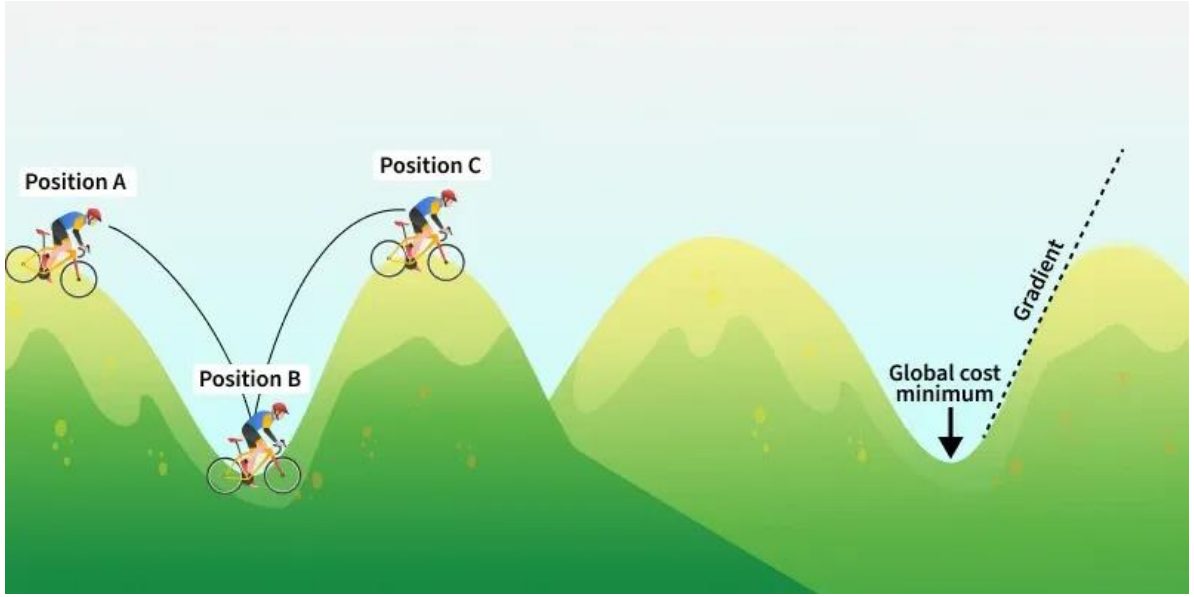
- Neural Networks
- **Gradient Descent**
- Backpropagation

# Gradient Descent

Update rule:

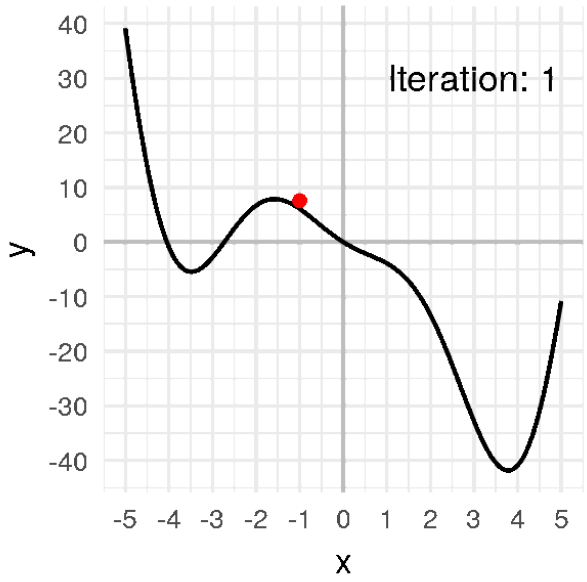
$$w := w - \eta \frac{\partial L}{\partial w}$$

- $\eta$ = Learning Rate
- $\frac{\partial L}{\partial w}$ = Gradient
- Move in direction of steepest descent




https://www.geeksforgeeks.org/machine-learning/what-is-gradient-descent/

# Gradient Descent Illustration

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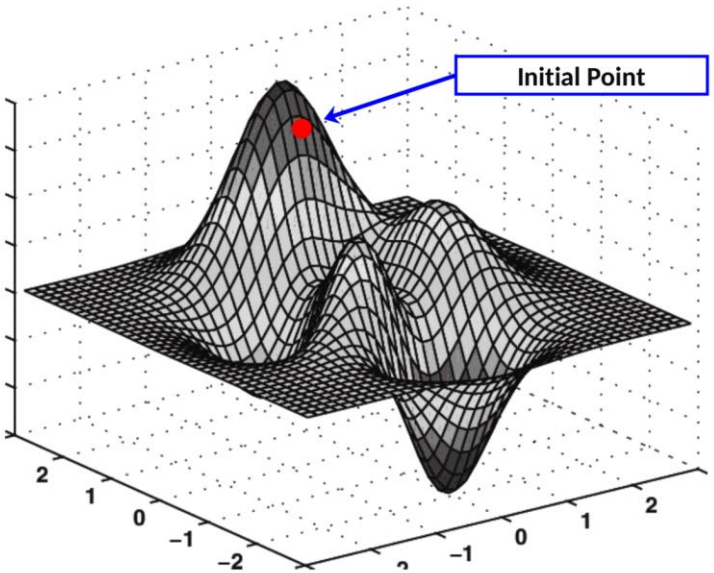


# What is a Gradient?

- Gradient = slope of loss surface
- Points to direction of steepest increase
- We go in negative direction

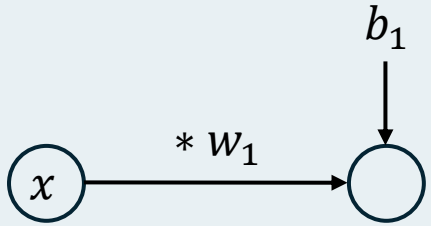
$$w := w - \eta \boxed{\frac{\partial L}{\partial w}}$$

But neural networks are:

- Deep
- Nested
- Composed of many functions

*How do we compute gradients efficiently?*

E.g., simple neural network with input $x$:




$$a(x) = w_1 x + b_1$$

This goes into an activation function $g(\cdot)$ (e.g. ReLU, Sigmoid, …):

$$g(\mathrm{a}) = \sigma(\mathrm{a}) = \frac{1}{1 + e^{-\mathrm{a}}}$$

# What is a Gradient?

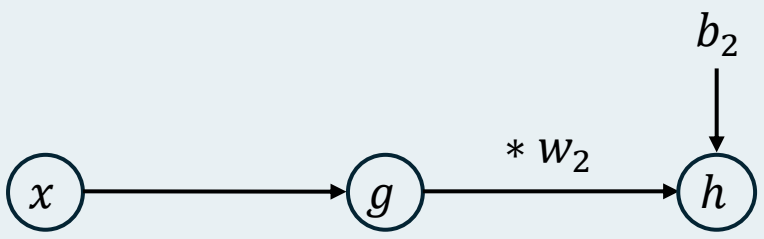
- Gradient = slope of loss surface
- Points to direction of steepest increase
- We go in negative direction

$$w := w - \eta \boxed{\frac{\partial L}{\partial w}}$$

But neural networks are:

- Deep
- Nested
- Composed of many functions

*How do we compute gradients efficiently?*




$$\hat{y} = h(g) = w_2 g + b_2$$

If we wrap this up, we'll get this function for **prediction**:

$$\hat{y} = h(s) = h\Big(g\big(a(x)\big)\Big) = w_2, g(w_1 x + b_1) + b_2$$

Nested:

$$\hat{y}(x) = w_2 g(w_1 x + b_1) + b_2$$

Composed:

$$\hat{y}(x) = (h \circ g \circ a)(x)$$

# What is a Gradient?

We are still interested in this:

$$w := w - \eta \frac{\partial L}{\partial w}$$

and have this now as a prediction:

$$\hat{y}(x) = (h \circ g \circ a)(x)$$

We also know:

$$L(\hat{y}, y) = \frac{1}{2}(y - \hat{y})^2$$

Which becomes to:

$$L(x, y) = \frac{1}{2}\left(y - (w_2, g(w_1 x + b_1) + b_2)\right)^2$$

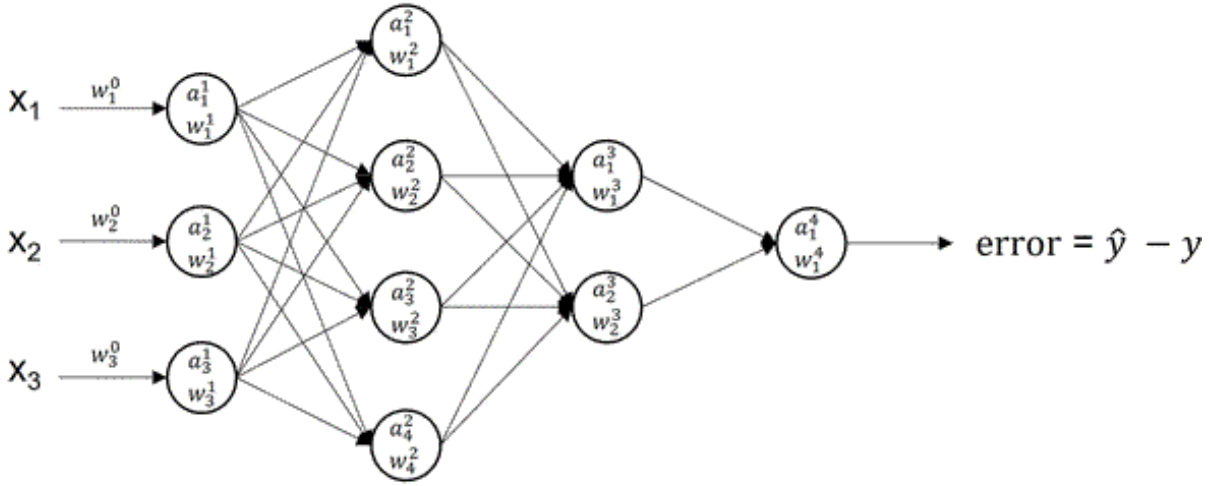
# Chain Rule

*How do we compute the gradient?*

By using the **Chain Rule!**

$$\left[f(g(h(i(j(x))))\right]'$$
$$=$$
$$f'(g(h(i(j(x)))))$$
$$* g'(h(i(j(x))))$$
$$* h'(i(j(x)))$$
$$* i'(j(x))$$
$$* j'(x)$$
$$* 1$$

https://en.wikipedia.org/wiki/Chain_rule



# Agenda

- Neural Networks
- Gradient Descent
- **Backpropagation**

# Simple Chain Rule Example

Let:

$$g(x) = 3x + 1$$
$$f(u) = u^2$$

Then our full function is:

$$y = f\big(g(x)\big) = (3x + 1)^2$$

→ **Key idea: multiply local derivatives**

So we have two derivatives:

$$\frac{dy}{du} = 2u$$
$$\frac{du}{dx} = 3$$

# Chain Rule Formula

$$\boxed{\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}}$$

Now substitute:

$$\frac{dy}{dx} = (2u) \cdot 3$$

Replace $u = 3x + 1$:

$$\frac{dy}{dx} = 2(3x + 1) \cdot 3$$
$$= 6(3x + 1)$$

**Important**

- The outer function changes with respect to the inner function
- The inner function changes with respect to $x$

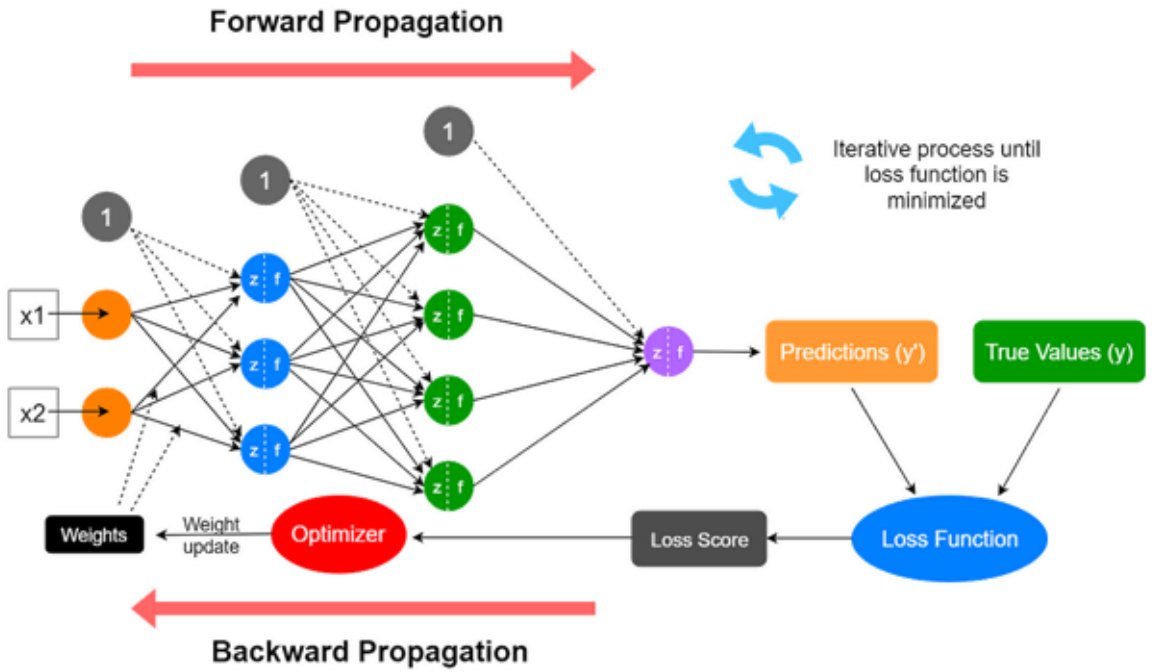
The total change = *outer change* × *inner change*

→ You multiply the sensitivities

# Forward & Backward Pass

Each node:

- Stores value (forward pass)
- Stores derivative (backward pass)

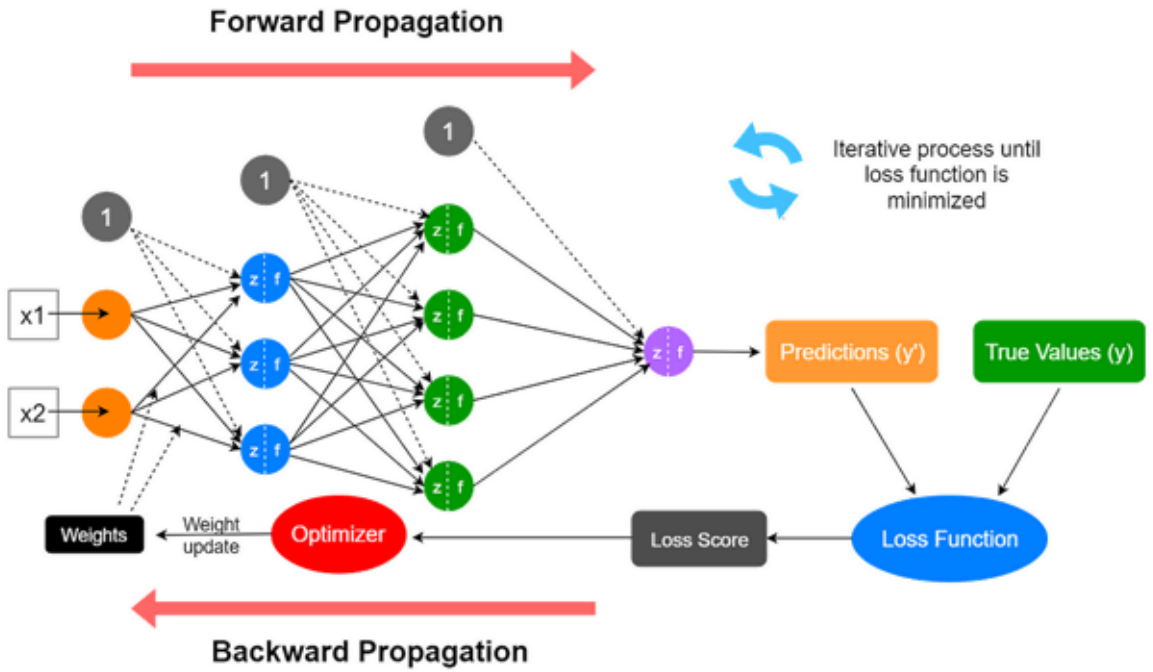



https://wandb.ai/mostafaibrahim17/ml-articles/reports/Decoding-Backpropagation-and-Its-Role-in-Neural-Network-Learning--Vmlldzo3MTY5MzM1

# Forward Pass

1. Initialize random $w$, $b$
2. Compute:
   - $z = w^T x + b$
   - $a = \sigma(z)$
   - $\hat{y}$
3. Compute loss

We now have **all intermediate values**




https://wandb.ai/mostafaibrahim17/ml-articles/reports/Decoding-Backpropagation-and-Its-Role-in-Neural-Network-Learning--Vmlldzo3MTY5MzM1

# Backward Pass

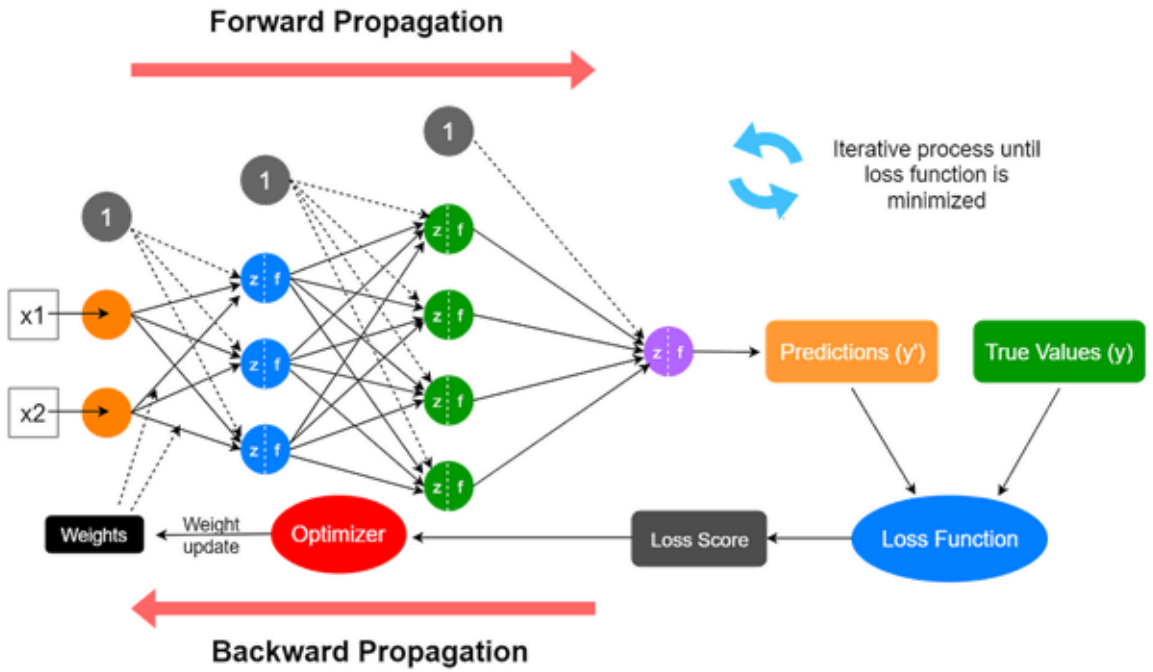
We compute:

$$\frac{\partial L}{\partial w}$$

Using chain rule:

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z} \cdot \frac{\partial z}{\partial w}$$

→ We move backwards layer by layer

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# Why Backprop Is Efficient

Without backprop:

- Recompute many partial derivatives repeatedly

With backprop:

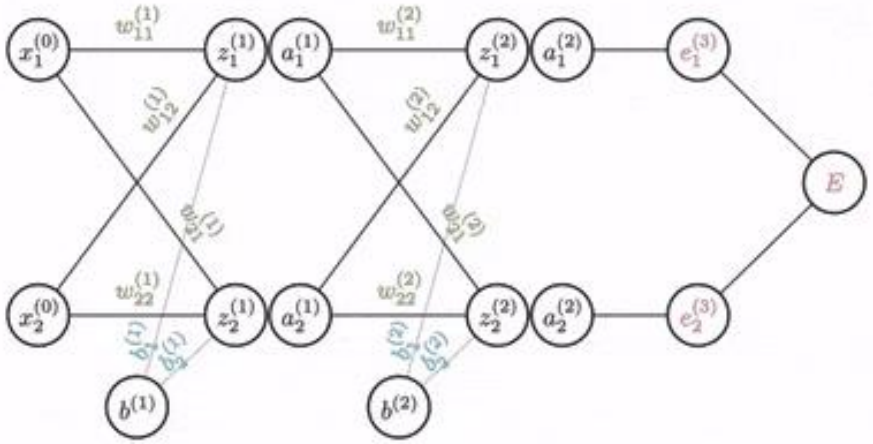
- Reuse intermediate derivatives
- Dynamic programming
- Complexity proportional to forward pass

→ Cost ≈ 2 × forward pass

# Matrix Form

$$\frac{\partial E}{\partial w_{11}^{(2)}} = \frac{\partial e_1^{(3)}}{\partial a_1^{(2)}} \frac{\partial a_1^{(2)}}{\partial z_1^{(2)}} \frac{\partial z_1^{(2)}}{\partial w_{11}^{(2)}} \qquad \frac{\partial E}{\partial w_{12}^{(2)}} = \frac{\partial e_1^{(3)}}{\partial a_1^{(2)}} \frac{\partial a_1^{(2)}}{\partial z_1^{(2)}} \frac{\partial z_1^{(2)}}{\partial w_{12}^{(2)}}$$

$$\frac{\partial E}{\partial w_{21}^{(2)}} = \frac{\partial e_2^{(3)}}{\partial a_2^{(2)}} \frac{\partial a_2^{(2)}}{\partial z_2^{(2)}} \frac{\partial z_2^{(2)}}{\partial w_{21}^{(2)}} \qquad \frac{\partial E}{\partial w_{22}^{(2)}} = \frac{\partial e_2^{(3)}}{\partial a_2^{(2)}} \frac{\partial a_2^{(2)}}{\partial z_2^{(2)}} \frac{\partial z_2^{(2)}}{\partial w_{22}^{(2)}}$$




For one layer:

$$Z = W X + b$$

$$A = \sigma(Z)$$

Backward:

$$dW = \delta X^T$$

$$\delta = dA \odot \sigma'(Z)$$

$$\begin{bmatrix} \frac{\partial E}{\partial w_{11}^{(2)}} & \frac{\partial E}{\partial w_{12}^{(2)}} \\ \frac{\partial E}{\partial w_{21}^{(2)}} & \frac{\partial E}{\partial w_{22}^{(2)}} \end{bmatrix} = \begin{bmatrix} \frac{\partial e_1^{(3)}}{\partial a_1^{(2)}} \\ \frac{\partial e_2^{(3)}}{\partial a_2^{(2)}} \end{bmatrix} \odot \begin{bmatrix} \frac{\partial a_1^{(2)}}{\partial z_1^{(2)}} \\ \frac{\partial a_2^{(2)}}{\partial z_2^{(2)}} \end{bmatrix} \odot \begin{bmatrix} \frac{\partial z_1^{(2)}}{\partial w_{11}^{(2)}} & \frac{\partial z_1^{(2)}}{\partial w_{12}^{(2)}} \\ \frac{\partial z_2^{(2)}}{\partial w_{21}^{(2)}} & \frac{\partial z_2^{(2)}}{\partial w_{22}^{(2)}} \end{bmatrix}$$

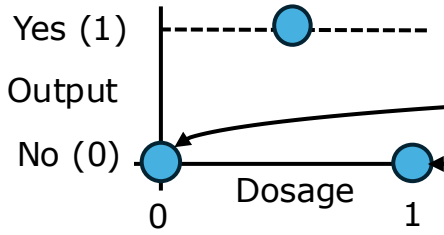
$$\frac{\partial E}{\partial W^{(2)}} \qquad \underbrace{\frac{\partial E}{\partial A^{(2)}} \quad \frac{\partial A^{(2)}}{\partial Z^{(2)}}}_{\delta^L} \qquad \frac{\partial Z^{(2)}}{\partial W^{(2)}}$$

$$\boxed{\frac{\partial E}{\partial W^{(2)}} = \delta^L \odot \frac{\partial Z^{(2)}}{\partial W^{(2)}}}$$

Where:

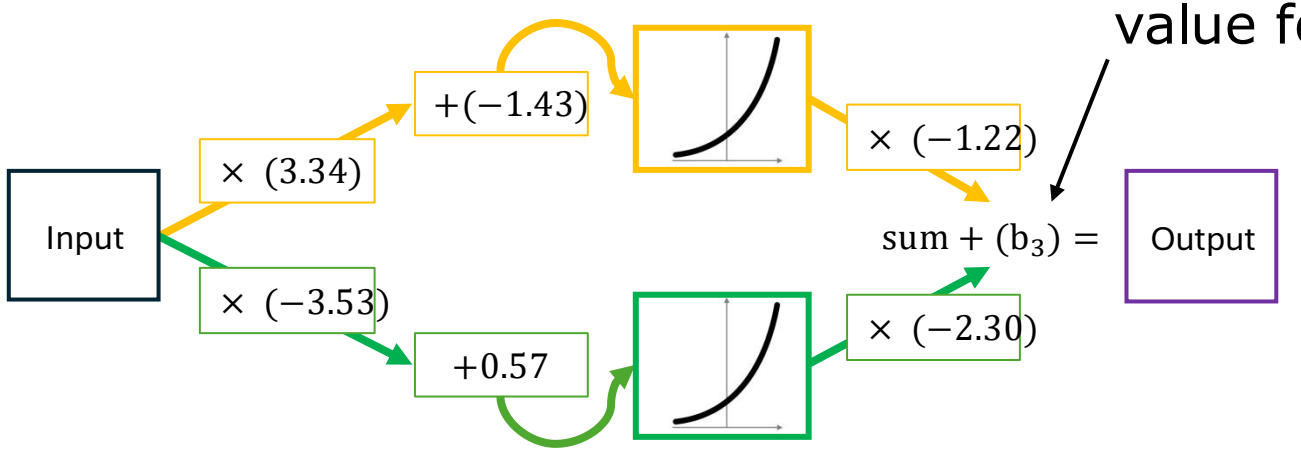
- $\odot$= elementwise product
- Efficient vectorized implementation

# Backpropagation Example

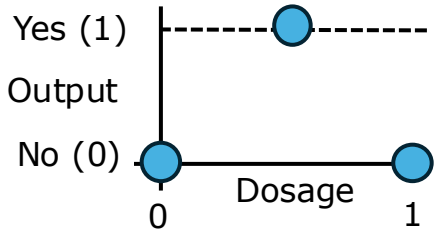



Our **Dosage** example again: Let's keep it simple with 3 data points and let's assume Dosages go from **0** (low) to **1** (high)

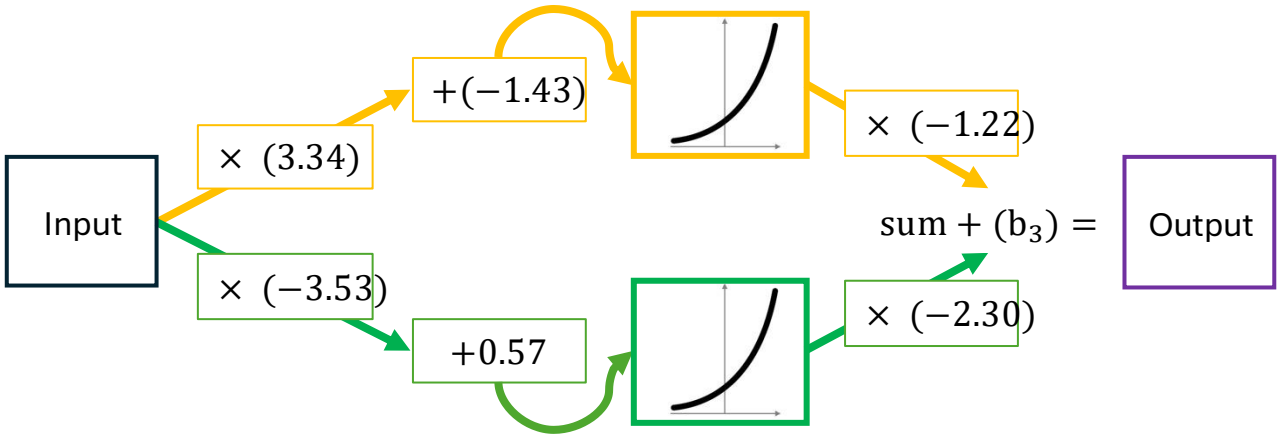
**Goal** is to find the optimal value for $b_3$



# Backpropagation Example




Because Bias terms are frequently initialized to 0, we will set $b_3 = 0$



# Backpropagation Example

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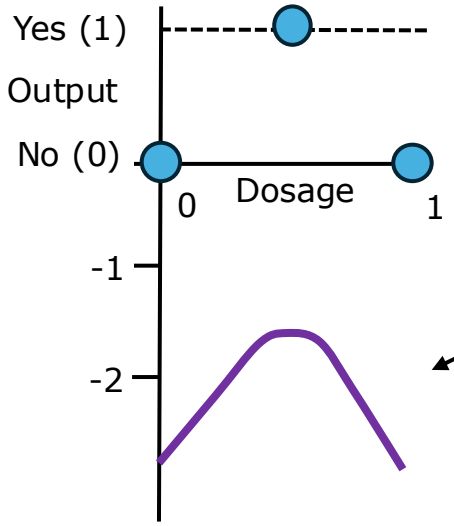
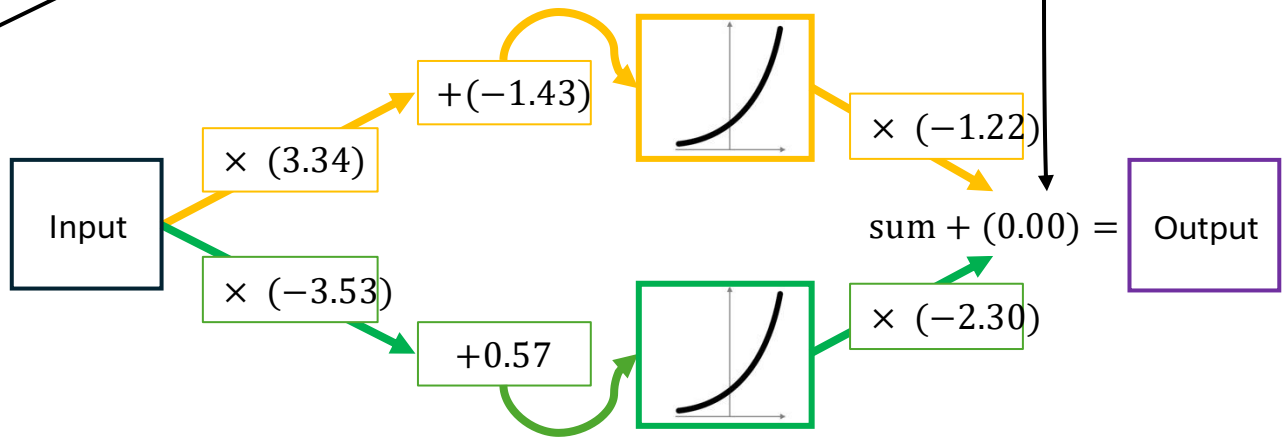
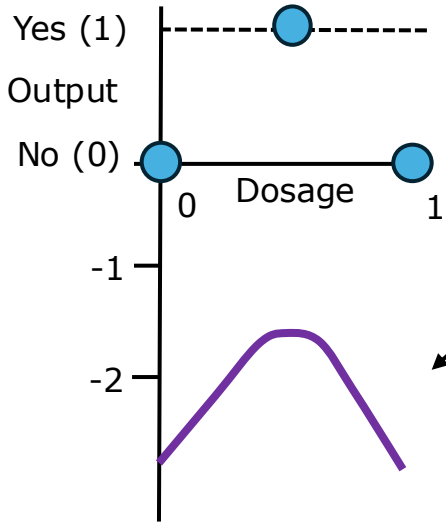
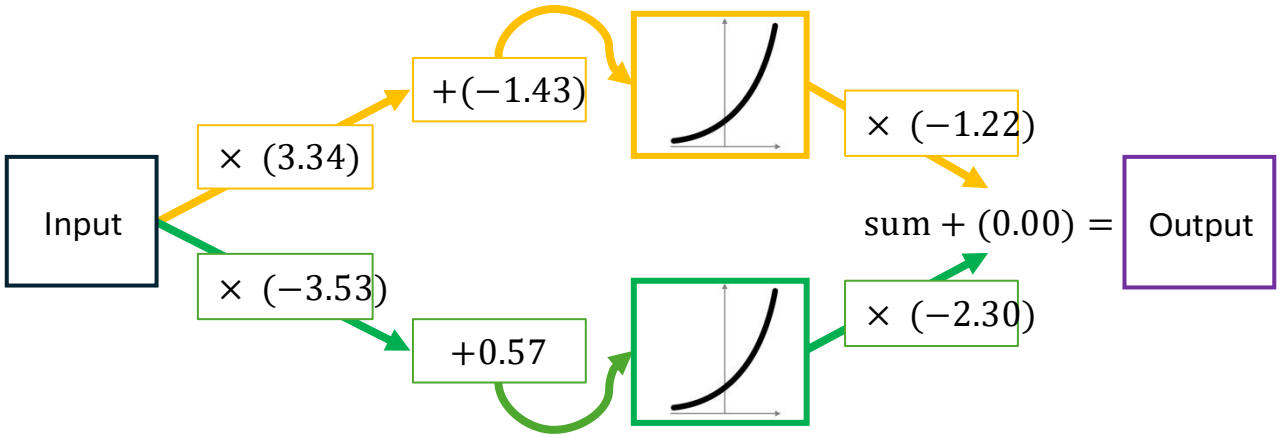


# Backpropagation Example

… the **purple squiggle** is pretty far from the data that we observed



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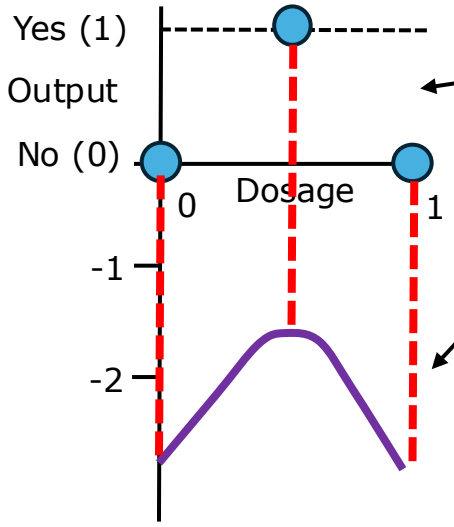
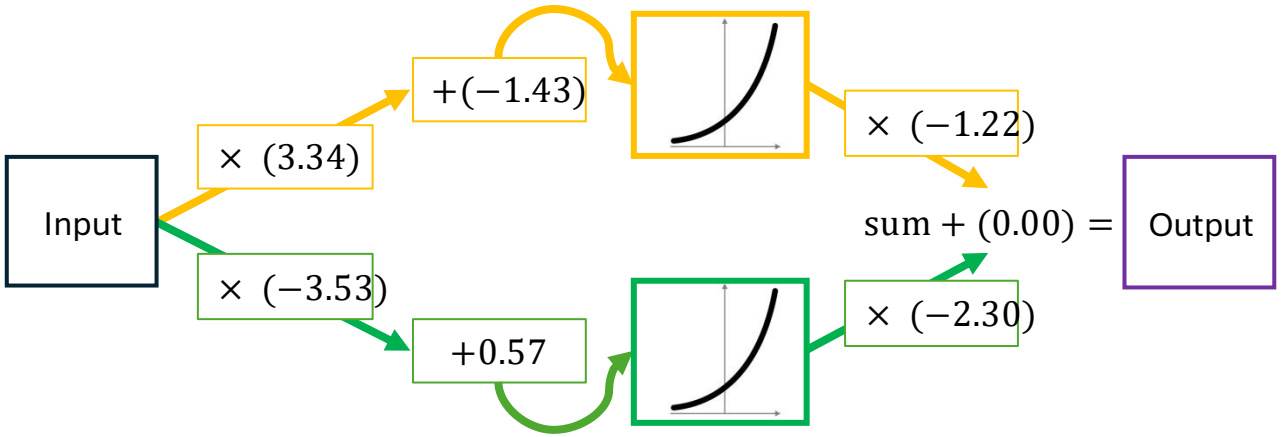
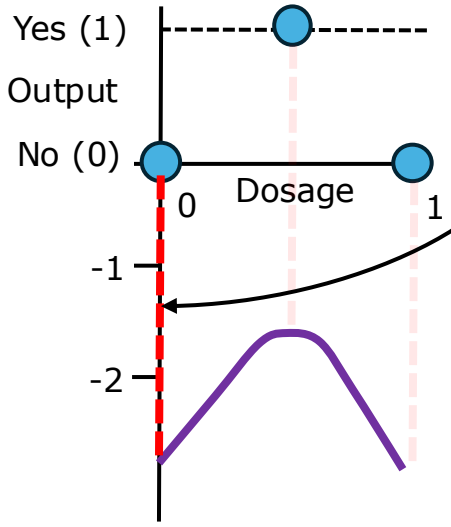



We can quantify how good the **purple squiggle** fits the data by calculating the **Sum of the Squared Residuals**

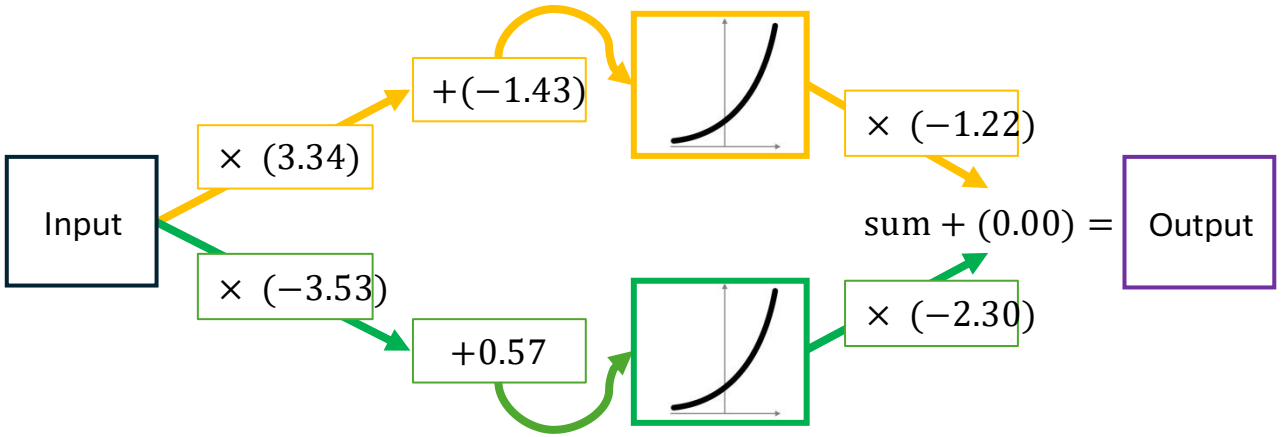


# Backpropagation Example

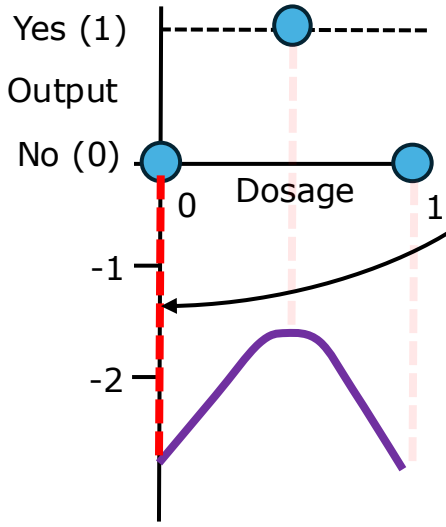



For example this **Residual** …

$$Residual = (Observed - Predicted)$$

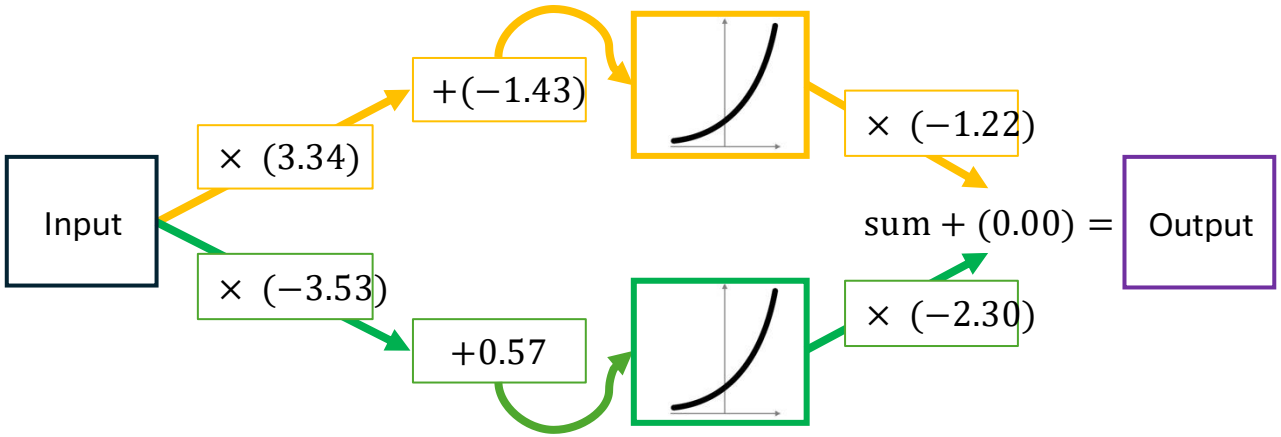


# Backpropagation Example

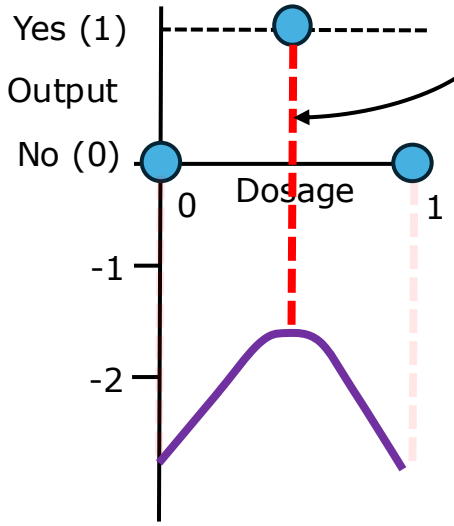



For example this **Residual** …

$$Residual_1 = (0 - (-2.6))$$



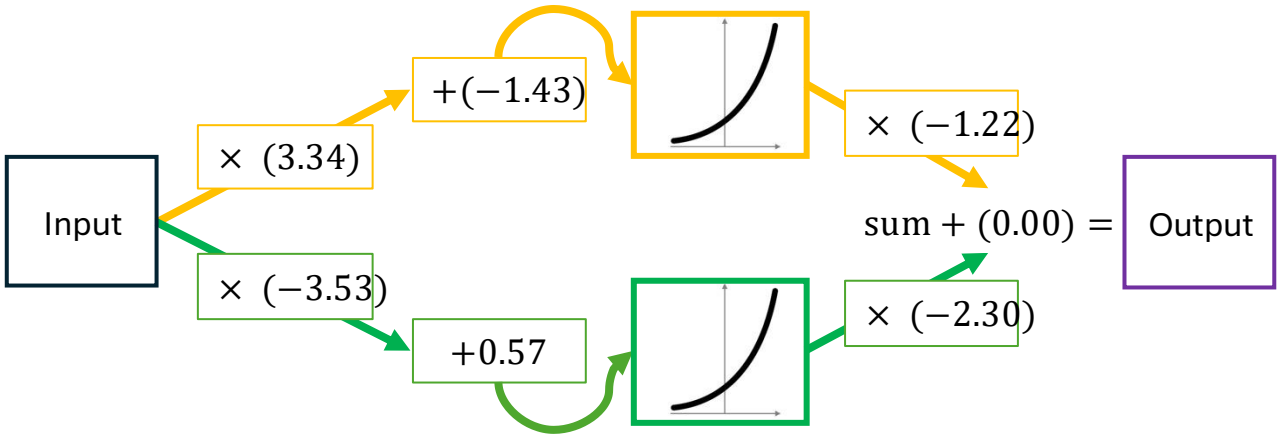
# Backpropagation Example



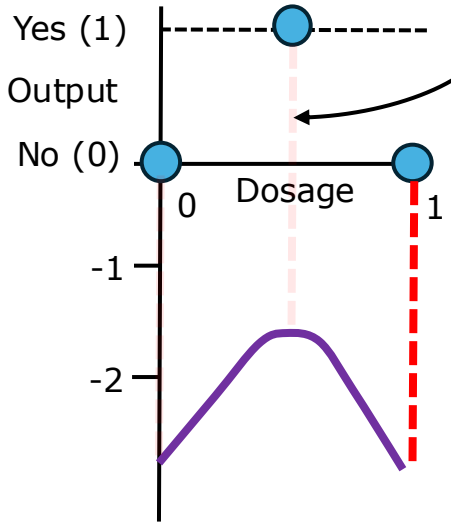

For example this **Residual** …

$$Residual_1 = (0 - (-2.6))$$

$$Residual_2 = (1 - (-1.6))$$



# Backpropagation Example

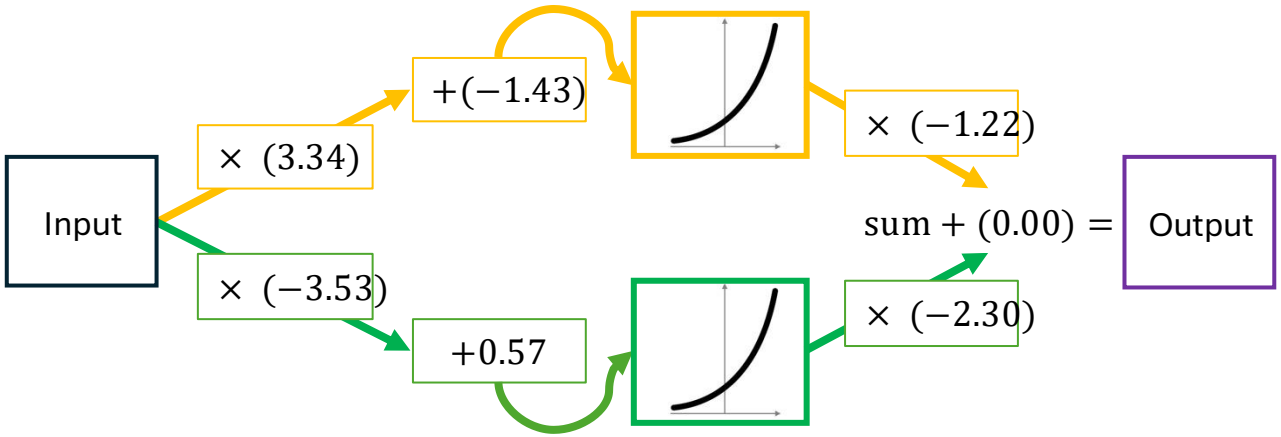



For example this **Residual** …

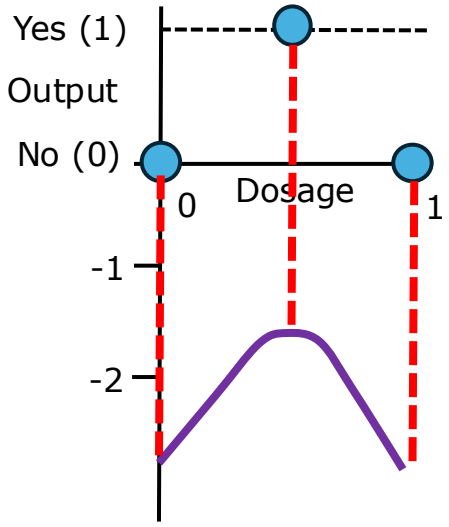
$$Residual_1 = (0 - (-2.6))$$

$$Residual_2 = (1 - (-1.6))$$

$$Residual_3 = (0 - (-2.6))$$

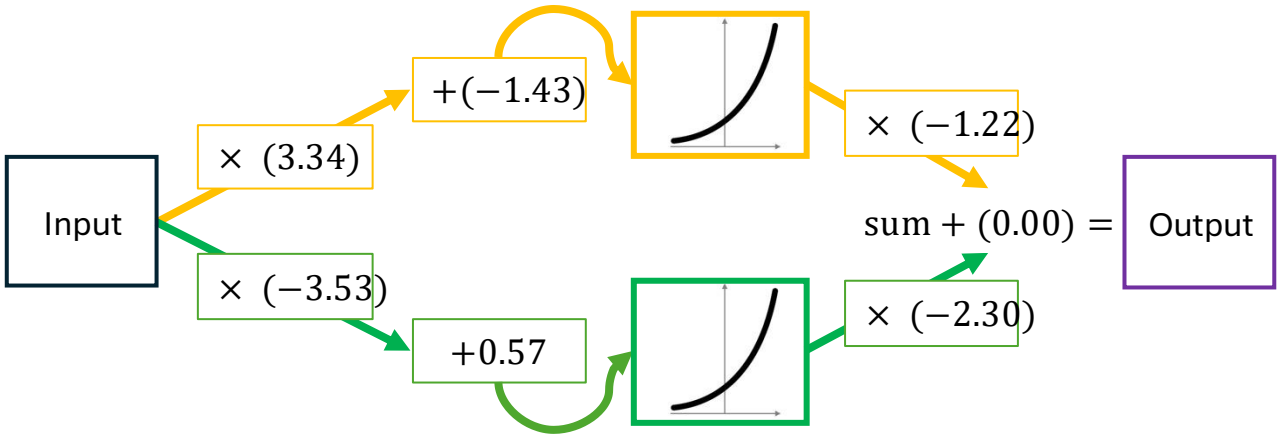


# Backpropagation Example

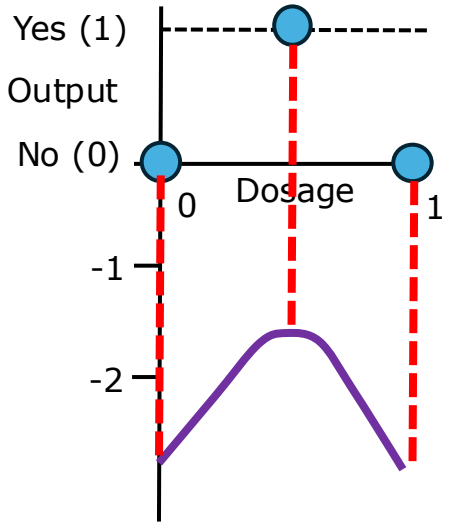



Now we square each **Residual** …

$$Residual = \left(0 - (-2.6)\right)^2 + \left(1 - (-1.6)\right)^2 + \left(0 - (-2.6)\right)^2$$

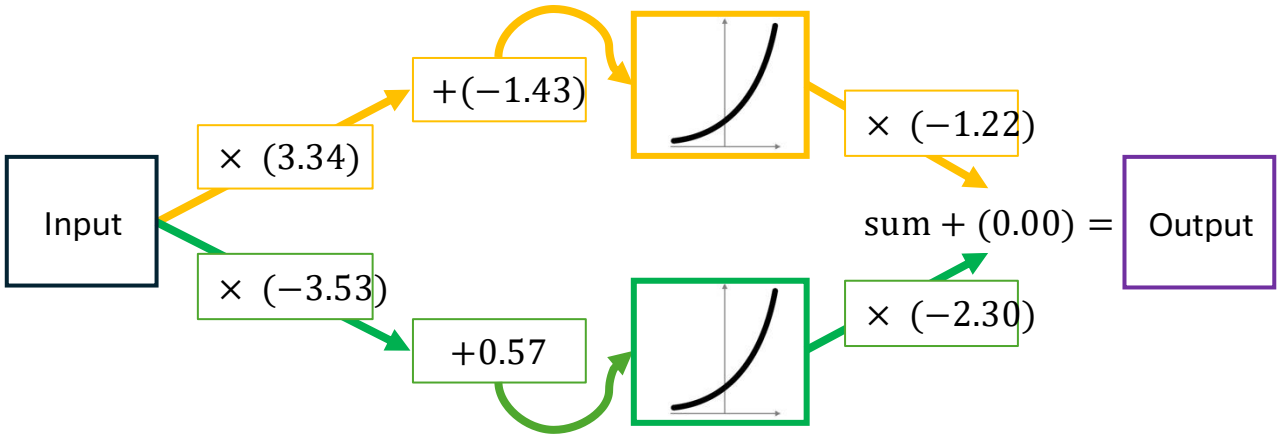


# Backpropagation Example



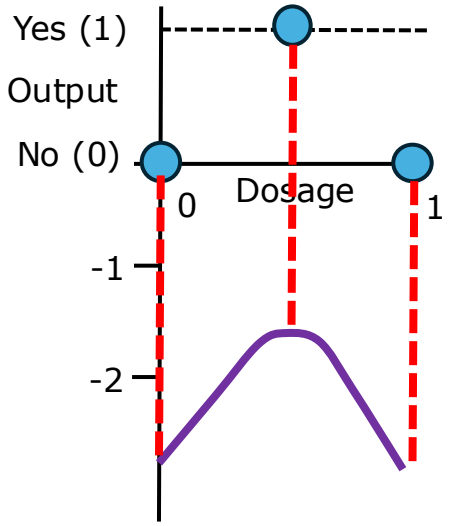

… to get $\mathbf{20.4}$ for the **Sum of the Squared Residuals ($L$)**:

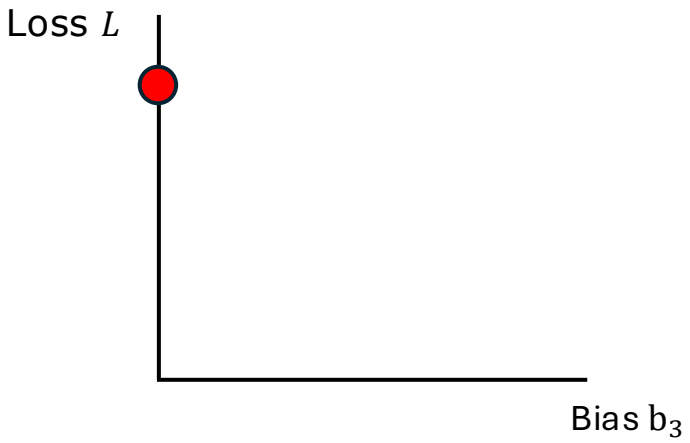
$$Residual = \left(0 - (-2.6)\right)^2 + \left(1 - (-1.6)\right)^2 + \left(0 - (-2.6)\right)^2 = 20.4$$

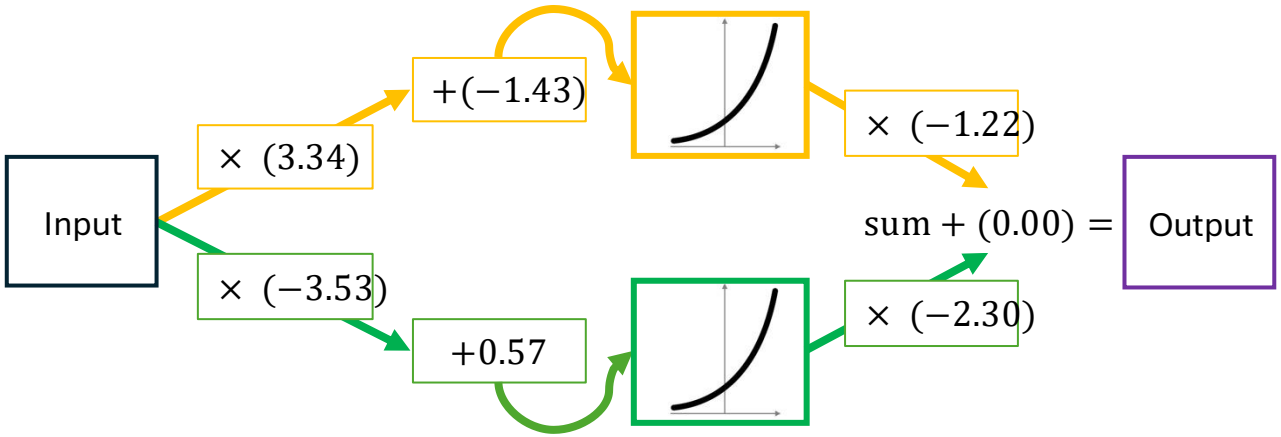


# Backpropagation Example

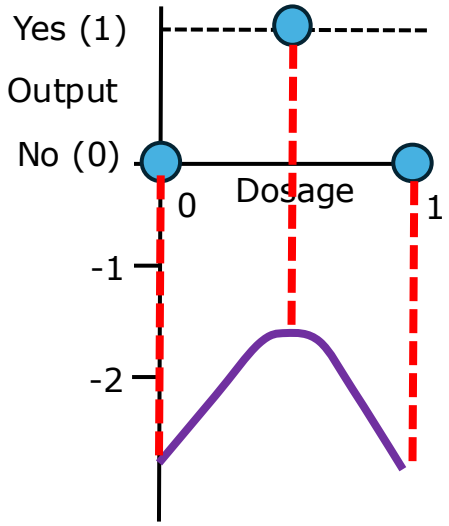
… plot the $\mathbf{20.4}$ for $\boldsymbol{L}$ on the y-axis



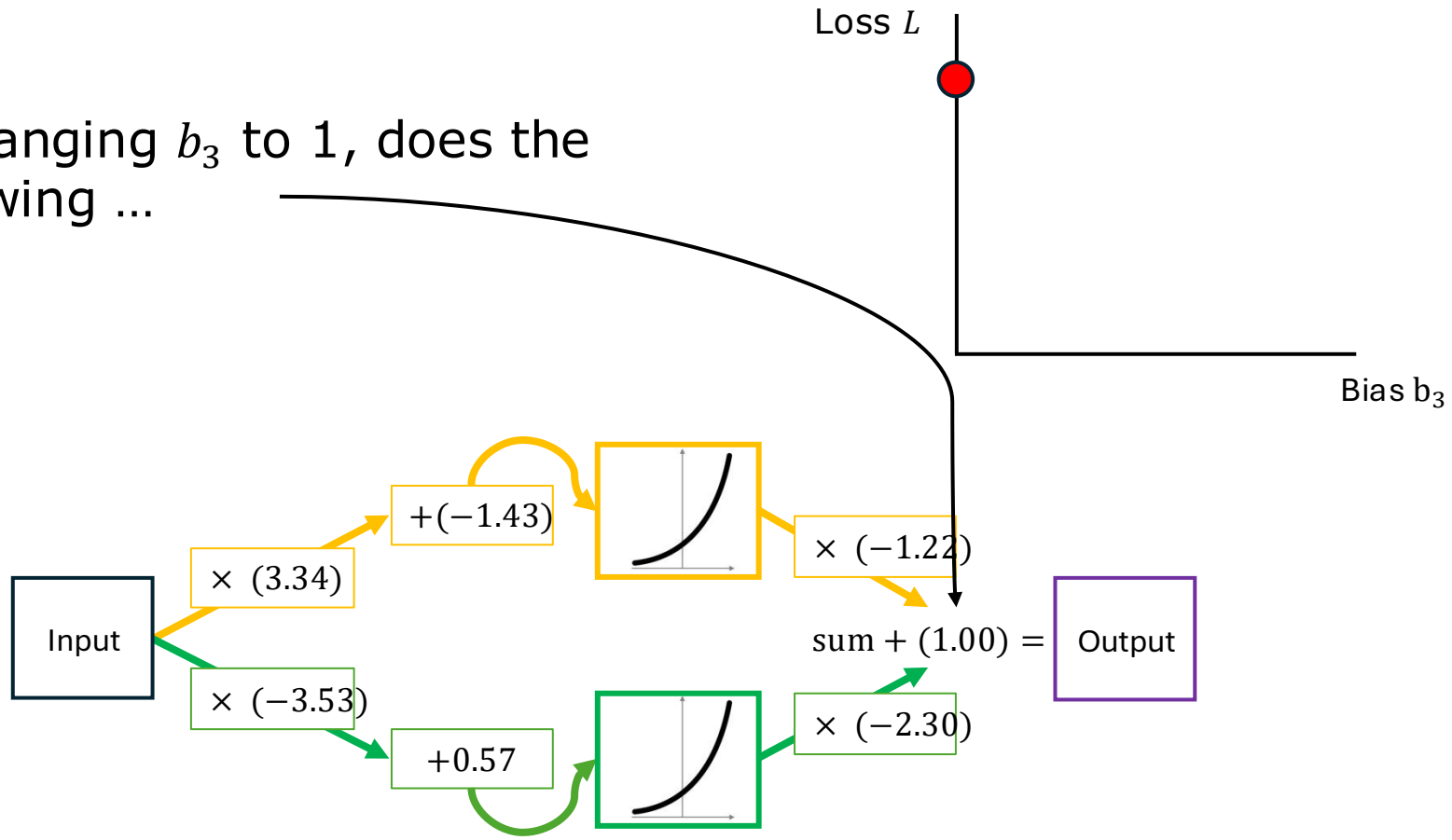



# Backpropagation Example

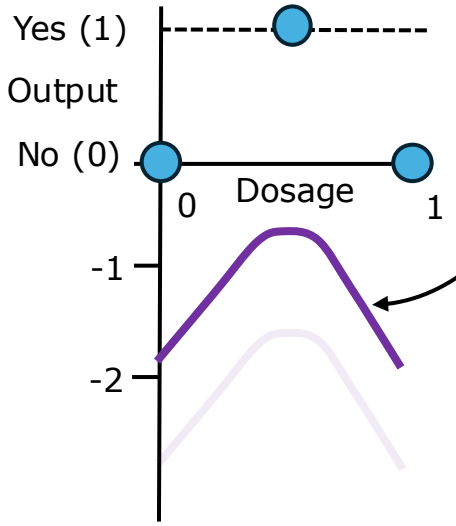


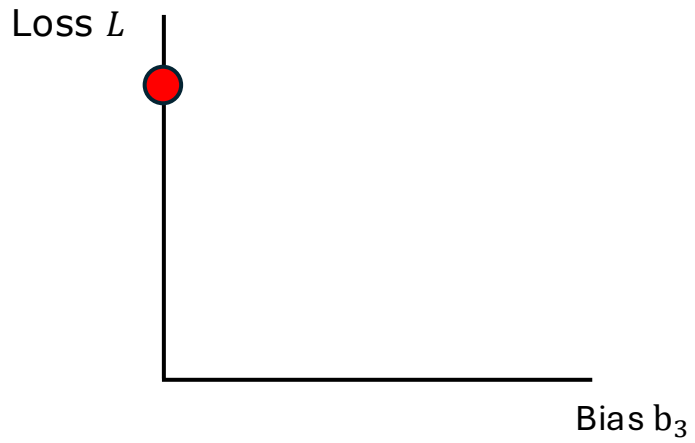

… changing $b_3$ to 1, does the following …

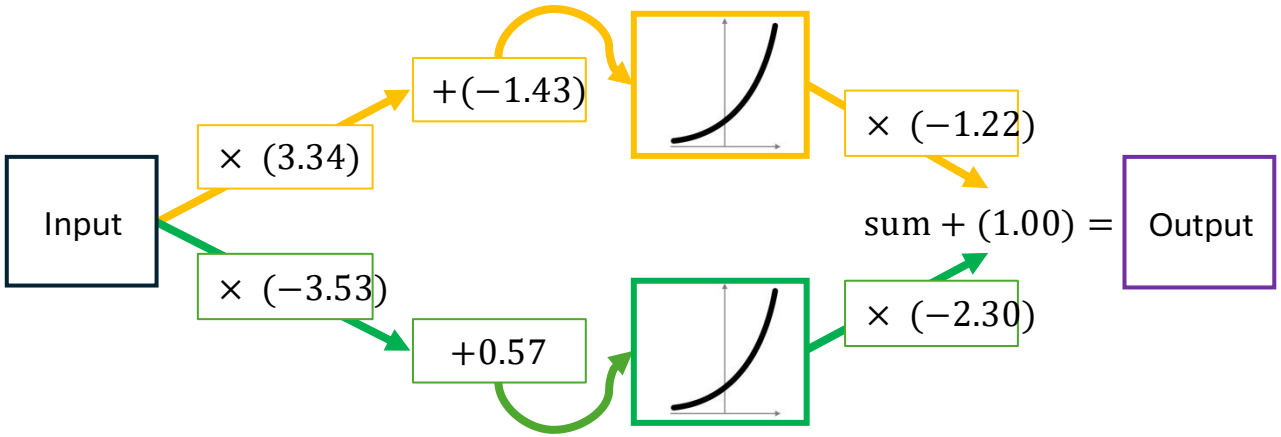


# Backpropagation Example

… changing $b_3$ to 1, does the following …



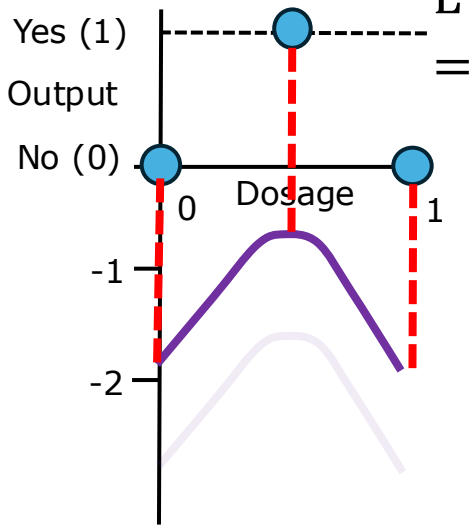



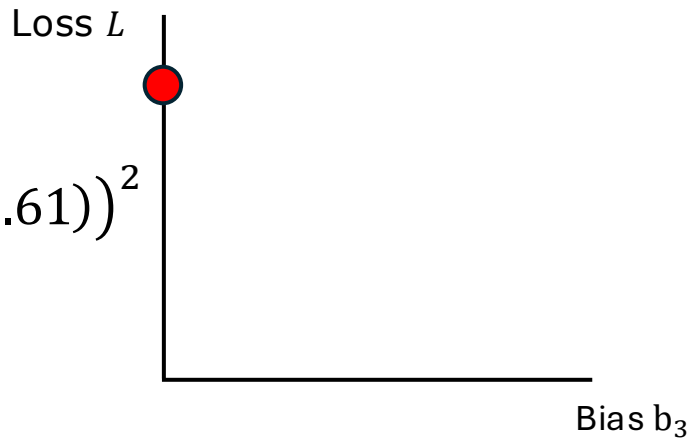



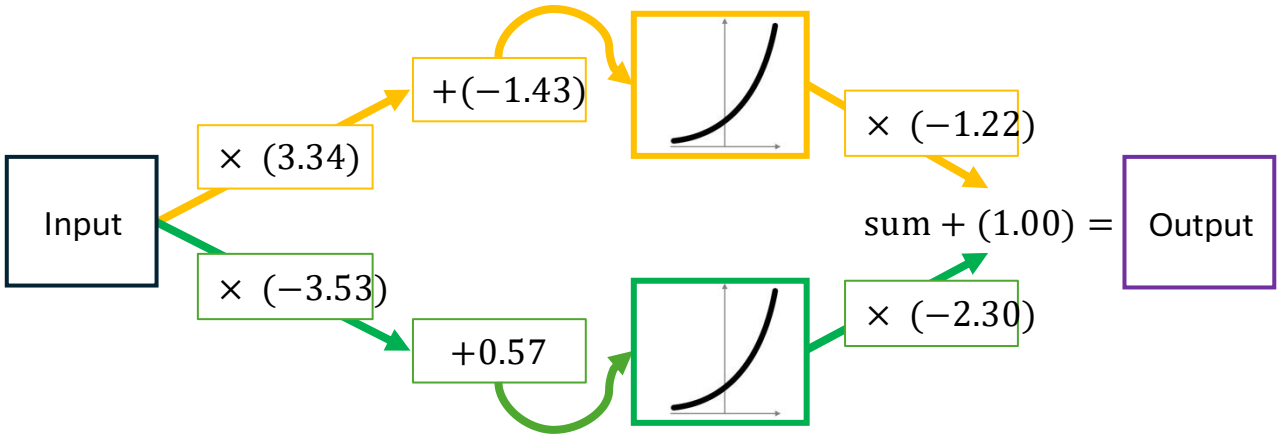
# Backpropagation Example

… computing the Loss:

$$L = \left(0 - (-1.6)\right)^2 + \left(1 - (-0.6)\right)^2 + \left(0 - (-1.61)\right)^2 = 7.8$$



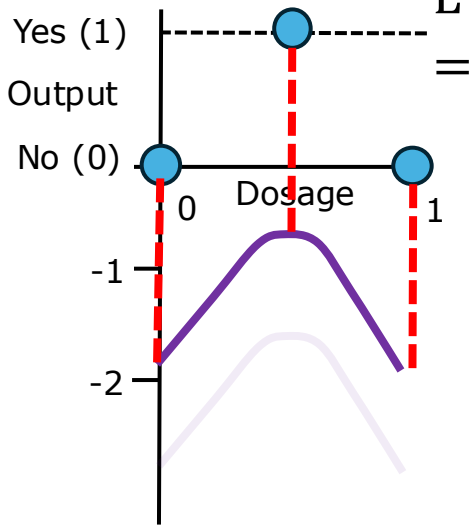



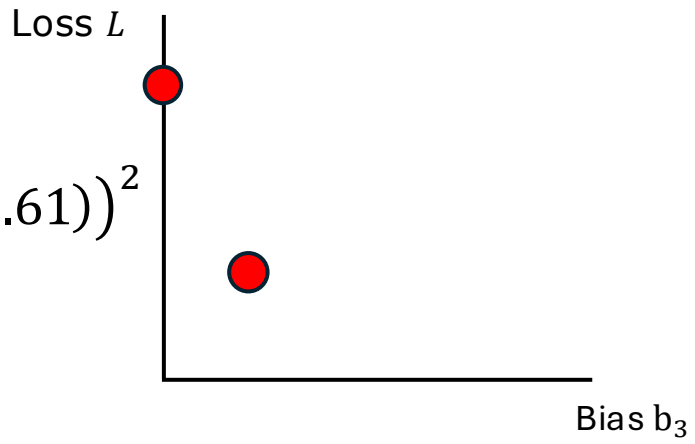



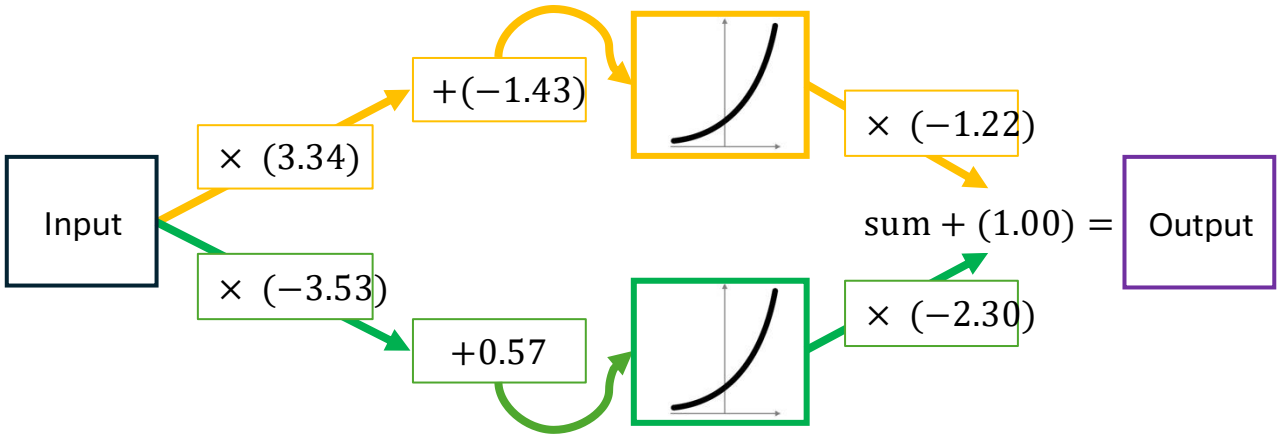
# Backpropagation Example

… computing the Loss:

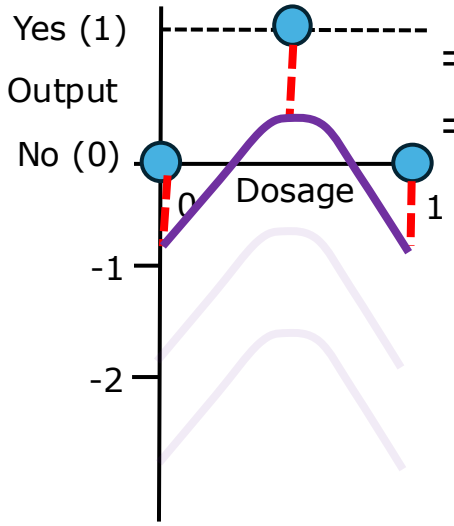
$$L = \left(0 - (-1.6)\right)^2 + \left(1 - (-0.6)\right)^2 + \left(0 - (-1.61)\right)^2 = 7.8$$



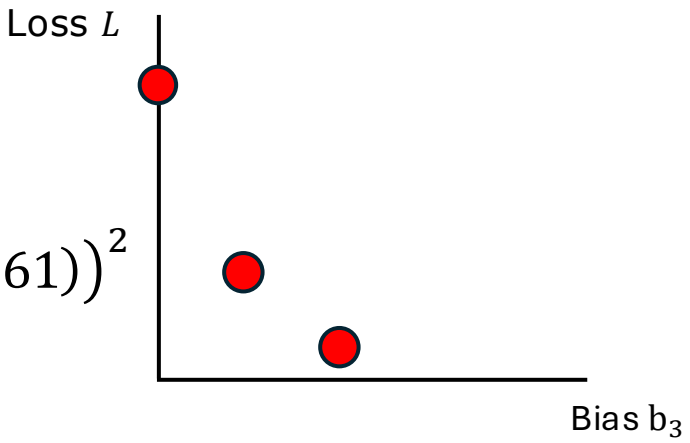



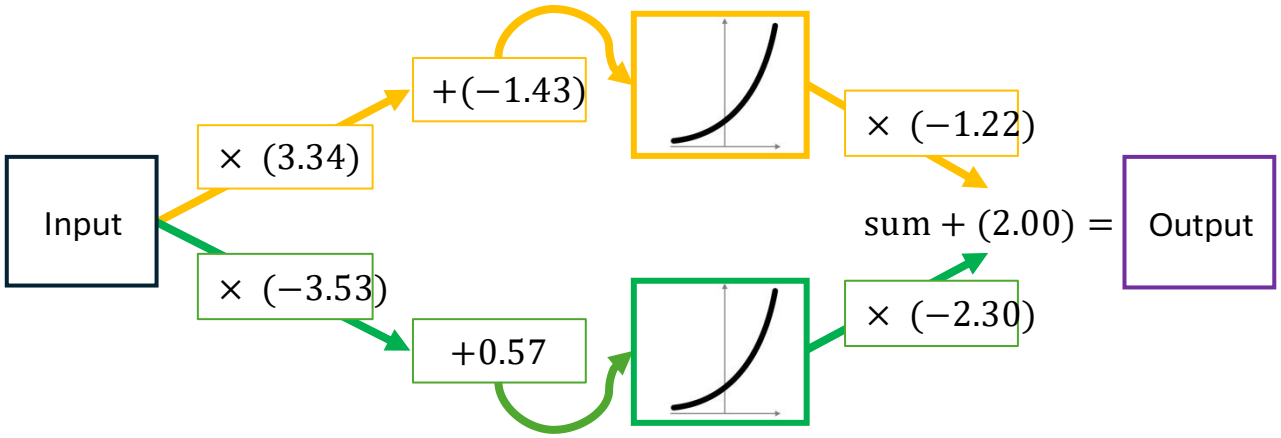
# Backpropagation Example




… increase $b_3$ to 2:

$$L = \left(0-(-0.6)\right)^2 + \left(1-(-0.39)\right)^2 + \left(0-(-0.61)\right)^2 = 1.11$$

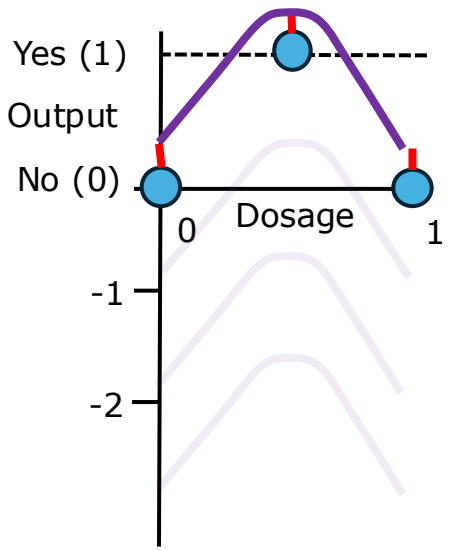


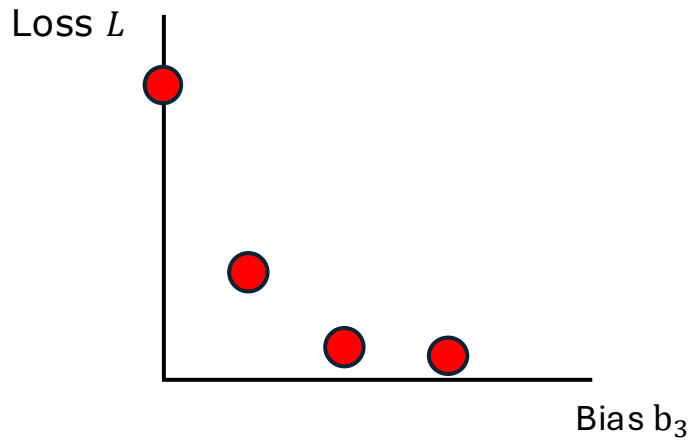



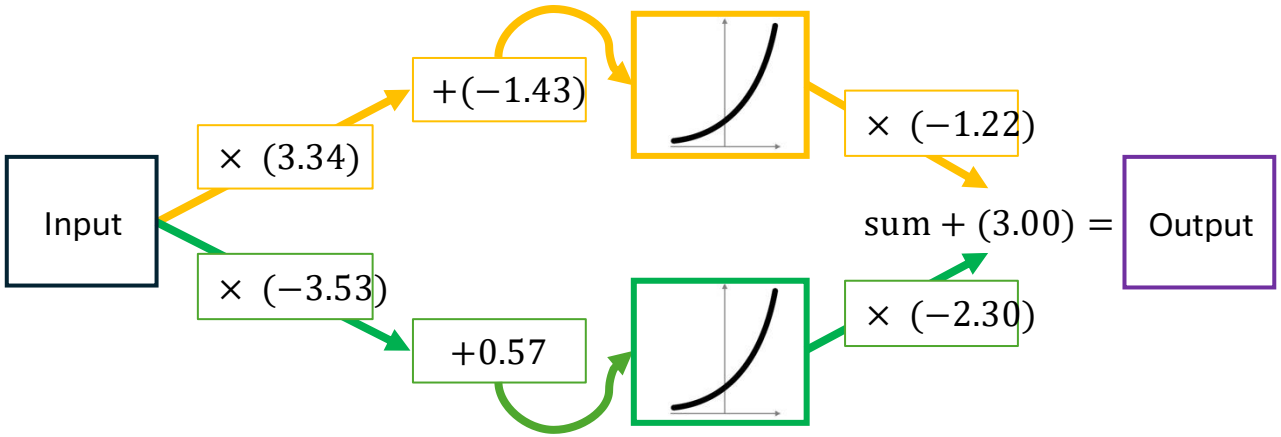
# Backpropagation Example

… increase $b_3$ to 3:

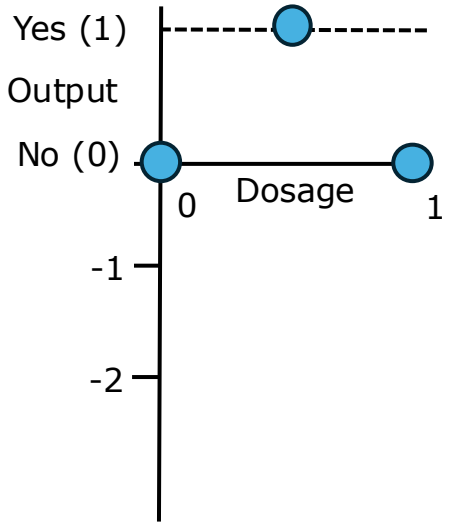
$$L = 0.46$$



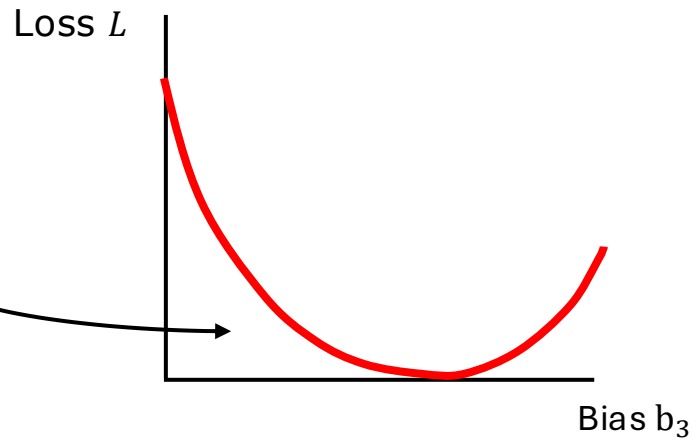



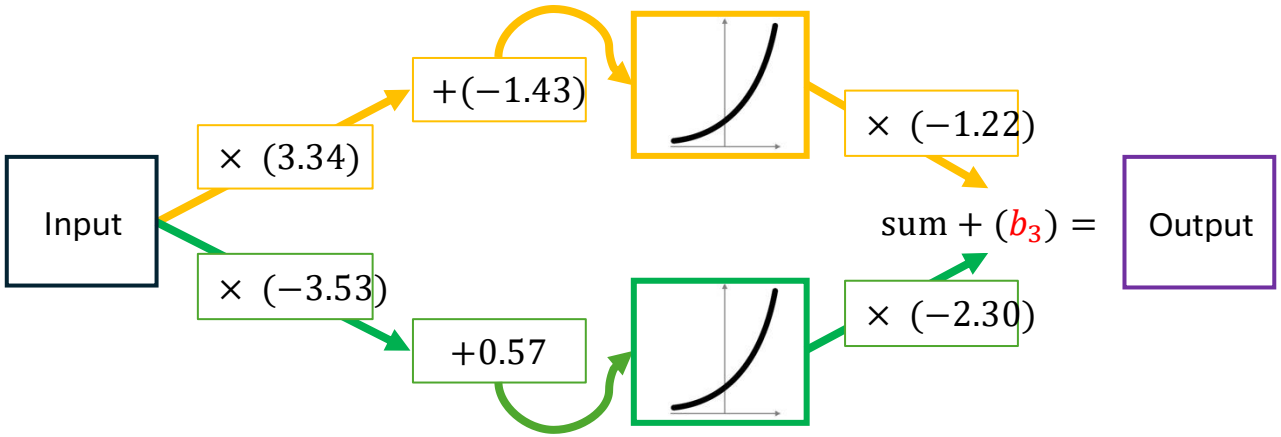



# Backpropagation Example

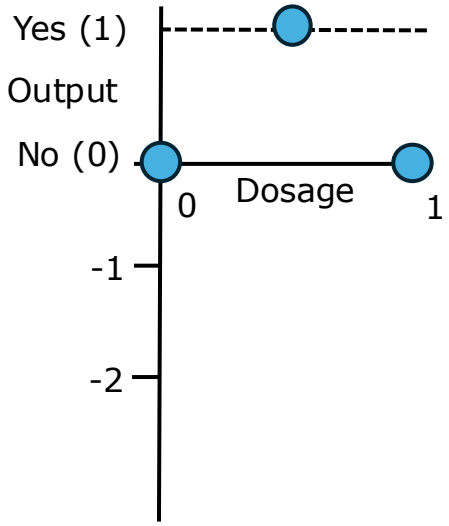



… if we had time to plug in tons of values for $b_3$, we would get this curve

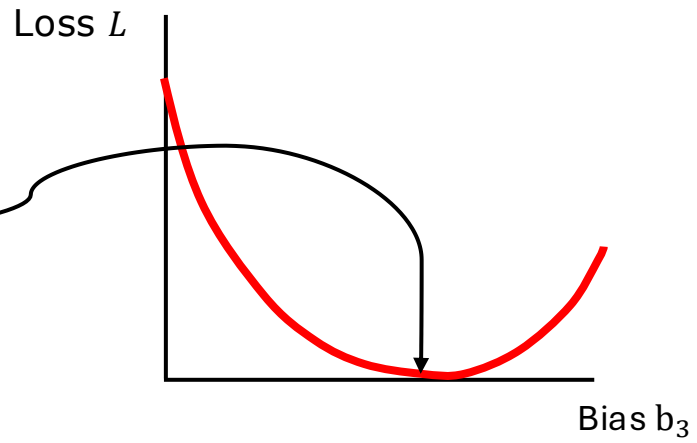


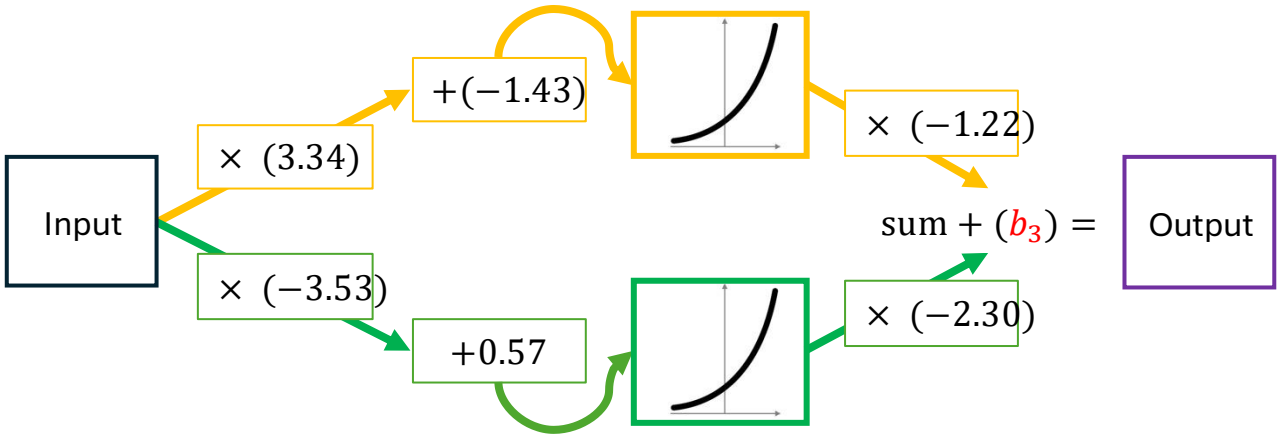



# Backpropagation Example

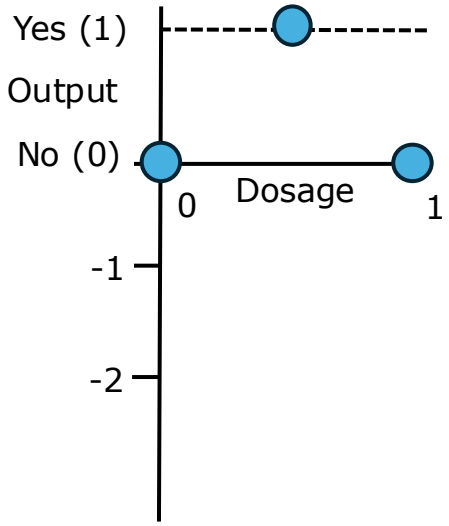



Aiming for the lowest Loss $L$ means aiming for the lowest point in this graph

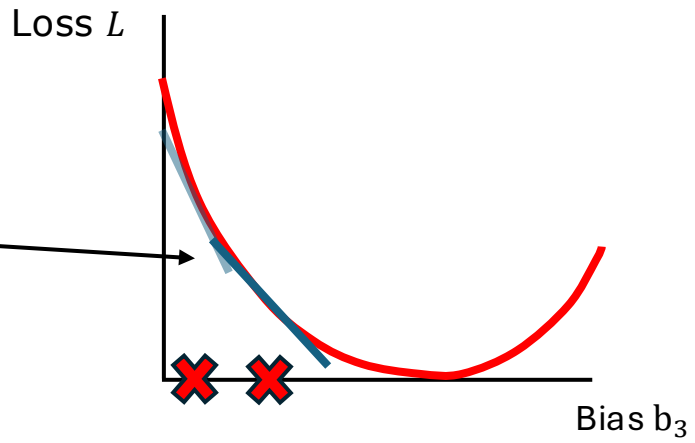


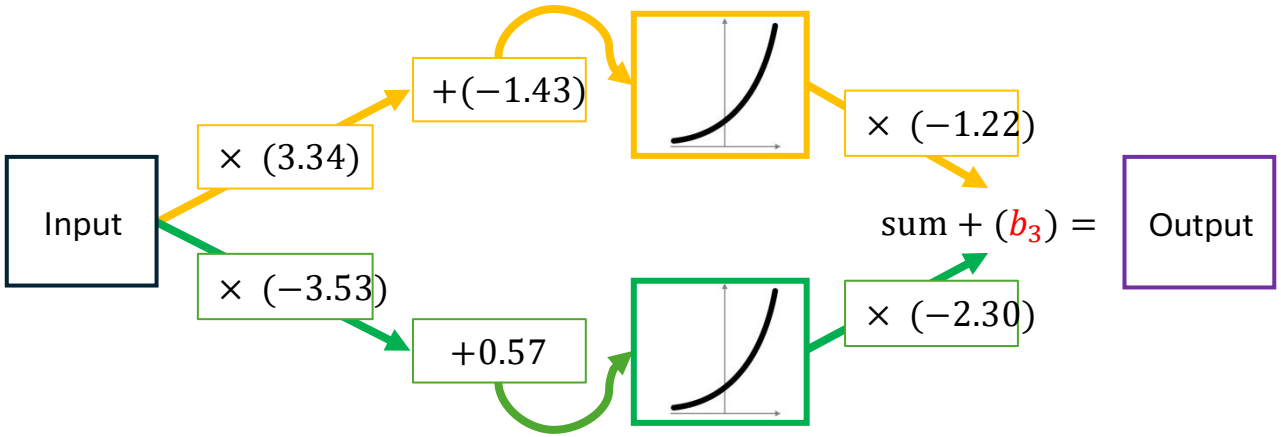



# Backpropagation Example

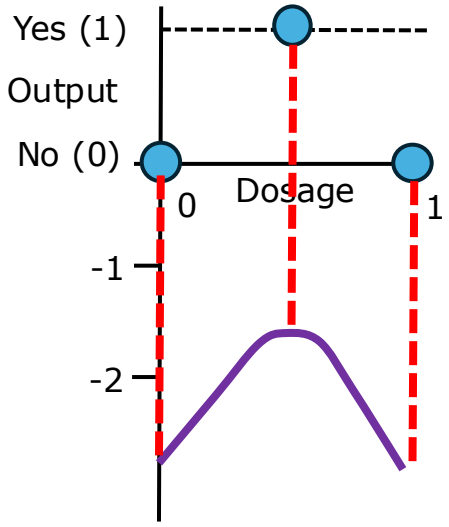



Instead of plugging in tons of values to find the lowest point, we use **Gradient Descent**



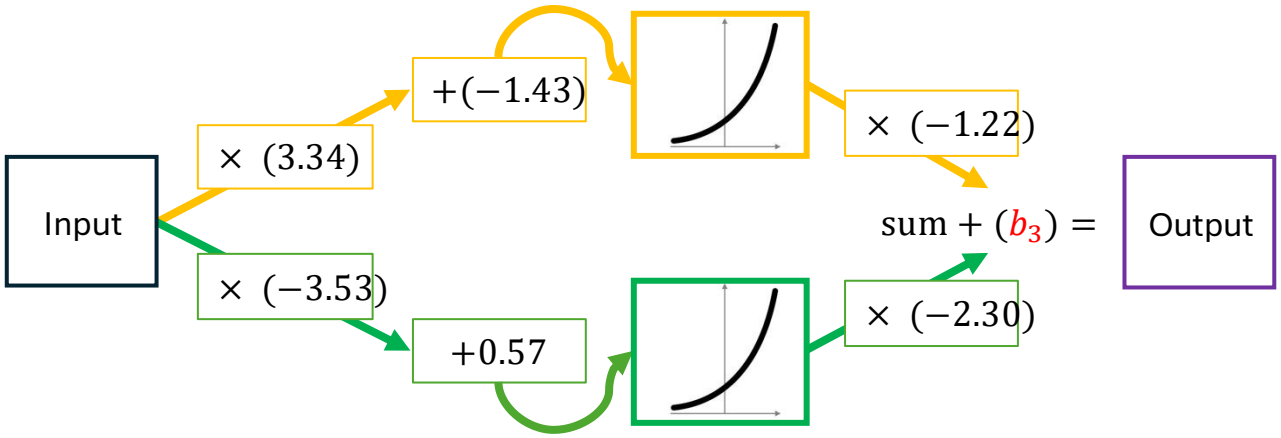



# Backpropagation Example

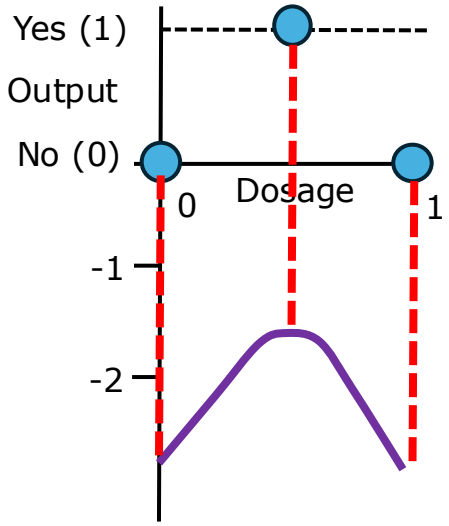



Remember the Loss:

$$\boldsymbol{L} = (\text{Oberserved}_1 - \text{Predicted}_1)^2 + (\text{Oberserved}_2 - \text{Predicted}_2)^2 + (\text{Oberserved}_3 - \text{Predicted}_3)^2$$

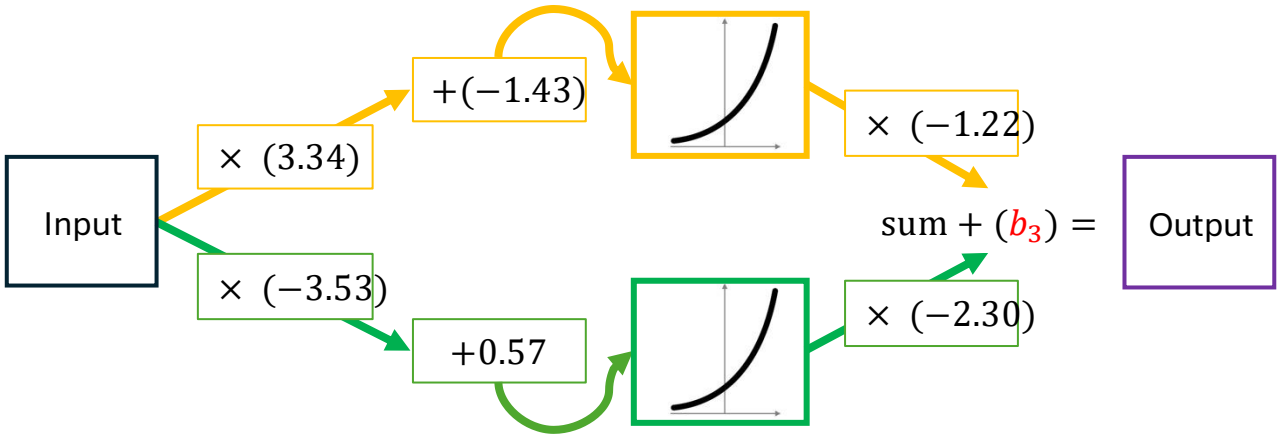


# Backpropagation Example

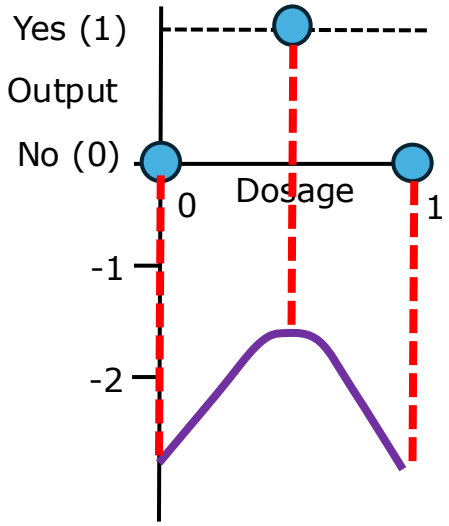



We could also write:

$$\boldsymbol{L} = \sum_{i=1}^{n=3} (\text{Observed}_i - \text{Predicted}_i)^2$$



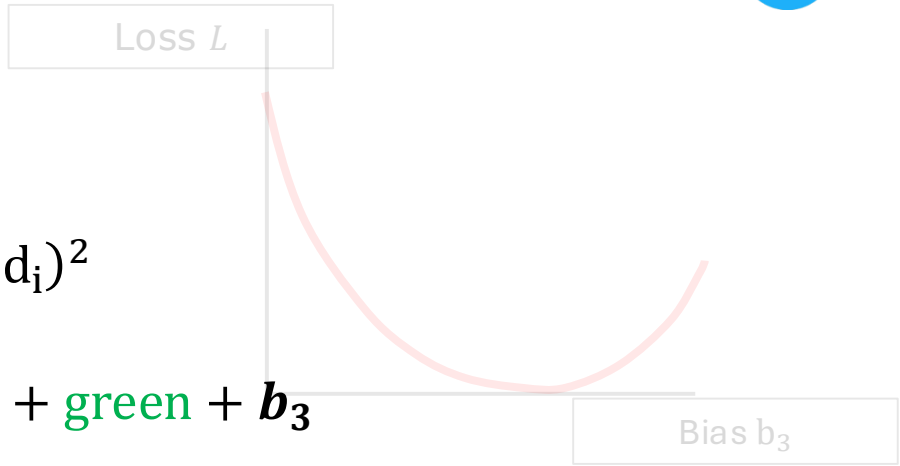
# Backpropagation Example

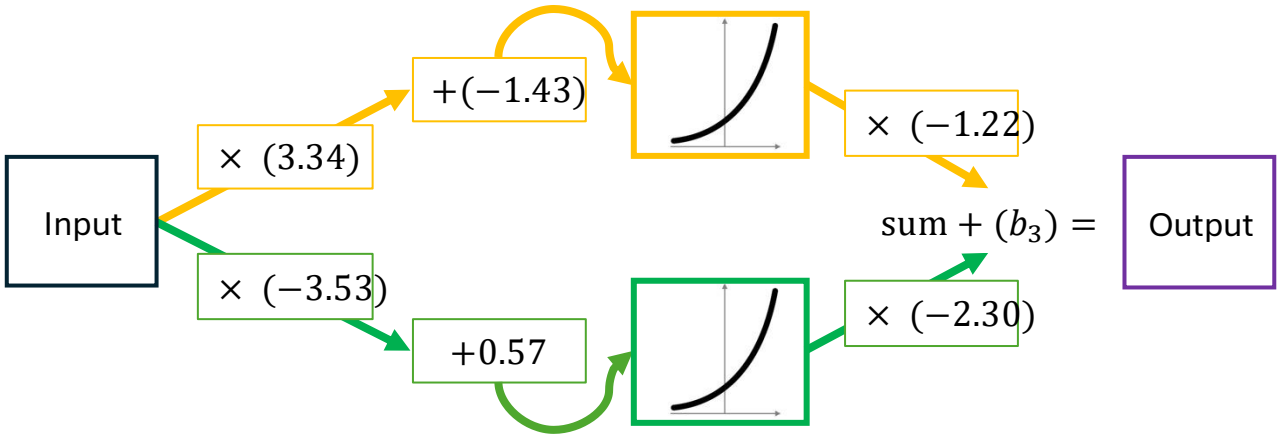



The Loss is linked to $b_3$:

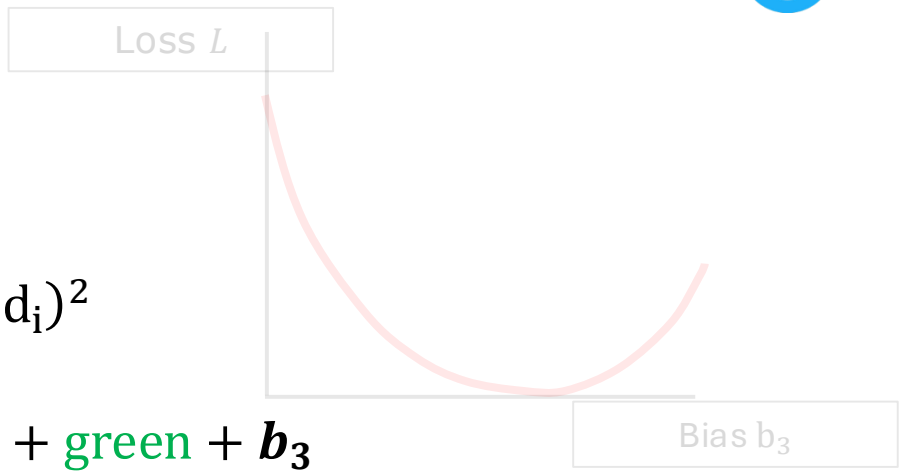
$$\boldsymbol{L} = \sum_{i=1}^{n=3} (\text{Observed}_i - \text{Predicted}_i)^2$$

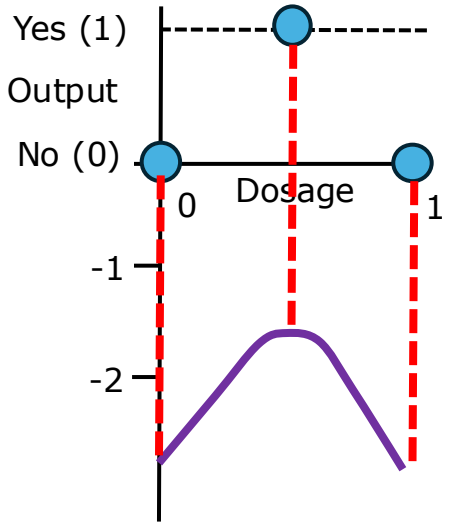
$$\text{Predicted}_i = \text{purple squiggle}_i = \text{yellow} + \text{green} + \boldsymbol{b_3}$$



# Backpropagation Example

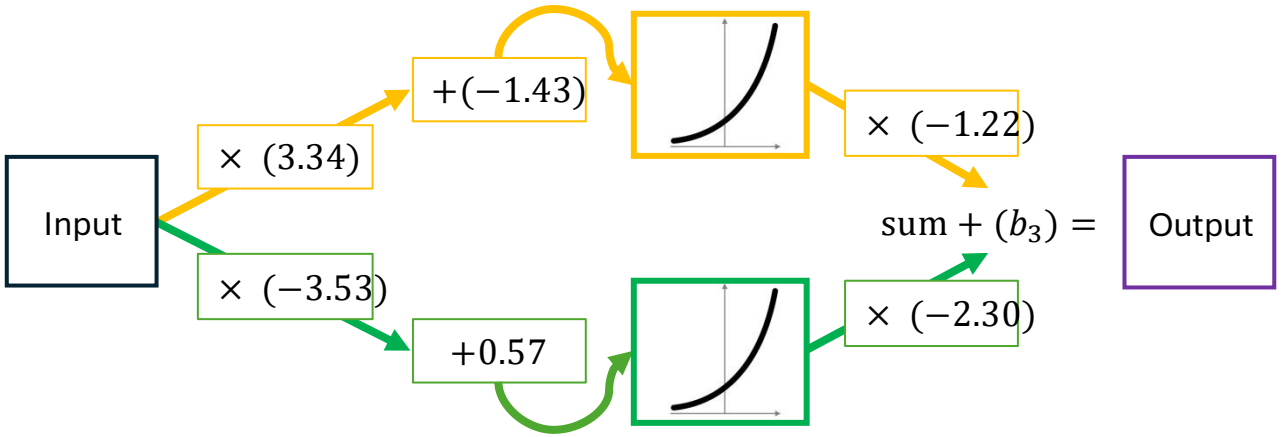


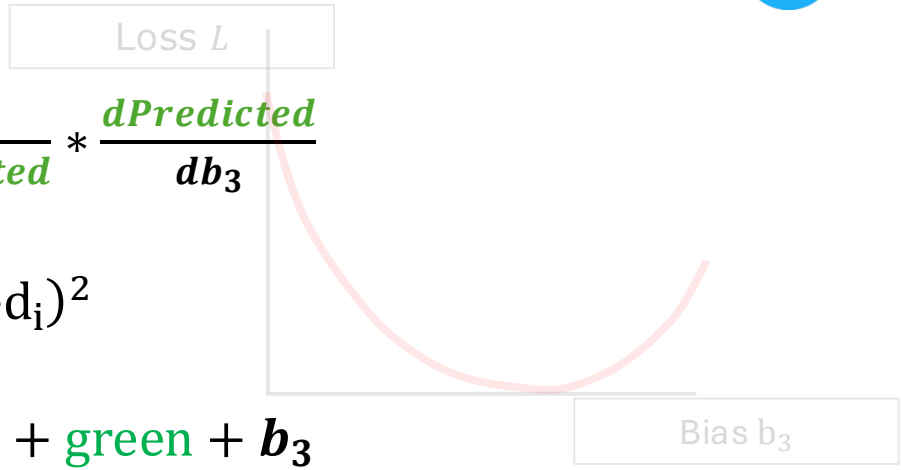

We want: $\frac{dL}{db_3}$

$$L = \sum_{i=1}^{n=3} (\text{Observed}_i - \text{Predicted}_i)^2$$

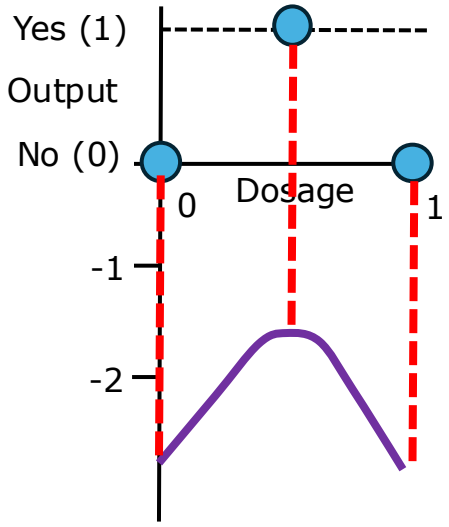
$$\text{Predicted}_i = \text{purple squiggle}_i = \text{yellow} + \text{green} + b_3$$



# Backpropagation Example

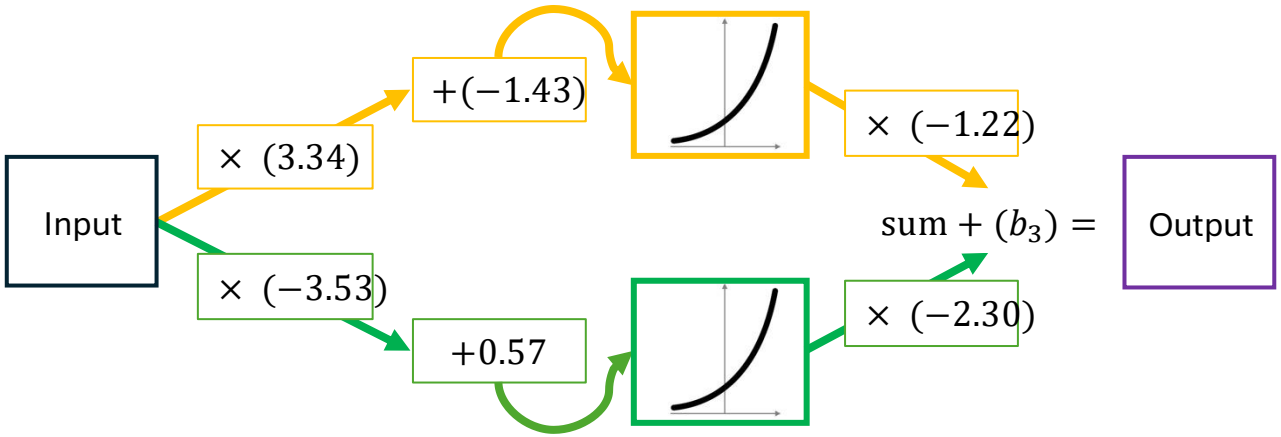



The **chain rule** says: $\frac{dL}{db_3} = \frac{dL}{dPredicted} * \frac{dPredicted}{db_3}$




$$L = \sum_{i=1}^{n=3} (\text{Observed}_i - \text{Predicted}_i)^2$$

$$\text{Predicted}_i = \text{purple squiggle}_i = \text{yellow} + \text{green} + b_3$$



# Backpropagation Example

$$\frac{dL}{db_3} = \frac{dL}{dPredicted} * \frac{dPredicted}{db_3}$$

Now, we can solve the derivative of $L$ with respect to the **Predicted** values by first substituting in the equation and then the **chain rule**

$$\frac{dL}{dPredicted} = \frac{d}{dPredicted} \sum_{i=1}^{n=3} (\text{Observed}_i - \text{Predicted}_i)^2$$

$$L = \sum_{i=1}^{n=3} (\text{Observed}_i - \text{Predicted}_i)^2$$

$$\text{Predicted}_i = \text{purple squiggle}_i = \text{yellow} + \text{green} + b_3$$

# Backpropagation Example

$$\frac{dL}{db_3} = \frac{dL}{dPredicted} * \frac{dPredicted}{db_3}$$

Now, we can solve the derivative of $L$ with respect to the **Predicted** values by first substituting in the equation and then the **chain rule**

$$\frac{dL}{dPredicted} = \frac{d}{dPredicted} \sum_{i=1}^{n=3} (\text{Observed}_i - \text{Predicted}_i)^2 = \sum_{i=1}^{n=3} 2 * (\text{Observed}_i - \text{Predicted}_i) * (-1)$$

$$L = \sum_{i=1}^{n=3} (\text{Observed}_i - \text{Predicted}_i)^2$$

$$\text{Predicted}_i = \text{purple squiggle}_i = \text{yellow} + \text{green} + b_3$$

# Backpropagation Example

$$\frac{dL}{db_3} = \frac{dL}{dPredicted} * \frac{dPredicted}{db_3}$$

Now, we can solve the derivative of $L$ with respect to the **Predicted** values by first substituting in the equation and then the **chain rule**

$$\frac{dL}{dPredicted} = \frac{d}{dPredicted} \sum_{i=1}^{n=3} (\text{Observed}_i - \text{Predicted}_i)^2 = \sum_{i=1}^{n=3} -2 * (\text{Observed}_i - \text{Predicted}_i)$$

$$L = \sum_{i=1}^{n=3} (\text{Observed}_i - \text{Predicted}_i)^2$$

$$\text{Predicted}_i = \text{purple squiggle}_i = \text{yellow} + \text{green} + b_3$$

# Backpropagation Example

$$\frac{dL}{db_3} = \underbrace{\frac{dL}{dPredicted}} * \frac{dPredicted}{db_3}$$

$$= \sum_{i=1}^{n=3} -2 * (\text{Observed}_i - \text{Predicted}_i)$$

We are done with the first part

$$L = \sum_{i=1}^{n=3} (\text{Observed}_i - \text{Predicted}_i)^2$$

$$\text{Predicted}_i = \text{purple squiggle}_i = \text{yellow} + \text{green} + b_3$$

# Backpropagation Example

$$\frac{dL}{db_3} = \underbrace{\frac{dL}{dPredicted}} * \frac{dPredicted}{db_3}$$

$$= \sum_{i=1}^{n=3} -2 * (\text{Observed}_i - \text{Predicted}_i)$$

We are done with the first part

$$L = \sum_{i=1}^{n=3} (\text{Observed}_i - \text{Predicted}_i)^2$$

$$\text{Predicted}_i = \text{purple squiggle}_i = \text{yellow} + \text{green} + b_3$$

# Backpropagation Example

$$\frac{dL}{db_3} = \frac{dL}{dPredicted} * \frac{dPredicted}{db_3}$$

Now, let's proceed with the second part, the **Predicted** values with respect to $b_3$

$$= \sum_{i=1}^{n=3} -2 * (\text{Observed}_i - \text{Predicted}_i)$$

$$L = \sum_{i=1}^{n=3} (\text{Observed}_i - \text{Predicted}_i)^2$$

$$\text{Predicted}_i = \text{purple squiggle}_i = \text{yellow} + \text{green} + b_3$$

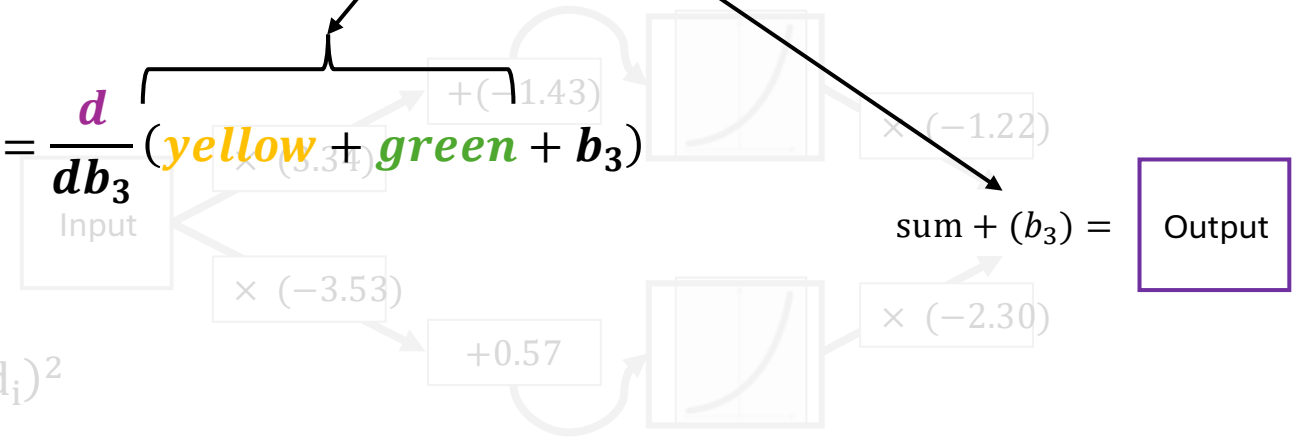
# Backpropagation Example

$$\frac{dL}{db_3} = \frac{dL}{dPredicted} * \frac{dPredicted}{db_3}$$

$$= \sum_{i=1}^{n=3} -2 * (\text{Observed}_i - \text{Predicted}_i)$$

$$\frac{dPredicted}{db_3} = \frac{d}{db_3} purple\ squiggle = \frac{d}{db_3}(yellow + green + b_3)$$

Remember, the **yellow** and **green** curves were created before we got to $b_3$




$$L = \sum_{i=1}^{n=3} (\text{Observed}_i - \text{Predicted}_i)^2$$

$$\text{Predicted}_i = \text{purple squiggle}_i = \text{yellow} + \text{green} + b_3$$

# Backpropagation Example

$$\frac{dL}{db_3} = \frac{dL}{dPredicted} * \frac{dPredicted}{db_3}$$

... which leads us to ...

$$= \sum_{i=1}^{n=3} -2 * (\text{Observed}_i - \text{Predicted}_i)$$

$$\frac{dPredicted}{db_3} = \frac{d}{db_3} purple\ squiggle = \frac{d}{db_3}(yellow + green + b_3) = 0 + 0 + 1 = 1$$

$$L = \sum_{i=1}^{n=3} (\text{Observed}_i - \text{Predicted}_i)^2$$

$$\text{Predicted}_i = \text{purple squiggle}_i = \text{yellow} + \text{green} + b_3$$

# Backpropagation Example

$$\frac{dL}{db_3} = \frac{dL}{dPredicted} * \frac{dPredicted}{db_3}$$

... which leads us to ...

$$= \sum_{i=1}^{n=3} -2 * (\text{Observed}_i - \text{Predicted}_i) * 1$$

$$\frac{dPredicted}{db_3} = \frac{d}{db_3} purple\ squiggle = \frac{d}{db_3}(yellow + green + b_3) = 1$$

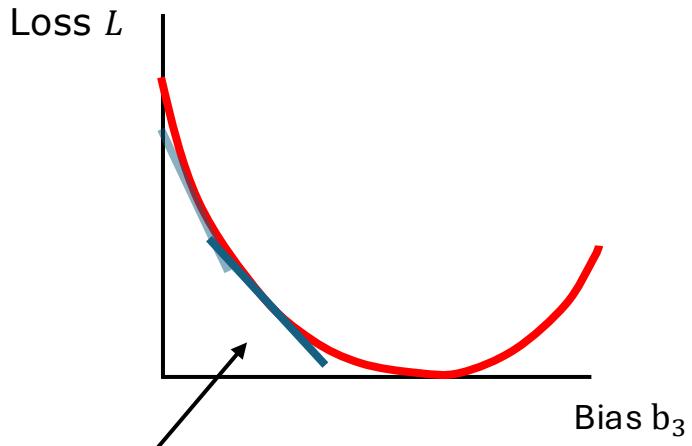
$$L = \sum_{i=1}^{n=3} (\text{Observed}_i - \text{Predicted}_i)^2$$

$$\text{Predicted}_i = \text{purple squiggle}_i = \text{yellow} + \text{green} + b_3$$

# Backpropagation Example

$$\frac{dL}{db_3} = \frac{dL}{dPredicted} * \frac{dPredicted}{db_3}$$

$$= \sum_{i=1}^{n=3} -2 * (\text{Observed}_i - \text{Predicted}_i) * 1$$




Now, we have the derivative of the Loss $\boldsymbol{L}$ with respect to $\boldsymbol{b_3}$

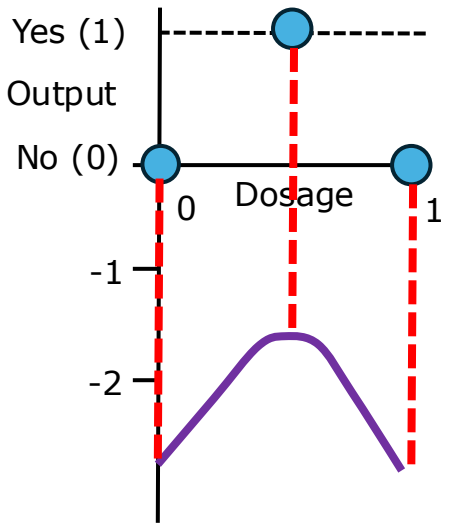
… and that means we can plug the derivative into **Gradient Descent** to find the optimal value for $b_3$

# Backpropagation Example

$$\frac{dL}{db_3} = \frac{dL}{dPredicted} * \frac{dPredicted}{db_3}$$

$$= \sum_{i=1}^{n=3} -2 * (\text{Observed}_i - \text{Predicted}_i) * 1$$

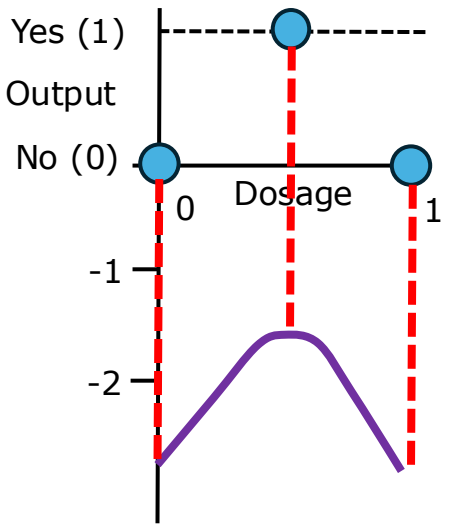
$$= -\mathbf{2} * (0 - \text{Predicted}_1) * \mathbf{1}$$
$$+ (-\mathbf{2} * (1 - \text{Predicted}_2) * \mathbf{1})$$
$$+ (-\mathbf{2} * (0 - \text{Predicted}_3) * \mathbf{1})$$



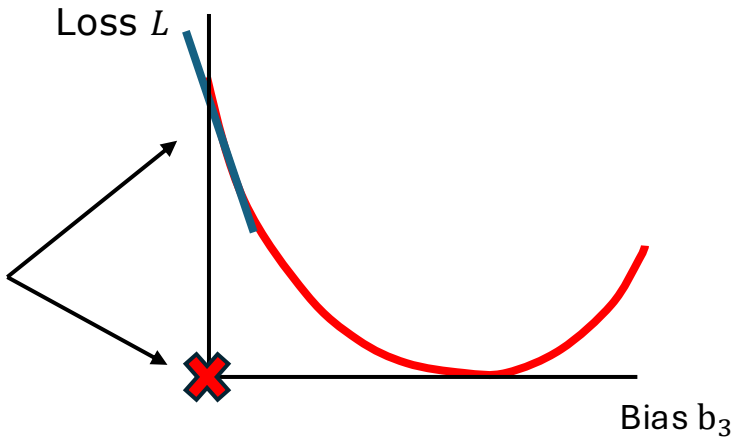

$$= -\mathbf{2} * \big(0 - (-2.6)\big) * \mathbf{1} + \big(-\mathbf{2} * \big(1 - (-1.6)\big) * \mathbf{1}\big)$$
$$+ \big(-\mathbf{2} * \big(0 - (-2.6)\big) * \mathbf{1}\big) = -15.7$$

# Backpropagation Example

$$\frac{dL}{db_3} = \boxed{-15.7}$$




That corresponds to the slope for $\boldsymbol{b_3 = 0}$



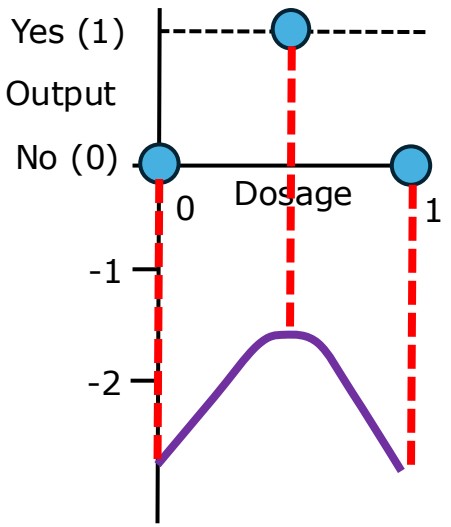
# Backpropagation Example

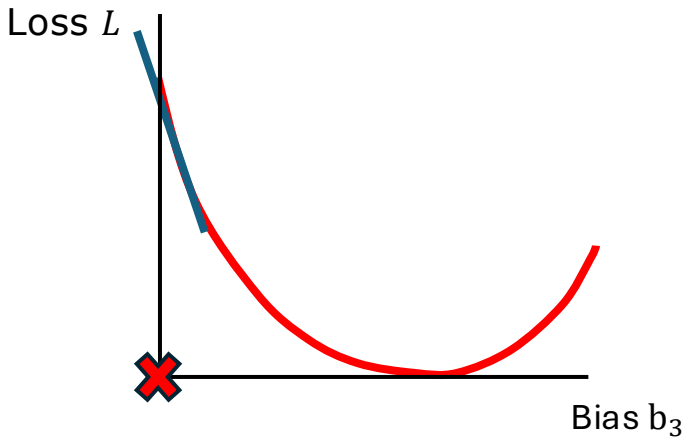
$$\frac{dL}{db_3} = \boxed{-15.7}$$

Remember
**Gradient Descent:**

$$w := w - \eta \frac{\partial L}{\partial w}$$

$$\text{Step Size} = 0 - \text{Learning Rate} \cdot (-15.7)$$



# Backpropagation Example

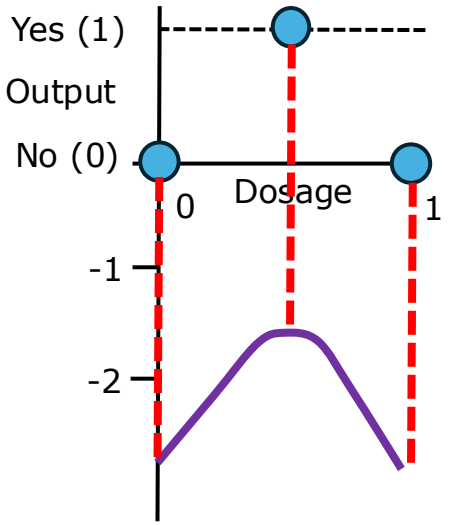
$$\frac{dL}{db_3} = \boxed{-15.7}$$

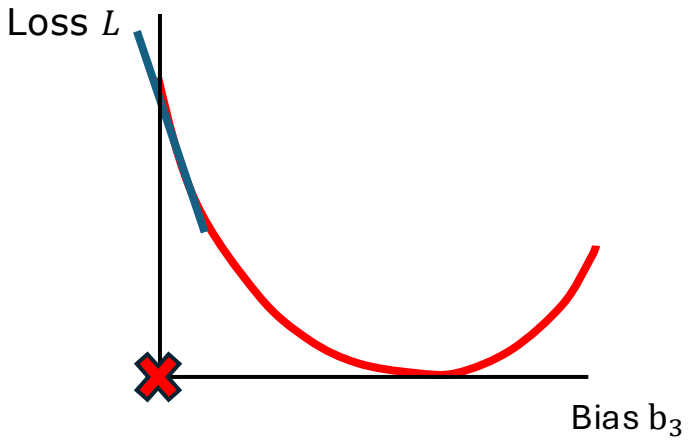
Remember
**Gradient Descent:**

$$w := w - \eta \frac{\partial L}{\partial w}$$

$$\boldsymbol{Step\ Size = 0 - 0.1 \cdot (-15.7) = 1.57}$$

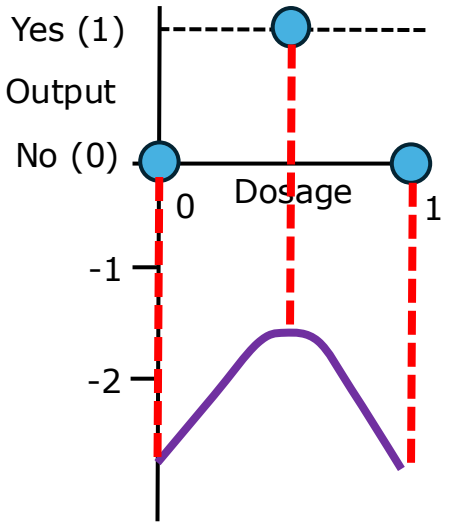
E.g. Learning Rate 0.1



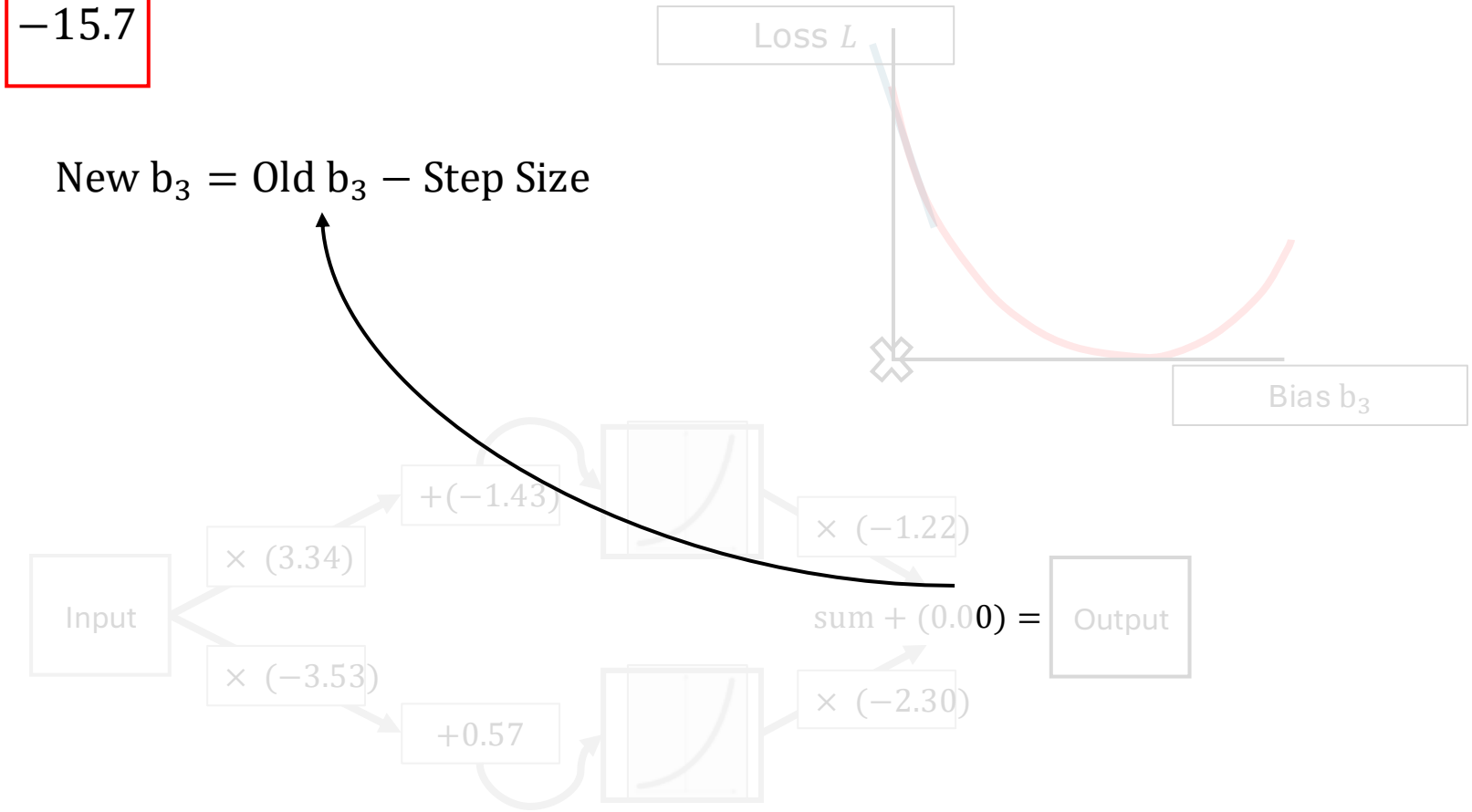



# Backpropagation Example

$$\frac{dL}{db_3} = \boxed{-15.7}$$

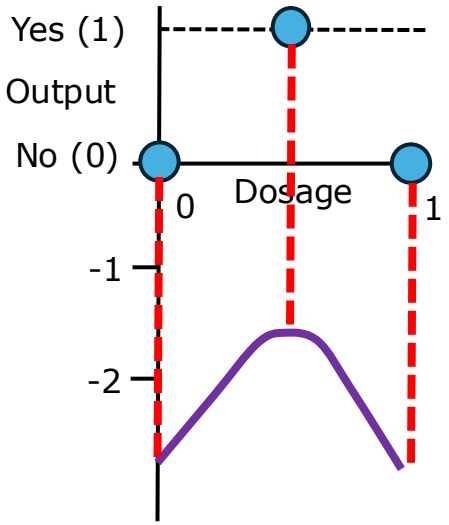



$$\text{New } b_3 = \text{Old } b_3 - \text{Step Size}$$

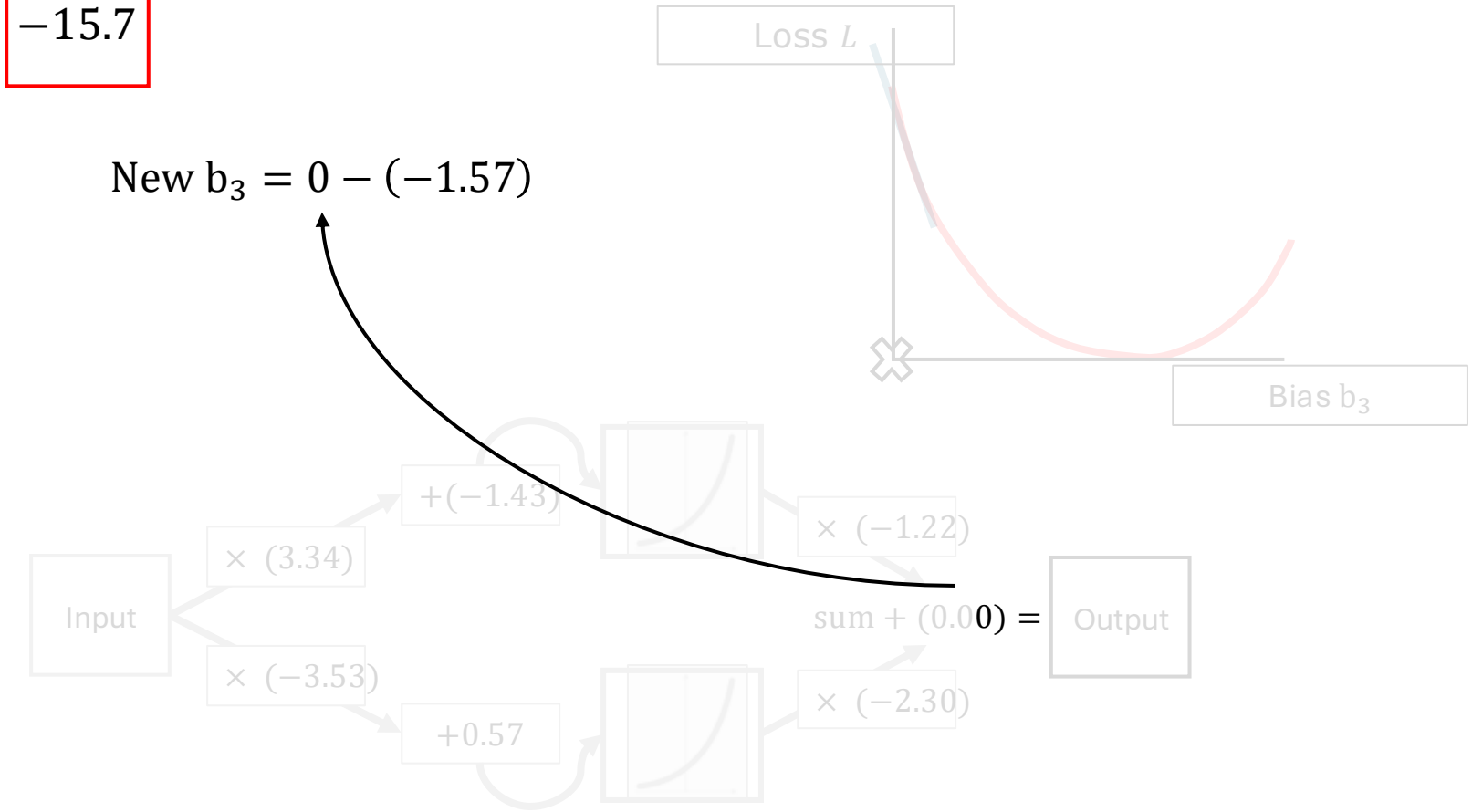


# Backpropagation Example

$$\frac{dL}{db_3} = \boxed{-15.7}$$

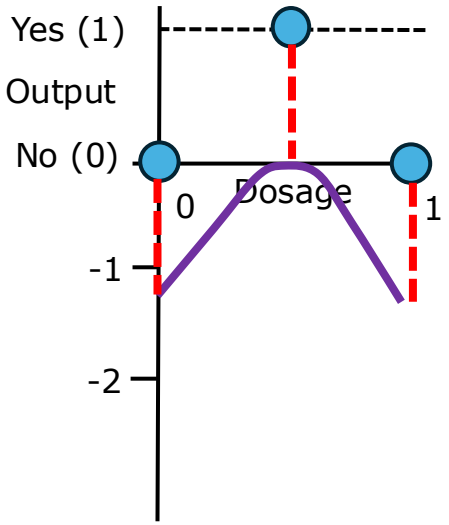



$$\text{New } b_3 = 0 - (-1.57)$$

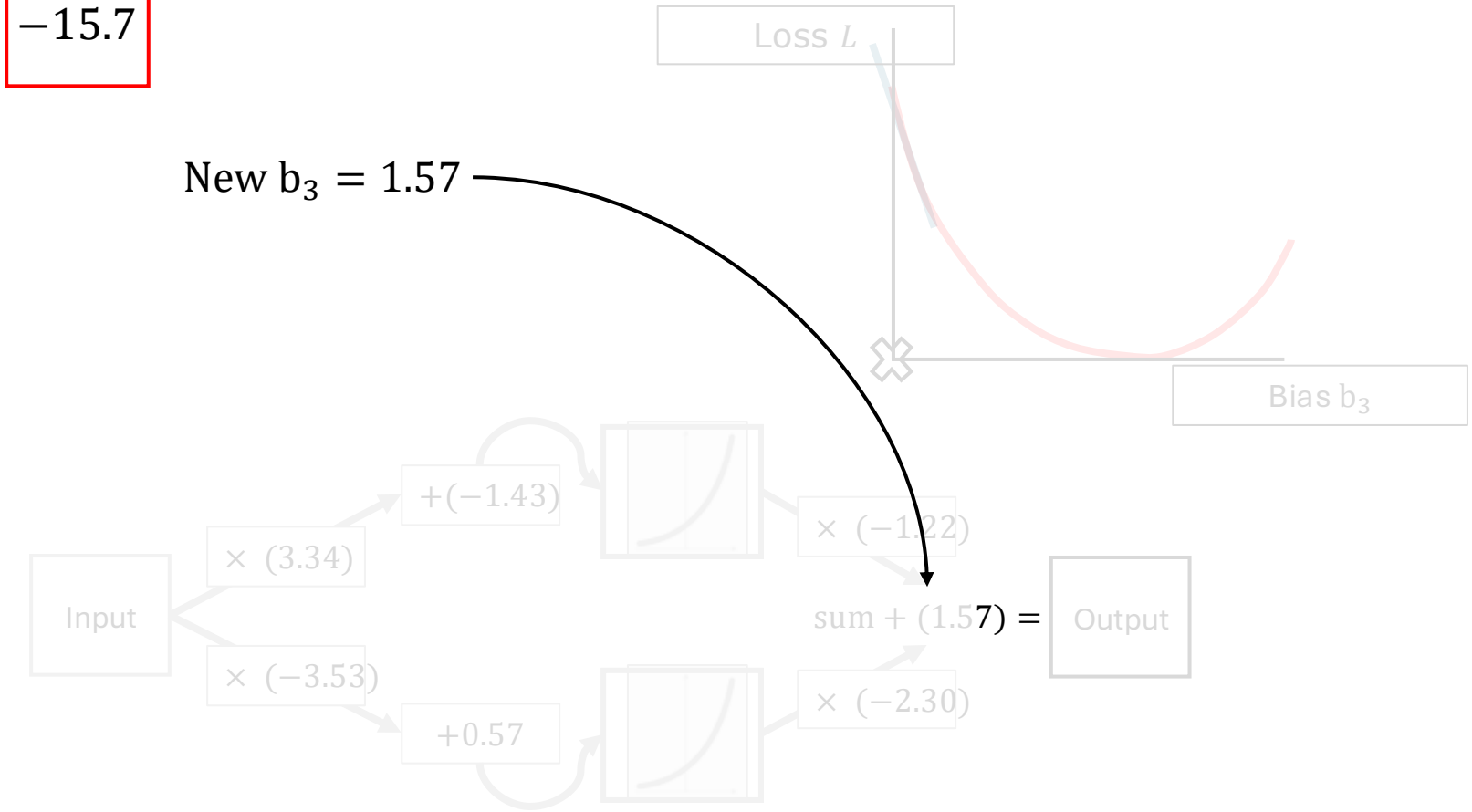


# Backpropagation Example

$$\frac{dL}{db_3} = \boxed{-15.7}$$

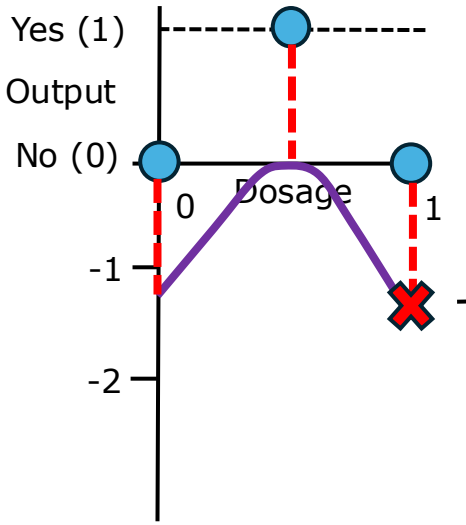



New $b_3 = 1.57$

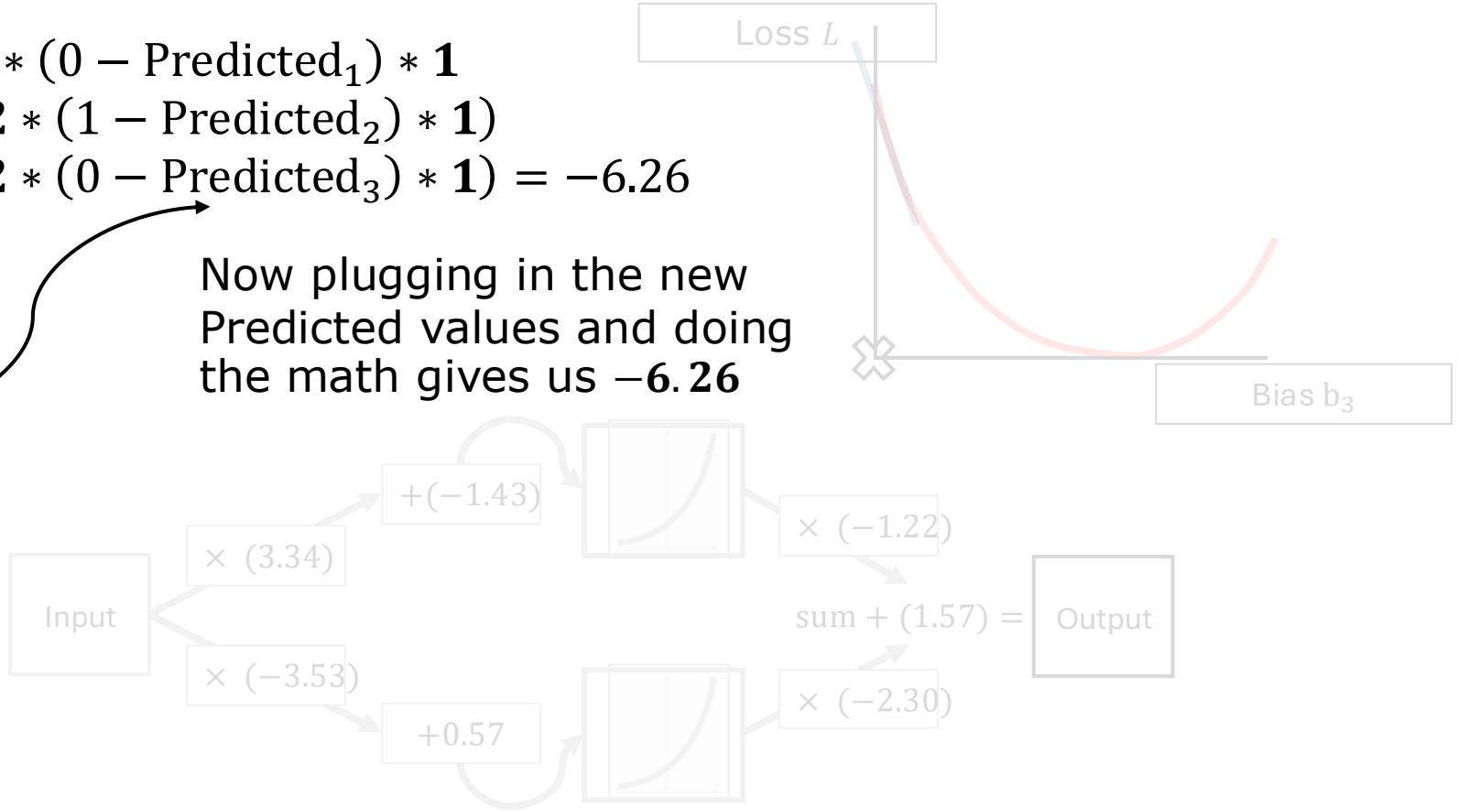


# Backpropagation Example

$$\frac{dL}{db_3} = -2 * (0 - \text{Predicted}_1) * 1 + (-2 * (1 - \text{Predicted}_2) * 1) + (-2 * (0 - \text{Predicted}_3) * 1) = -6.26$$

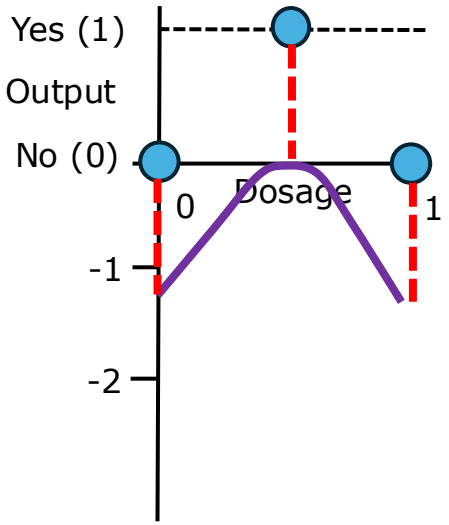


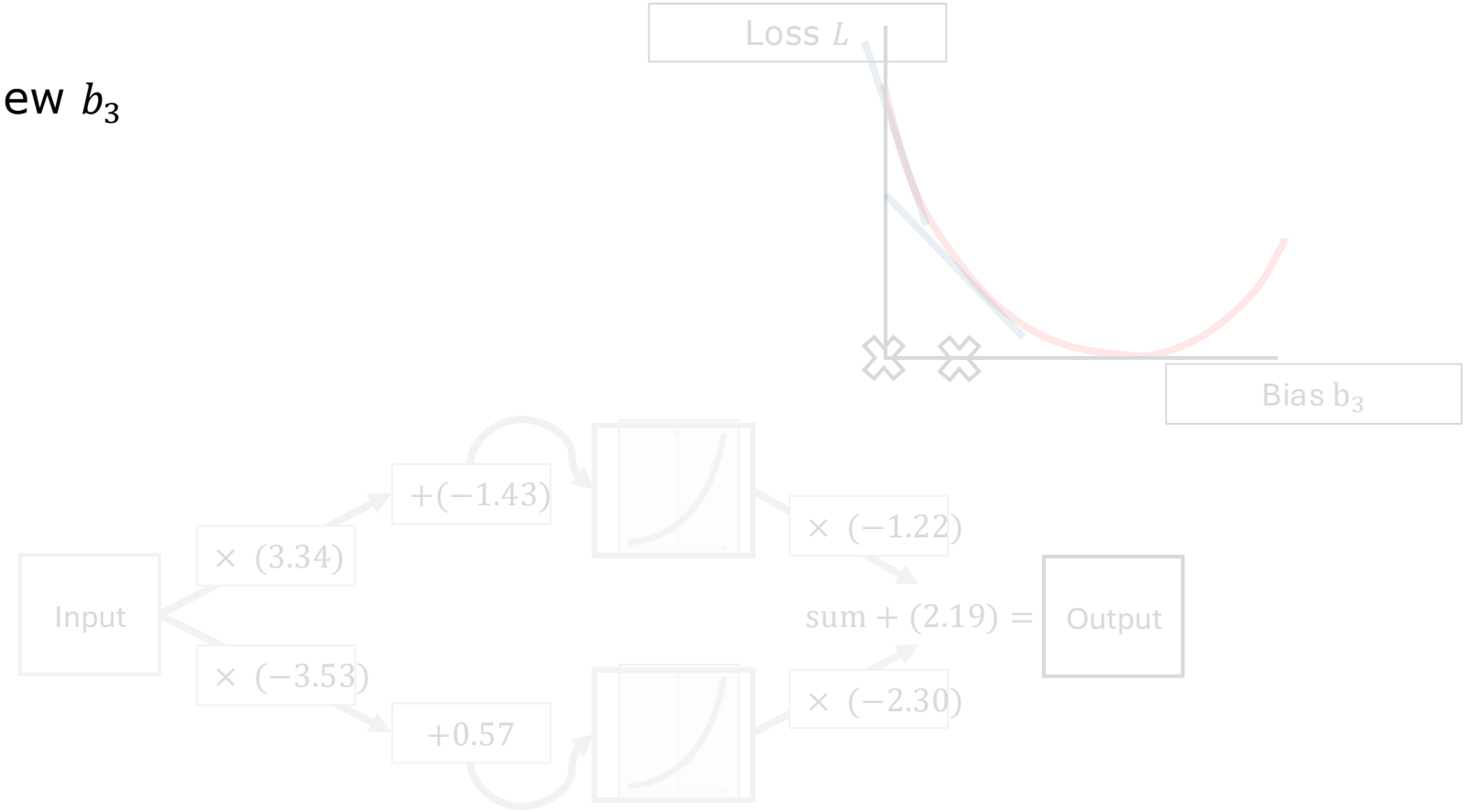

Now plugging in the new Predicted values and doing the math gives us $-6.26$



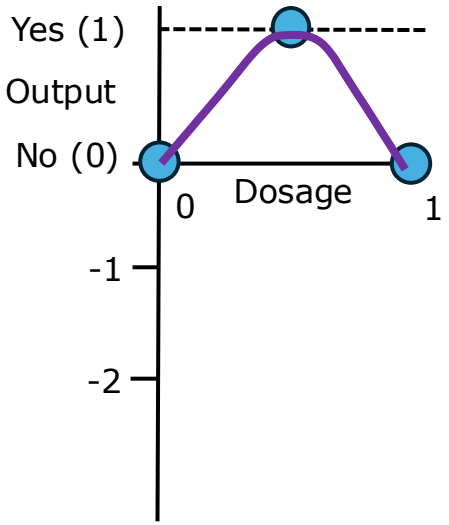
# Backpropagation Example

New $b_3$



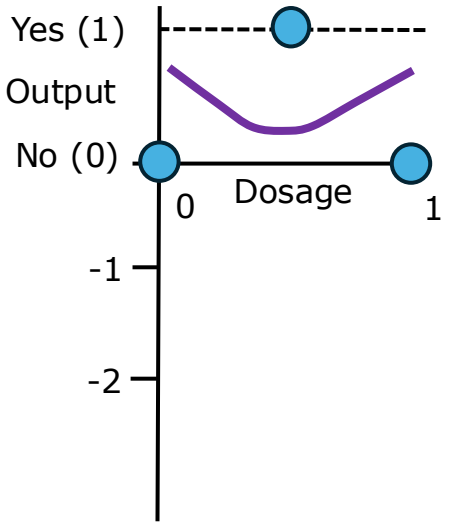
# Backpropagation Example

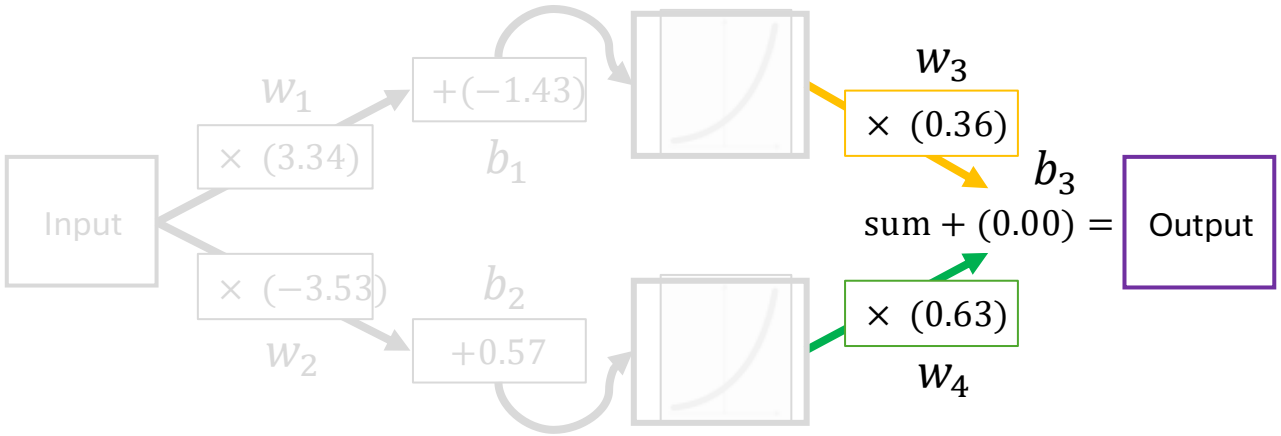



By proceeding we'll always get a new $b_3$ till convergence

# Backpropagation Example

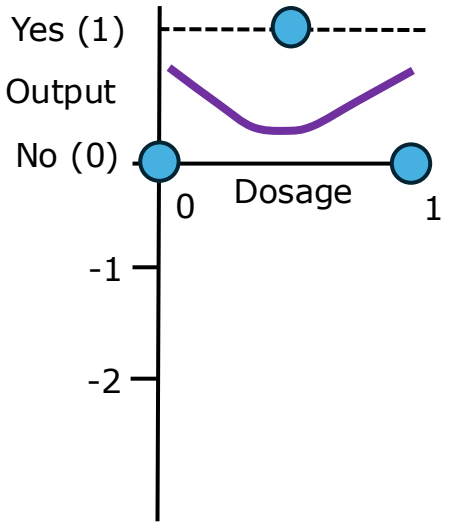
Now, we would like to optimize all other parameters as well, e.g. $w_3$ and $w_4$



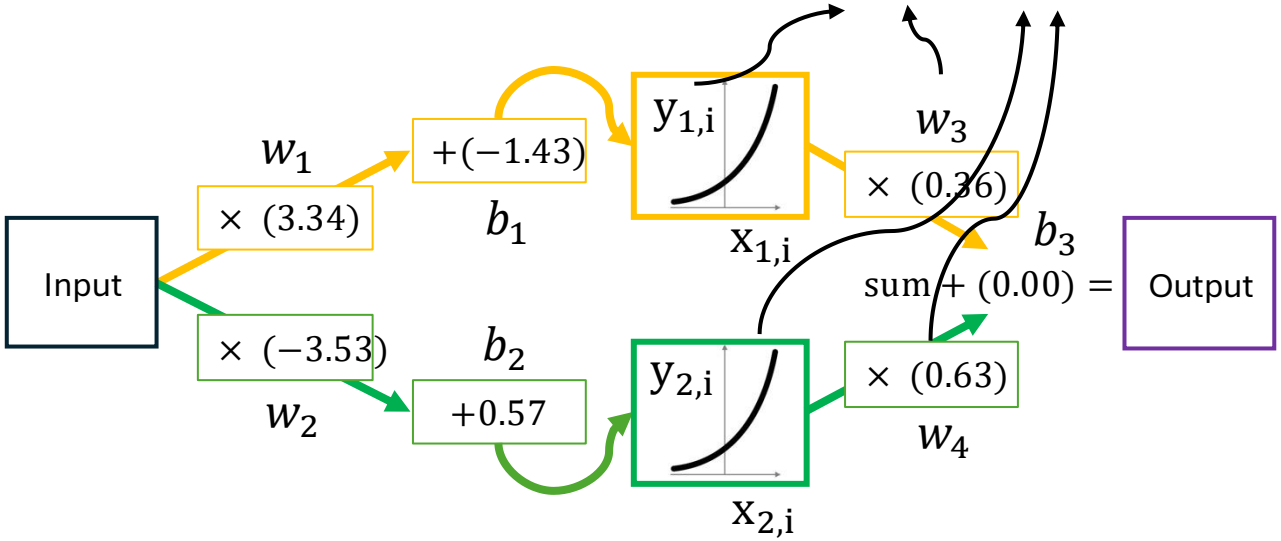



# Backpropagation Example

Now, we would like to optimize all other parameters as well, e.g. $w_3$ and $w_4$




$$\text{Predicted}_i = \text{purple squiggle}_i = y_{1,i}w_3 + y_{2,i}w_4 + b_3$$



# Backpropagation Example

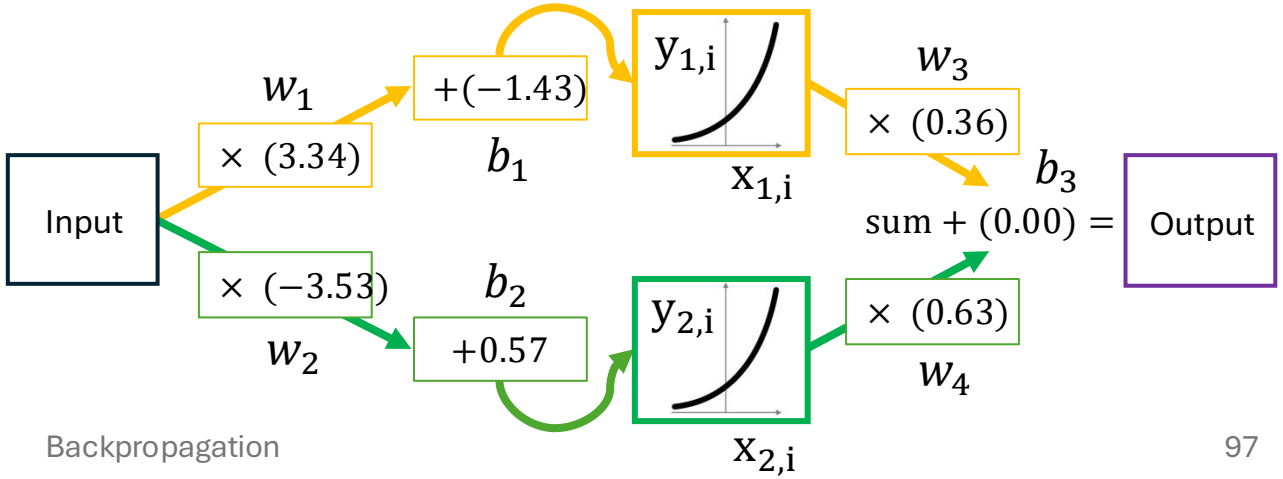
$$\frac{dL}{dw_3} = \frac{dL}{dPredicted} * \frac{dPredicted}{dw_3}$$

$$\frac{dL}{dw_4} = \frac{dL}{dPredicted} * \frac{dPredicted}{dw_4}$$

← Goal is to compute this

$$L = \sum_{i=1}^{n=3} (\text{Observed}_i - \text{Predicted}_i)^2$$

$$\text{Predicted}_i = \text{purple squiggle}_i = y_{1,i} w_3 + y_{2,i} w_4 + b_3$$



# Backpropagation Example

$$\frac{dL}{dw_3} = \boxed{\frac{dL}{dPredicted}} * \frac{dPredicted}{dw_3}$$

$$\frac{dL}{dw_4} = \boxed{\frac{dL}{dPredicted}} * \frac{dPredicted}{dw_4}$$

*How?*

$$\frac{dL}{db_3} = \frac{dL}{dPredicted} * \frac{dPredicted}{db_3}$$

… is the exact same as the derivative used for $b_3$

# Backpropagation Example

$$\frac{dL}{dw_3} = \boxed{\frac{dL}{dPredicted}} * \frac{dPredicted}{dw_3}$$

$$\frac{dL}{dw_4} = \boxed{\frac{dL}{dPredicted}} * \frac{dPredicted}{dw_4}$$

*How?*

$$\frac{dL}{db_3} = \frac{dL}{dPredicted} * \frac{dPredicted}{db_3} = \sum_{i=1}^{n=3} -2 * (\text{Observed}_i - \text{Predicted}_i) * 1$$

# Backpropagation Example

$$\frac{dL}{dw_3} = \frac{dL}{dPredicted} * \frac{dPredicted}{dw_3} = \sum_{i=1}^{n=3} -2 * (\text{Observed}_i - \text{Predicted}_i)$$

$$\frac{dL}{dw_4} = \frac{dL}{dPredicted} * \frac{dPredicted}{dw_3} = \sum_{i=1}^{n=3} -2 * (\text{Observed}_i - \text{Predicted}_i)$$

$$\frac{dPredicted}{dw_3} = \frac{d}{dw_3}\text{purple squiggle} = \frac{d}{dw_3}(y_{1,i}w_3 + y_{2,i}w_4 + b_3) = y_{1,i}$$

$$\frac{dPredicted}{dw_4} = \frac{d}{dw_4}\text{purple squiggle} = \frac{d}{dw_4}(y_{1,i}w_3 + y_{2,i}w_4 + b_3) = y_{2,i}$$

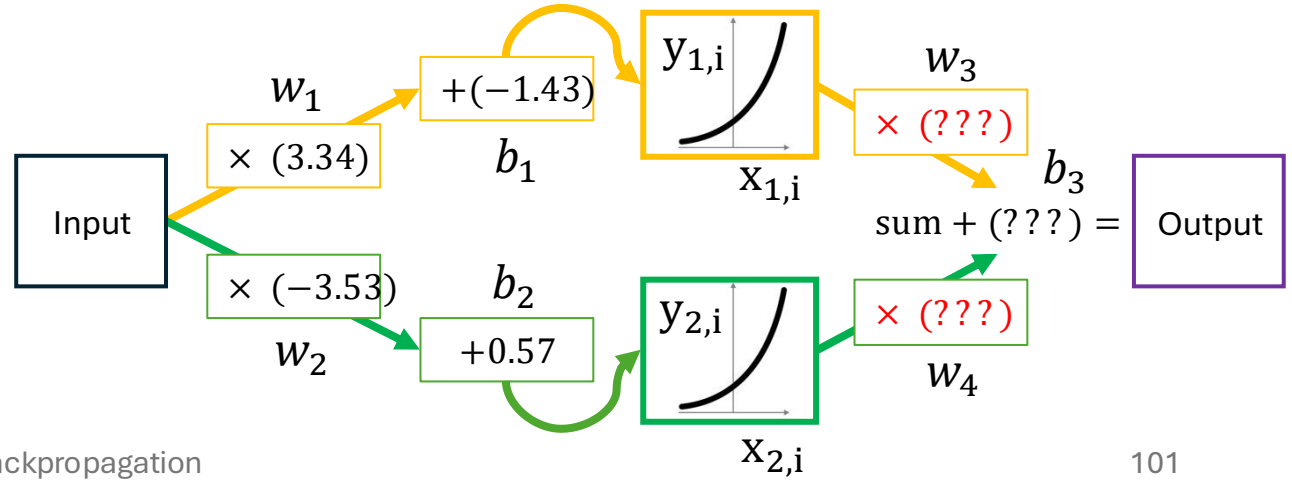
# Backpropagation Example

$$\frac{dL}{dw_3} = \frac{dL}{dPredicted} * \frac{dPredicted}{dw_3} = \sum_{i=1}^{n=3} -2 * (\text{Observed}_i - \text{Predicted}_i) * y_{1,i}$$

$$\frac{dL}{dw_4} = \frac{dL}{dPredicted} * \frac{dPredicted}{dw_3} = \sum_{i=1}^{n=3} -2 * (\text{Observed}_i - \text{Predicted}_i) * y_{2,i}$$

$$\frac{dL}{db_3} = \frac{dL}{dPredicted} * \frac{dPredicted}{db_3} = \sum_{i=1}^{n=3} -2 * (\text{Observed}_i - \text{Predicted}_i) * 1$$

… we can plug them into Gradient Descent to optimize $w_3$, $w_4$, and $b_3$



# Backpropagation Example

$$\frac{dL}{dw_3} = \frac{dL}{dPredicted} * \frac{dPredicted}{dw_3} = \sum_{i=1}^{n=3} -2 * (\text{Observed}_i - \text{Predicted}_i) * y_{1,i}$$

$$\frac{dL}{dw_4} = \frac{dL}{dPredicted} * \frac{dPredicted}{dw_3} = \sum_{i=1}^{n=3} -2 * (\text{Observed}_i - \text{Predicted}_i) * y_{2,i}$$

$$\frac{dL}{db_3} = \frac{dL}{dPredicted} * \frac{dPredicted}{db_3} = \sum_{i=1}^{n=3} -2 * (\text{Observed}_i - \text{Predicted}_i) * 1$$

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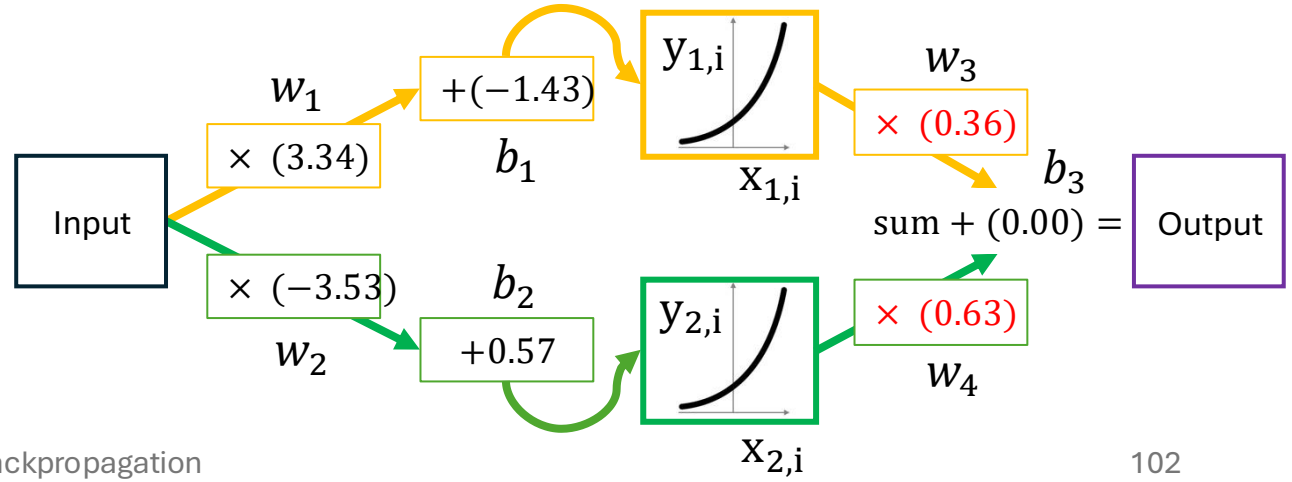
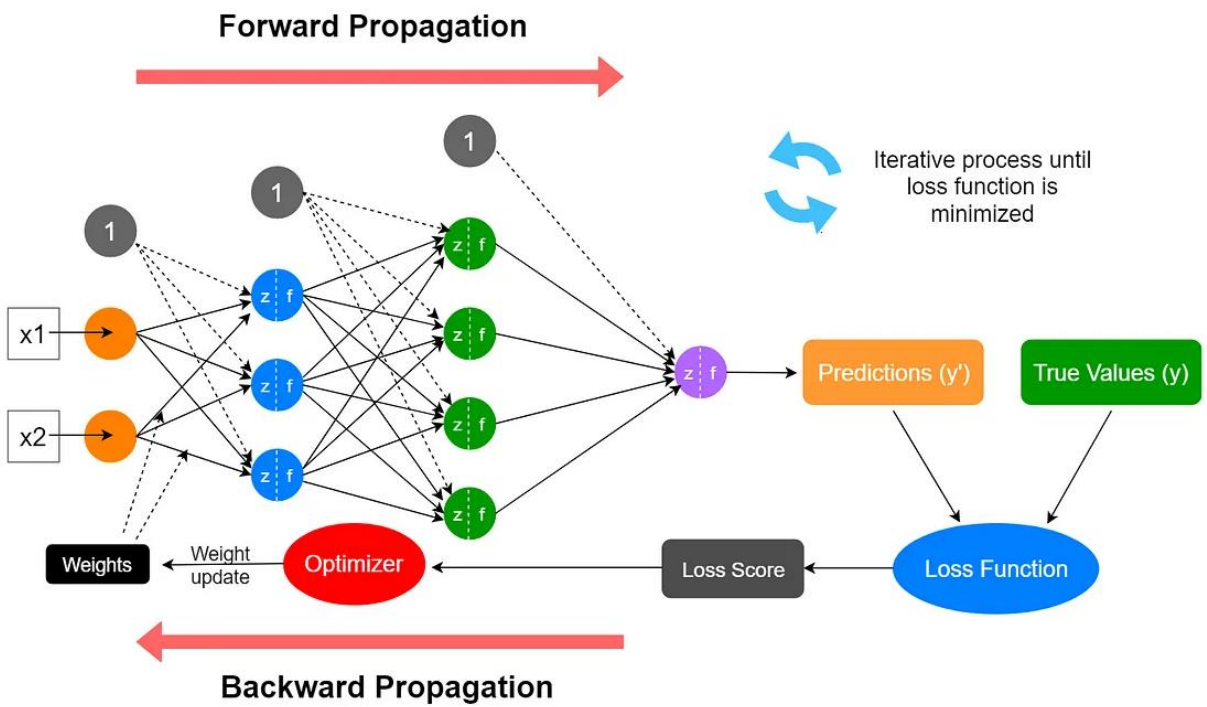


# What to Take Home

Backpropagation is:

- Efficient gradient computation
- Based on chain rule
- Uses stored forward values
- Enables gradient descent in deep networks




https://wandb.ai/mostafaibrahim17/ml-articles/reports/Decoding-Backpropagation-and-Its-Role-in-Neural-Network-Learning--Vmlldzo3MTY5MzM1

# Additional Sources

**Videos**

- The Most Important Algorithm in Machine Learning (Artem Kirsanov ~ 40 min)
- Neural Networks Pt. 2: Backpropagation Main Ideas (StatQuest with Josh Starmer ~ 18 min)
- Backpropagation, intuitively | Deep Learning Chapter 3 (3Blue1Brown ~ 13 min)

**Slides**

- Lecture 4: Backpropagation and Neural Networks (Stanford CS231n: Deep Learning for Computer Vision)

**Books**

- The StatQuest Illustrated Guide to Neural Networks and AI: With hands-on examples in PyTorch!!! (Josh Starmer)

**Contact**

**Mail:**
**konak@uni-potsdam.de**

**Slides from Today**

# Intuition

- If prediction too small → gradient negative
- Weight increases
- Model improves

Learning = repeated small corrections

# Pitfalls

**Vanishing Gradients** $\sigma'(z) \approx 0$:

- Gradients shrink layer by layer
- Deep layers stop learning

**Exploding Gradients (**If derivatives > 1 repeatedly**):**

- Gradients grow exponentially

**Solutions:**

- ReLU
- Proper initialization
- BatchNorm
- Gradient clipping
- Residual connections